

The Gaussian free field limit of the balanced six-vertex model with variance multiplier $1/\arcsin(c/2)$

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Abstract

We prove that the height function of the square-lattice six-vertex model with weights $a = b = 1$ and $0 < c \leq 2$, in the plane state obtained from balanced tori, converges to a multiple of the Gaussian free field. For unit height jumps and Green kernel $-(2\pi)^{-1} \log |x - y|$, the squared multiplier is $1/\arcsin(c/2)$.

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1 Introduction

The six-vertex model assigns an arrow to each edge of the square lattice, subject to the rule that two arrows enter and two arrows leave each vertex. Its height function is a discrete random surface. We prove the Gaussian free field limit of this surface throughout the critical range $a = b = 1$, $0 < c \leq 2$, for the plane state specified below.

1.1 The plane law and the height convention

The two vertices having both horizontal arrows directed inward, or both directed outward, have weight c ; the other four vertices have weight one. A finite configuration has the product of its vertex weights. On

$$T_{M,L} = (\mathbb{Z}/M\mathbb{Z}) \times (\mathbb{Z}/L\mathbb{Z}), \quad M, L \geq 4, \quad L \text{ even},$$

condition every vertical column of horizontal edges to contain $L/2$ left arrows and $L/2$ right arrows. This is a conserved sector of the row transfer matrix. We first take $M \rightarrow \infty$, and then even $L \rightarrow \infty$, in the local weak topology of arrow configurations. We denote the resulting plane law by $\mathbb{P}_c$. Existence in the previously untreated range is part of the theorem.

Give the face on the left of an arrow height one more than the face on its right. The ice rule makes this a well-defined height h , modulo an additive constant, on the faces of the plane. For $x = (x_1, x_2) \in \mathbb{R}^2$, let $h(x)$ be the height of the face with bottom-left corner $(\lfloor x_1 \rfloor, \lfloor x_2 \rfloor)$, and set

$$h_\delta(x) = h(x/\delta), \quad \delta > 0.$$

This convention also fixes the value on grid lines.

Let Γ be the real plane Gaussian free field modulo constants, normalized by the Green kernel

$$G(x, y) = -\frac{1}{2\pi} \log |x - y|. \quad (1.1)$$

Thus its covariance on smooth compactly supported tests of total integral zero is the integral of G against their product. For compactly supported finite signed measures μ, ν of total mass zero, write

$$\mathcal{E}(\mu, \nu) = \iint G(x, y) \mu(dx) \nu(dy).$$

We say that μ has finite energy if

$$\iint \log_+ \frac{1}{|x - y|} |\mu|(dx) |\mu|(dy) < \infty, \quad \log_+ t = \max(\log t, 0).$$

For compactly supported finite signed measures, this is equivalent to requiring $\mathcal{E}(\mu, \mu)$ to be an ordinary finite signed-measure integral. In particular, the definition does not allow cancellation between infinite positive and negative energies.

Theorem 1.1. *For every fixed $c \in (0, 2]$, the prescribed iterated balanced-torus limit $\mathbb{P}_c$ exists. Define*

$$\Delta = 1 - \frac{c^2}{2}, \quad \sigma(c)^2 = \frac{2}{\arccos \Delta} = \frac{1}{\arcsin(c/2)}. \quad (1.2)$$

As $\delta \downarrow 0$ through arbitrary positive real values, the fields h_δ converge to $\sigma(c)\Gamma$ in each of the following senses.

1. *For every finite family $\mu_1, \dots, \mu_m$ of compactly supported finite signed measures of total mass zero and finite energy,*

$$(\langle h_\delta, \mu_j \rangle)_{j=1}^m \implies \mathcal{N}\left(0, (\sigma(c)^2 \mathcal{E}(\mu_i, \mu_j))_{i,j=1}^m\right).$$

2. *For every bounded open $U \subset \mathbb{R}^2$ and $-1 < \alpha < 0$, there is convergence in law, with tightness, in the ordinary restriction spaces*

$$\mathcal{C}_{\text{res}}^\alpha(U) := B_{\infty, \infty}^\alpha(U), \quad B_{p, q}^\alpha(U), \quad W^{\alpha, p}(U), \quad 1 \leq p < \infty, \quad 1 \leq q \leq \infty,$$

modulo constants. Convergence also holds in the outward completion (1.4) defined below. No regularity of ∂U is required. One may fix a smooth compactly supported ρ of integral one and replace each field X by $X - \langle X, \rho \rangle$, using the same pinning for the limit.

3. For every $k \geq 1$ and every tuple $(u_i, u'_i)_{i=1}^k$ with $\{u_i, u'_i\} \cap \{u_j, u'_j\} = \emptyset$ for $i \neq j$,

$$\mathbb{E}_c \prod_{i=1}^k (h_\delta(u'_i) - h_\delta(u_i)) \longrightarrow \sigma(c)^k \sum_{\pi} \prod_{\{i,j\} \in \pi} C_{ij}, \quad (1.3)$$

where the sum is over pairings of $\{1, \dots, k\}$, is zero for odd k , and

$$C_{ij} = G(u'_i, u'_j) - G(u'_i, u_j) - G(u_i, u'_j) + G(u_i, u_j).$$

The convergence is uniform on compact sets of such tuples.

The theorem concerns this particular zero-slope plane state. It makes no uniqueness assertion among all Gibbs states or all slopes, and no estimate uniform as $c \downarrow 0$.

Regularity conventions. The restriction spaces in part 2 are restrictions of the corresponding whole-plane distribution spaces, with their quotient norms. For $W^{\alpha,p}$ we include both the Bessel-potential and the negative Slobodeckij realizations. The Hölder laws are supported on the separable closure of smooth functions. This choice makes the statement meaningful on every bounded open set without an extension theorem for its boundary. We also prove convergence in the intrinsic dual spaces $(W^{s,p'}(U))^*$, for $0 < s < 1$ and $1 \leq p < \infty$, where the fractional seminorm uses only pairs of points of U ; see Proposition 7.11.

The outward convention is the literal seminorm of [7, Definition 2.6(iii)]. Let $\mathcal{T}_1(Q)$, with $Q = (-1, 1)^2$, consist of compactly supported signed densities in Q , of integral zero, whose zero extensions are 1-Lipschitz. Set

$$\|f\|_{\text{out}, \alpha, U} = \sup_{\substack{0 < \varepsilon \leq 1, \\ \varphi \in \mathcal{T}_1(Q) \\ \text{supp } \varphi \subset U/\varepsilon}} \varepsilon^{-\alpha} \left| \int f(x/\varepsilon) \varphi(x) \, dx \right|. \quad (1.4)$$

Here U indexes the allowed tests of the global plane field; the sampled points x/ε need not lie in U . The space is the completion of $C_c^\infty(\mathbb{R}^2)$ modulo the kernel of this seminorm. The absolute value leaves the source convention unchanged because the test family is symmetric. Its convergence requires a separate argument from convergence in restriction spaces.

1.2 Prior work and the new ingredients

At $c = 1$, Lieb computed the exact square-ice entropy [22]. The Gaussian description of critical six-vertex heights and the prediction of its coupling belong to the Coulomb-gas approach. In the F-model regime $1 < c < 2$, Di Francesco, Saleur and Zuber [3, Section 2.3] give a Gaussian-action prediction which, after converting their height steps of $\pi/2$ to unit steps, agrees with (1.2). This physics prediction is distinct from convergence in the measure classes and distribution topologies of Theorem 1.1.

At the free-fermion point $c = \sqrt{2}$, a dimer representation makes the correlations determinantal. Kenyon's conformal-invariance and GFF theorems for domino tilings [15, 16] are foundational scaling-limit results in this setting. The periodic dimer Gibbs measures, surface tension and logarithmic fluctuations studied by Kenyon, Okounkov and Sheffield [17] describe the corresponding plane-state picture. Near the free-fermion point, Falco [8] derived arrow-correlation asymptotics and logarithmic height variance for the isotropic six-vertex model through an interacting-fermion representation. Giuliani, Mastropietro and Toninelli proved GFF convergence for a symmetric weakly interacting

dimer model [11] and the relation between its height amplitude and correlation exponent [10]. Their later treatment of tilted profiles [12, Section 2.3 and Theorem 2] includes the six-vertex interaction and establishes height-covariance and amplitude identities near the free-fermion point; its Remark 5 leaves the higher-cumulant extension needed for a full GFF theorem undeveloped.

Duminil-Copin, Kozłowski, Krachun, Manolescu and Tikhonovskaia [6] proved existence and condensation of Bethe roots and obtained the free energy and its expansion near half filling. Duminil-Copin, Karrila, Manolescu and Oulamara [4] proved logarithmic delocalization for $1 \leq c \leq 2$, using crossing estimates and free-energy information. Lis [24] developed the Baxter–Kelland–Wu representation for multi-point spin correlations and proved infinite-volume and delocalization results on subintervals of this range. Glazman and Lammers [13] subsequently obtained delocalization, including logarithmic bounds in the isotropic case, by percolation methods. These results establish the scale of fluctuations; identifying their joint law requires additional information. The variational framework of Sheffield [26] and the strict convexity theorem of Lammers and Tassy [21] provide the equilibrium tools used below when $c \geq 1$.

Duminil-Copin, Kozłowski, Lammers and Manolescu [7] give a nonperturbative GFF theorem on $\sqrt{3} \leq c \leq 2$, with the exact multiplier (1.2). Their proof combines transfer-matrix spectral representations and reflection positivity with rotational invariance and regularity estimates. The rotational-invariance input is listed as in preparation in the cited version. We retain the use of transfer spectra and reflection, but for $0 < c < 2$ construct the two Gaussian currents from finite row identities and uniform thermodynamic estimates. At $c = 2$, Appendix A supplies the needed rotational-invariance statement from the independent fair orientations of FK loops, the plane loop correspondence of [24], and the homotopy coupling underlying the FK rotation theorem of [5]. The remaining endpoint argument uses [7]. Its conditional amplitude theorem is also used on $1 \leq c < 2$, after convergence has been established. Thus the full-range statement concerns precisely the balanced plane law defined above, with its normalization fixed in physical height units.

The finite algebra is the algebraic Bethe ansatz. The norm determinant goes back to Gaudin, McCoy and Wu [9] and Korepin [20]; the scalar-product formula is due to Slavnov [27]. We use its trigonometric and diagonally twisted forms as developed by Kitanine, Maillet and Terras [19] and Kitanine, Maillet, Slavnov and Terras [18]. These exact finite identities are the algebraic input. Uniform control as the row length grows is a separate analytical problem, addressed in Section 3.

Three estimates connect this finite algebra to the limiting field. Root-tail control and an edge maximum principle give uniform inverses for Gaudin matrices and for mixed discrete–continuous quadrature operators. Finite-dimensional residue conditions then control simultaneous escapes of several rapidities in the determinant formulas. Next, bounds on the zeros of angular spectral functions outside two separated scale windows justify the limiting holomorphic and antiholomorphic currents. Finally, when $c < 1$, a homogeneous stiffness lower bound and a staggered vacuum-overlap upper bound identify the same covariance coefficient. This last step supplies the normalization where the conditional probabilistic input is unavailable. Moment estimates and desmoothing pass from the currents to the test classes and separated increments in Theorem 1.1.

1.3 Proof overview

For $0 < c < 2$, introduce

$$\lambda = \arccos(-\Delta) \in (0, \pi), \quad \eta = \lambda/\pi.$$

Section 2 defines commuting row operators and their diagonal marks, which record height increments. Section 3 constructs their limiting matrix elements, proves uniform tail bounds, and shows that words with widely separated angular parameters decorrelate. A positive full-line resolvent gives uniform density comparisons for every fixed positive λ , including the small- λ range.

These matrix elements define a Hilbert space with commuting angular rows in Section 4. Their joint spectral values are inner functions of a complex row parameter. The finite folded-row continuation argument in Lemma 4.2 gives a common analytic neighborhood; canonical inner-function factorization describes the possible zeros and boundary singularities [1]. A mark acts as a current. The finite mass κ of the spectrum transverse to common angular dilation is the coefficient of the logarithmic height covariance. Section 5 computes κ directly when $c < 1$; for $c \geq 1$ the cited conditional normalization theorem applies once convergence has been established.

Section 6 separates the angular spectrum into two scales and proves the Wick rules for the limiting currents. Section 7 passes from these currents to smooth tests, separated point increments and finite-energy measures, then proves convergence in negative regularity spaces. Its wavelet argument uses compactly supported wavelets [2] and the classical Besov coefficient characterization in the form of [14, Proposition 2.3]; restriction from the plane avoids any regularity assumption on the boundary of U . The final proof combines the interval $0 < c < 2$ with the endpoint argument of Appendix A. Each thermodynamic passage specifies which size tends to infinity first.

2 Finite rows and their scalar products

The first step is finite-dimensional. We express every fixed word of marked rows as a finite sum of determinants on the Bethe roots, with all normalizing factors specified. The construction of those roots and the estimates needed to pass to infinite volume are deferred to Section 3. In particular, no completeness assertion for Bethe eigenvectors is used in this section.

2.1 The physical row and the standard monodromy

The algebra below holds for $0 < \lambda < \pi$, corresponding to the open parameter interval $0 < c < 2$. The endpoint $c = 2$ is treated separately in Appendix A. Set

$$\lambda = \arccos(-\Delta), \quad \eta = \lambda/\pi, \quad v = \pi/2, \quad x = i\eta z, \quad w = e^{iz}. \quad (2.1)$$

The physical time angle is $z = 0$, increasing time points east, and the space angle is v , increasing space points north. Thus a common imaginary translation of the z arguments is a common real translation of their x arguments. At a path step, the spin is $+1$ when the arrow points along its left normal; the height increment is minus that spin. These conventions fix the sign of every marked row.

Let L be even and let $b_0, \dots, b_{L-1}$ be real numbers. On $\mathcal{H}_L = (\mathbb{C}^2)^{\otimes L}$ use the orthonormal spin basis, and put

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

An auxiliary copy of $\mathbb{C}^2$ is denoted by the subscript a . Products over the sites below have increasing site index from left to right. Define

$$R(u) = \begin{pmatrix} \sin(\lambda - u)/\sin \lambda & 0 & 0 & 0 \\ 0 & \sin u/\sin \lambda & 1 & 0 \\ 0 & 1 & \sin u/\sin \lambda & 0 \\ 0 & 0 & 0 & \sin(\lambda - u)/\sin \lambda \end{pmatrix}, \quad (2.2)$$

$$M_{\text{ph}}(x) = \prod_{j=0}^{L-1} R_{aj}(\lambda/2 + i(x - b_j)), \quad T^D(x) = \text{tr}_a(D_a M_{\text{ph}}(x)), \quad (2.3)$$

where D is an arbitrary complex 2×2 matrix. Its row index is on the right tile of the horizontal seam and its column index on the left. The four indices in (2.2) send the top and left step spins to the bottom and right step spins. When $b_j = 0$ and $x = 0$, dividing all vertex weights by $\sin(\lambda/2)/\sin \lambda$ gives $a = b = 1$ and $c = 2 \cos(\lambda/2)$, hence $\Delta = -\cos \lambda$.

Write $[u] = \sinh u$ and $s_\lambda = [i\lambda] = i \sin \lambda$. The standard monodromy used in the algebraic Bethe ansatz is

$$L_{aj}(x - b_j) = \begin{pmatrix} [x - b_j + i\lambda/2]/s_\lambda & 0 & 0 & 0 \\ 0 & [x - b_j - i\lambda/2]/s_\lambda & 1 & 0 \\ 0 & 1 & [x - b_j - i\lambda/2]/s_\lambda & 0 \\ 0 & 0 & 0 & [x - b_j + i\lambda/2]/s_\lambda \end{pmatrix},$$

$$M(x) = \prod_{j=0}^{L-1} L_{aj}(x - b_j) = \begin{pmatrix} A(x) & B(x) \\ C(x) & L(x) \end{pmatrix}_a. \quad (2.4)$$

In particular, the physical mixed diagonal entry has the opposite sign. Let

$$\mathcal{P}_L = \prod_{j \text{ even}} Z_j, \quad \mathcal{Q}_L = \prod_{j=0}^{L-1} Z_j. \quad (2.5)$$

Lemma 2.1 (Finite row algebra). *For even L , real b_j , and all complex spectral arguments,*

$$M_{\text{ph}}(x) = \mathcal{P}_L M(x) \mathcal{P}_L \mathcal{Q}_L, \quad T^D(x)^* = T^{X\bar{D}X}(\bar{x}). \quad (2.6)$$

The rows with a common diagonal twist commute. More generally, put

$$\mathcal{R}(u) = \begin{pmatrix} [u + i\lambda]/s_\lambda & 0 & 0 & 0 \\ 0 & [u]/s_\lambda & 1 & 0 \\ 0 & 1 & [u]/s_\lambda & 0 \\ 0 & 0 & 0 & [u + i\lambda]/s_\lambda \end{pmatrix}, \quad \mathcal{R}_{\text{ph}}(u) = Z_a \mathcal{R}(u) Z_b.$$

Then

$$\mathcal{R}_{\text{ph},ab}(p - q) M_{\text{ph},a}(p) M_{\text{ph},b}(q) = M_{\text{ph},b}(q) M_{\text{ph},a}(p) \mathcal{R}_{\text{ph},ab}(p - q). \quad (2.7)$$

Where the intertwiner is invertible, exchanging two rows conjugates their joint auxiliary mark by $\mathcal{R}_{\text{ph}}(p - q)$. For the homogeneous chain, $T^I(i\lambda/2)$ is the cyclic spin translation and $T^I(-i\lambda/2)$ is its inverse.

Proof. The local identity is $R_{aj}(\lambda/2 + i(x - b_j)) = Z_a L_{aj}(x - b_j) Z_j$. Both local matrices commute with $Z_a Z_j$. Moving the auxiliary Z 's through the product, and using that L is even, gives (2.6). This also proves that the gauge is independent of the roots, spectral arguments, and inhomogeneities.

If $N_{\alpha\beta}(x)$ is one auxiliary block of a physical local matrix, direct conjugation of its entries gives $N_{\alpha\beta}(x)^* = N_{\bar{\alpha}\bar{\beta}}(\bar{x})$, where a bar on a spin index exchanges its two values. In an entry of a monodromy, different local factors act on different quantum sites. They therefore commute when the adjoint reverses their order. Consequently

$$(M_{\text{ph},\alpha\beta}(x))^* = M_{\text{ph},\bar{\alpha}\bar{\beta}}(\bar{x}).$$

Taking the marked trace proves the second identity in (2.6); no reversal of the spatial inhomogeneities is involved.

The standard six-vertex Yang–Baxter identity gives $\mathcal{R}_{ab}(p-q)M_a(p)M_b(q) = M_b(q)M_a(p)\mathcal{R}_{ab}(p-q)$. It is a local identity of the three displayed six-vertex matrices: spin conservation reduces its nonconstant entries to the addition formula for $\sinh$. Moreover,

$$M_{\text{ph},a}(p)M_{\text{ph},b}(q) = Z_b \mathcal{P}_L M_a(p)M_b(q)\mathcal{P}_L Z_b,$$

and the analogous reversed product has Z_a in place of Z_b . Substitution yields (2.7). Taking auxiliary traces proves the assertion about marks. For a common diagonal matrix D , $D \otimes D$ commutes with the intertwiner, so the two twisted rows commute. Finally $R(0)$ is the permutation matrix, and tracing a chain of such permutations gives the cyclic translation. The other endpoint follows by the adjoint identity. $\square$

2.2 Bethe vectors, counting functions, and finite regularity

Let $|0\rangle = |+\rangle^{\otimes L}$. It is annihilated by $\mathbf{C}(x)$, and its two diagonal eigenvalues are

$$a(x) = \prod_{j=0}^{L-1} \frac{[x - b_j + i\lambda/2]}{s_\lambda}, \quad l(x) = \prod_{j=0}^{L-1} \frac{[x - b_j - i\lambda/2]}{s_\lambda}. \quad (2.8)$$

The scalar functions in the creation-entry commutation rules are

$$f(u) = \frac{[u + i\lambda]}{[u]}, \quad g(u) = \frac{s_\lambda}{[u]}, \quad \beta(u) = \frac{g(u)}{f(u)} = \frac{s_\lambda}{[u + i\lambda]}. \quad (2.9)$$

In particular $\beta(0) = 1$ and $1/f(0) = 0$ by continuation. For a finite list U set $|U\rangle = \prod_{u \in U} \mathbf{B}(u)|0\rangle$. The creation entries commute, so this vector is symmetric in U . Fix an integer $0 \leq n \leq L$. For a list $Y = (y_1, \dots, y_n)$ of distinct roots define

$$F_Y(x) = \prod_{y \in Y} \frac{[y - x + i\lambda]}{[y - x]},$$

$$q_Y(x) = \frac{l(x)}{a(x)} \prod_{y \in Y} \frac{[x - y + i\lambda]}{[x - y - i\lambda]}, \quad (2.10)$$

$$\tau_t(x; Y) = e^{it} a(x) F_Y(x) (1 + e^{-2it} q_Y(x)). \quad (2.11)$$

Here t is real near zero. We say that Y is on shell for t when

$$q_Y(y_j) = -e^{2it} \quad (1 \leq j \leq n). \quad (2.12)$$

The apparently singular eigenvalue in (2.11) then has removable poles at the roots.

For real u put

$$s(u) = \frac{\sin \lambda}{\pi(\cosh(2u) - \cos \lambda)}, \quad K(u) = \frac{\sin(2\lambda)}{\pi(\cosh(2u) - \cos(2\lambda))}. \quad (2.13)$$

At $\lambda = \pi/2$, the second function is identically zero. Let S and Θ be their odd primitives, vanishing at zero. Define

$$\xi_Y(u) = \sum_{j=0}^{L-1} S(u - b_j) - \sum_{k=1}^n \Theta(u - y_k). \quad (2.14)$$

Consecutive real roots are solutions of

$$\xi_Y(y_j) = j - \frac{n+1}{2} + \frac{t}{\pi}, \quad y_1 < \dots < y_n. \quad (2.15)$$

Lemma 2.2 (Finite Bethe data). *Let L be even, b_j real, and $0 < \lambda < \pi$. A real solution of (2.15) satisfies (2.12). If its Bethe vector is nonzero, then $|Y\rangle$ is an eigenvector of $e^{it}\mathbf{A}(x) + e^{-it}\mathbf{L}(x)$ with eigenvalue $\tau_t(x; Y)$, and $\mathcal{P}_L|Y\rangle$ is an eigenvector of the physical twisted row with eigenvalue $(-1)^n \tau_t(x; Y)$. The Jacobian of (2.15) is*

$$G_{jk} = \xi'_Y(y_j)\delta_{jk} + K(y_j - y_k), \quad \xi'_Y(u) = \sum_{a=0}^{L-1} s(u - b_a) - \sum_{k=1}^n K(u - y_k). \quad (2.16)$$

If $\det G \neq 0$, the roots have a real analytic continuation for real t near the given value, and on that branch

$$G \frac{dY}{dt} = \frac{1}{\pi} \mathbf{1}. \quad (2.17)$$

For $\lambda \geq \pi/2$, G is positive definite at every real root list. Whenever $\xi'_Y(y_j) > 0$, write

$$\ell_j = \frac{1}{\xi'_Y(y_j)}, \quad (K\ell)_{jk} = K(y_j - y_k)\ell_k; \quad G \operatorname{diag}(\ell) = I + K\ell. \quad (2.18)$$

These assertions concern finite matrices; they give no uniform bound on $(I + K\ell)^{-1}$ as L increases.

Proof. Logarithmic differentiation gives $q'_Y/q_Y = 2\pi i \xi'_Y$. Comparing the limits as $u \rightarrow +\infty$ fixes the integration constant:

$$q_Y(u) = (-1)^n e^{2\pi i \xi_Y(u)} \quad (u \in \mathbb{R}). \quad (2.19)$$

Indeed $S(+\infty) = (1 - \eta)/2$ and $\Theta(+\infty) = (1 - 2\eta)/2$, while $q_Y(+\infty) = e^{-i\lambda L + 2i\lambda n}$ and L is even. Equation (2.15) now gives the Bethe equations. The two diagonal action formulas proved in Lemma 2.3 below give the coefficient replacing a root p by x as

$$g(x - p) \left(e^{it} a(p) \prod_{y \neq p} f(y - p) - e^{-it} l(p) \prod_{y \neq p} f(p - y) \right).$$

The Bethe equation, with the self factor -1 removed, makes the parenthesis zero. The unchanged coefficient is (2.11). The physical eigenvalue follows from (2.6), since $\mathcal{Q}_L = (-1)^n$ on the sector with n minus spins.

Differentiating the left side of (2.15) with respect to every root gives (2.16); the $K(0)$ term cancels the derivative of the self interaction. The analytic implicit-function theorem gives the local branch, and differentiation with respect to t gives (2.17). When $K \leq 0$,

$$G_{jj} = \sum_a s(y_j - b_a) + \sum_{k \neq j} |K(y_j - y_k)| > \sum_{k \neq j} |G_{jk}|.$$

The matrix is real symmetric, so strict diagonal dominance proves positive definiteness. The formula for $G \operatorname{diag}(\ell)$ follows entry by entry. $\square$

In the remainder of the paper, a *regular real on-shell list* means a list of distinct real roots satisfying the Bethe equations, with $\xi'_Y(y_j) > 0$ for every j and $\det G_Y \neq 0$. Section 3 constructs lists with these properties.

We also record the full-line kernels that will be used to compare the finite matrices with convolution. With $\hat{f}(p) = \int_{\mathbb{R}} e^{-ipu} f(u) du$,

$$\hat{s}(p) = \frac{\sinh((\pi - \lambda)p/2)}{\sinh(\pi p/2)}, \quad \widehat{K}(p) = \frac{\sinh((\pi - 2\lambda)p/2)}{\sinh(\pi p/2)}, \quad (2.20)$$

$$h := \|K\|_{L^1(\mathbb{R})} = |1 - 2\eta| < 1, \quad r(u) := (I + K*)^{-1}s(u) = \frac{1}{2\lambda \cosh(u/\eta)}. \quad (2.21)$$

Here $K*$ denotes convolution with K . To verify the transforms, integrate $e^{-ipu} \sin \theta / [\pi(\cosh(2u) - \cos \theta)]$ around a rectangle of height $i\pi$, for $0 < \theta < 2\pi$. The two residues at $i\theta/2$ and $i(\pi - \theta/2)$ give $\sinh((\pi - \theta)p/2) / \sinh(\pi p/2)$; coincident or zero values follow by continuity. Take $\theta = \lambda, 2\lambda$. The kernel K has constant sign, proving the L^1 formula, and $\widehat{s}/(1 + \widehat{K}) = 1/(2 \cosh(\lambda p/2))$ proves the formula for r . In particular $\int r = 1/2$. We write

$$R_0* = K*(I + K*)^{-1}; \quad R_0 + K*R_0 = K. \quad (2.22)$$

The inverse exists, for example on bounded functions, by the Neumann series since $h < 1$.

2.3 Exact actions on products of creation entries

For the next formulas all arguments are initially distinct and avoid the zeros of their displayed denominators. They are identities of meromorphic functions; coincident cases are obtained by taking the limit of the complete expression. We use the scalar functions f, g, β defined in (2.9).

Lemma 2.3 (Single-entry actions). *For a finite spectral set U , put $E = U \cup \{v'\}$. Then*

$$\begin{aligned} B(v')|U\rangle &= |E\rangle, \\ A(v')|U\rangle &= \sum_{p \in E} a(p)\beta(v' - p) \prod_{u \in E \setminus \{p\}} f(u - p) |E \setminus \{p\}\rangle, \end{aligned} \quad (2.23)$$

$$L(v')|U\rangle = \sum_{p \in E} l(p)\beta(p - v') \prod_{u \in E \setminus \{p\}} f(p - u) |E \setminus \{p\}\rangle, \quad (2.24)$$

$$C(v')|U\rangle = \sum_{\substack{p, q \in E \\ p \neq q}} a(p)l(q)\beta(v' - p)\beta(q - v')f(q - p) \prod_{u \in E \setminus \{p, q\}} f(u - p)f(q - u) |E \setminus \{p, q\}\rangle. \quad (2.25)$$

The last sum is over ordered pairs: p belongs to the upper diagonal factor and q to the lower diagonal factor. Thus a word of m entries is an explicit sum over paths that add the new argument at each step and remove zero, one, or two elements. If the final set has the original cardinality, its sets of removed original roots and surviving added arguments have equal size, at most m .

Proof. The standard RTT relation gives commuting creation entries and

$$\begin{aligned} A(p)B(u) &= f(u - p)B(u)A(p) + g(p - u)B(p)A(u), \\ L(p)B(u) &= f(p - u)B(u)L(p) + g(u - p)B(p)L(u), \\ [C(p), B(u)] &= g(p - u)(A(u)L(p) - A(p)L(u)). \end{aligned}$$

Move the diagonal entry successively through the creation entries and apply its vacuum value. Combining terms that replace the same root uses

$$f(u - h) - f(u - p) = g(p - u) \frac{[p - h]}{[u - h]}.$$

This gives (2.23); reversing differences gives (2.24). These formulas include the unchanged term $p = v'$ because $\beta(0) = 1$.

For completeness, the two-removal induction can be checked before any on-shell specialization. Induct on $|U|$. Strip one creation at u from the product, use the commutator, and apply the two diagonal formulas. For fixed upper and lower removals h, k , divide the coefficients by

$$a(h)l(k)f(k - h) \prod_{s \text{ remaining}} f(s - h)f(k - s).$$

The required addition of the three resulting contributions is

$$\begin{aligned}\beta(v' - h)\beta(k - v') &= \frac{\beta(v' - h)\beta(k - v')}{f(u - h)f(k - u)} \\ &\quad + g(v' - u) \left(\frac{\beta(u - h)\beta(k - v')}{f(k - u)} - \frac{\beta(v' - h)\beta(k - u)}{f(k - v')} \right).\end{aligned}$$

Substituting (2.9) and multiplying by the displayed denominators reduces this identity to the addition formula for $\sinh$. The initial case is $C(v')|0\rangle = 0$; for one creation the two ordered removals give the displayed commutator on the vacuum. The induction proves (2.25), including cases where p or q is the newly added argument. Finally, there are only m newly added arguments. If the final cardinality equals the initial one, the number of absent original roots equals the number of surviving added arguments and is therefore at most m . $\square$

We now have an exact expansion of a finite row word into off-shell Bethe vectors. The remaining finite-dimensional step is to take their scalar products with the initial vector and cancel the vacuum factors.

2.4 Slavnov's identity with its normalization factors

The scalar-product determinant below specializes to the norm determinant associated with Gaudin, McCoy and Wu and proved by Korepin [9, 20]. We use the twisted Slavnov identity to keep the exact prefactors in our convention visible.

Let Y be on shell for t and have n distinct real roots. For a list U of length n , define the algebraic scalar product

$$S(Y, U) = \langle 0 | \prod_{y \in Y} C(y) \prod_{u \in U} B(u) | 0 \rangle. \quad (2.26)$$

It is not defined using a Hermitian adjoint; that identification will be proved separately. Write $C(Y, U)_{jk} = 1/[U_k - y_j]$, and put

$$\begin{aligned}m_u^t(y) &= -\frac{1}{2\pi i} [(1 + e^{-2it} q_Y(u)) \coth(u - y) \\ &\quad - \coth(u - y - i\lambda) - e^{-2it} q_Y(u) \coth(u - y + i\lambda)],\end{aligned} \quad (2.27)$$

$$M^t(Y, U)_{jk} = m_{U_k}^t(y_j). \quad (2.28)$$

All root coincidences in these matrices mean analytic limits along the full formula. In particular

$$M^t(Y, Y) = G_Y. \quad (2.29)$$

The derivatives $\partial_{y_j} \tau_t$ below are taken with respect to the independent root coordinates before imposing the on-shell equations; they are not derivatives along a Bethe-root branch.

Lemma 2.4 (Exact scalar products and deleted columns). *In the normalization (2.4),*

$$S(Y, U) = e^{-int} \left(\prod_{y \in Y} l(y) \right) \frac{\det_{j,k} \partial_{y_j} \tau_t(U_k; Y)}{\det C(Y, U)}. \quad (2.30)$$

Consequently, if $\det G_Y \neq 0$, then $S(Y, Y) \neq 0$ and

$$S(Y, Y) = (2\pi i s_\lambda)^n \left(\prod_{y \in Y} a(y) l(y) \right) \left(\prod_{j \neq k} f(y_j - y_k) \right) \det G_Y. \quad (2.31)$$

Suppose U is obtained by replacing the roots indexed by $H = (p_1, \dots, p_k)$ by $V = (v'_1, \dots, v'_k)$, in those same slots. Set

$$A_Y(p) = a(p) \prod_{y \in Y \setminus \{p\}} f(y - p) \quad (p \in Y).$$

Then the normalized scalar product is exactly

$$\frac{S(Y, U)}{S(Y, Y)} = (-s_\lambda)^{-k} \frac{\prod_{v' \in V} a(v') F_Y(v')}{\prod_{p \in H} A_Y(p)} \frac{\det((G_Y^{-1} m_{v'}^t)_p)_{H, V}}{\det(1/[v' - p])_{H, V}}. \quad (2.32)$$

If the positive weights (2.18) are defined, the last numerator equals

$$\det((G_Y^{-1} m_{v'}^t)_p)_{H, V} = \left(\prod_{p \in H} \ell_p \right) \det(W_{v'}(p))_{H, V}, \quad W_{v'} = (I + K\ell)^{-1} m_{v'}^t. \quad (2.33)$$

Empty products and determinants have value one.

Proof. We apply the trigonometric scalar-product theorem to the fundamental six-vertex monodromy (2.4), with highest vector $|0\rangle$, vacuum functions a, l , and diagonal twist $\kappa = e^{-2it}$. To match the conventions of [18], take its anisotropy to be $i\lambda$ and its inhomogeneities to be $b_j + i\lambda/2$, with a relabeling of sites if necessary. Its monodromy is $s_\lambda^L M(x)$, so the scalar product and the vacuum-times-derivative determinant both acquire the same factor s_λ^{2Ln} . After cancelling this factor, the prefactor is $\prod_{y \in Y} l(y)$ for the eigenvalue $A + \kappa L$; this is [18, Proposition 3.1, equations (3.8)–(3.9)]. Replacing that eigenvalue by $\tau_t = e^{it} \tau_\kappa$ multiplies its n derivative columns by e^{it} and therefore gives the compensating factor e^{-int} in (2.30). The untwisted derivative-over-Cauchy version, with its equal-spin-normalized monodromy, is [19, Theorem 3.1, equations (3.4)–(3.5)]. These are finite scalar-product identities; they make no assertion about completeness or about a thermodynamic limit.

Differentiation of (2.11) gives

$$\partial_{y_j} \tau_t(u; Y) = e^{it} a(u) F_Y(u) (-2\pi i) m_u^t(y_j).$$

The factor e^{int} cancels that in (2.30). When $u \rightarrow y_k$, the product $(1 + e^{-2it} q_Y(u)) \coth(u - y_k)$ tends to $-2\pi i \xi_Y'(y_k)$, because $q_Y'(y_k) = -2\pi i e^{2it} \xi_Y'(y_k)$. The two remaining $\coth$ terms give $K(y_k - y_j)$ after division by $-2\pi i$. This proves (2.29), including its diagonal.

At a root p the residue of the extracted vacuum factor is

$$\lim_{u \rightarrow p} [u - p] a(u) F_Y(u) = -s_\lambda A_Y(p). \quad (2.34)$$

Cancel these poles with the corresponding Cauchy poles. If every column tends to its root, the remaining determinant is $\det G_Y$, giving (2.31). For partial coincidence first cancel the unchanged columns. The determinant of the matrix obtained from G_Y by replacing columns H by $m_{v'}^t$, divided by $\det G_Y$, is $\det((G_Y^{-1} m_{v'}^t)_p)_{H, V}$: multiply the matrix on the left by G_Y^{-1} and expand along its unchanged unit columns. Expanding the Cauchy determinant along its unchanged poles leaves exactly $\det(1/[v' - p])_{H, V}$ with the same slot order. Each deleted root loses the factor in (2.34). This proves (2.32), including its sign and constant. Finally $G_Y^{-1} = \text{diag}(\ell)(I + K\ell)^{-1}$ proves (2.33).

All factors in the norm outside $\det G_Y$ are nonzero: real y, b give $[y - b \pm i\lambda/2] \neq 0$, and distinct real roots give $[y - y'] \neq 0$ and $[y - y' \pm i\lambda] \neq 0$. The scalar-product theorem is first used at generic variables. At any regular on-shell specialization, the nonsingular counting Jacobian permits a local analytic perturbation of the roots along with the parameters. Meromorphic continuation and the pole cancellations above then give the asserted formulas there as well. In particular, no exclusion at a root of unity is necessary. $\square$

2.5 Hilbert duality and the finite-word formula

Lemma 2.5 (The Hilbert dual). *For even L and real inhomogeneities,*

$$\mathbf{B}(x)^* = -\mathbf{C}(\bar{x}). \quad (2.35)$$

Thus for a distinct real on-shell list with $\det G_Y \neq 0$, the physical unit vector

$$\Omega_Y = \frac{\mathcal{P}_L|Y\rangle}{\|\mathcal{P}_L|Y\rangle\|} \quad (2.36)$$

is well defined, and

$$\|\mathcal{P}_L|Y\rangle\|^2 = (-1)^n S(Y, Y) > 0.$$

If Y, Z are two such real on-shell lists of the same size, possibly for different twists, then

$$|\langle \Omega_Y, \Omega_Z \rangle|^2 = \frac{S(Y, Z)S(Z, Y)}{S(Y, Y)S(Z, Z)}. \quad (2.37)$$

These identities also hold for real staggered b_j , without a Perron interpretation of the vectors.

Proof. Polarize the adjoint identity in Lemma 2.1 over the four auxiliary marks. It gives $\mathbf{B}_{\text{ph}}(x)^* = \mathbf{C}_{\text{ph}}(\bar{x})$. Using the gauge identity and commuting $\mathcal{P}_L$ with $\mathcal{Q}_L$ gives $\mathbf{B}(x)^* = \mathcal{Q}_L \mathbf{C}(\bar{x}) \mathcal{Q}_L$. Since $\mathbf{C}$ changes the number of minus spins by one, it anticommutes with $\mathcal{Q}_L$, proving (2.35). Creation entries commute, so for real roots

$$(\mathcal{P}_L|Y\rangle)^* = (-1)^n \langle 0| \prod_{y \in Y} \mathbf{C}(y) \mathcal{P}_L.$$

Lemma 2.4 makes this norm nonzero, and positivity is then Hilbert-space positivity. Applying the same identity to both cross scalar products proves (2.37). Absorbing the twist into the monodromy merely replaces $\mathbf{B}$ by $e^{it}\mathbf{B}$ and $\mathbf{C}$ by $e^{-it}\mathbf{C}$, preserving the adjoint relation and the normalized ratio. $\square$

We can now combine the entry actions with the scalar-product formula. The result is a sum over the original roots removed by a row word. Its determinant size depends only on the word length, not on the chain length; this is the form needed for the thermodynamic limit. In the remainder of this subsection $t = 0$, Y is a regular real on-shell list, and the spectral arguments $v'_1, \dots, v'_m$ are distinct and generic. Define normalized physical rows by

$$A_L^D(x) = \frac{T^D(x)}{(-1)^n \tau_0(x; Y)}, \quad P_L = A_L^I, \quad J_L = A_L^Z, \quad (2.38)$$

on the open set where the denominator is nonzero. Then $P_L(x)\Omega_Y = \Omega_Y$. Holomorphic continuation and norm bounds for these rows will be proved using the estimates of Section 3.

For each path in Lemma 2.3, let T be the set remaining just after its current removal and define

$$U_Y(T, p) = \frac{\prod_{u \in T \setminus Y} f(u - p)}{\prod_{y \in Y \setminus (T \cup \{p\})} f(y - p)}, \quad L_Y(T, p) = \frac{\prod_{u \in T \setminus Y} f(p - u)}{\prod_{y \in Y \setminus (T \cup \{p\})} f(p - y)}, \quad (2.39)$$

$$\chi_Y(p) = \begin{cases} 1, & p \in Y, \\ q_Y(p), & p \notin Y. \end{cases} \quad (2.40)$$

Associate with an action at v' the correction factor

$$\gamma = \begin{cases} 1, & \text{B (no removal),} \\ \beta(v' - p)U_Y(T, p), & \text{A (remove } p), \\ \chi_Y(p)\beta(p - v')L_Y(T, p), & \text{L (remove } p), \\ \chi_Y(q)\beta(v' - p)\beta(q - v')f(q - p)U_Y(T, p)L_Y(T, q), & \text{C (remove } p, q). \end{cases} \quad (2.41)$$

These are explicit rational functions of hyperbolic sines. The only dependence on the original root set outside the factors displayed in (2.39) is through q_Y at added arguments.

Corollary 2.6 (Exact finite-word expansion). *Expand each auxiliary mark as*

$$T^{D_j} = \sum_{\alpha, \beta} (D_j)_{\beta\alpha} M_{\text{ph}, \alpha\beta}.$$

For each resulting entry word, let n_j be the number of minus spins after its j th action, with $n_0 = n$. Entry words with $n_m \neq n$ have zero expectation in Ω_Y . For a path of a word with $n_m = n$, let $H \subset Y$ be the roots absent at the end and let V be its surviving added arguments, ordered in the same replacement slots. Then $|H| = |V| =: k \leq m$, and its contribution to the normalized physical expectation is

$$\left(\prod_{j=1}^m (D_j)_{\beta_j \alpha_j} \right) (-1)^{\sum_{j=1}^m (n_{j-1} - n)} (-s_\lambda)^{-k} \frac{\prod_{j=1}^m \gamma_j}{\prod_{j=1}^m (1 + q_Y(v'_j))} \frac{\det(W_{v'}(p))_{H,V}}{\det(1/[v' - p])_{H,V}} \prod_{p \in H} \ell_p. \quad (2.42)$$

The expectation is the sum of these contributions. In particular, the determinant size is bounded by the word length. For diagonal entry words the parity sign in (2.42) is one. Among distinct removed original roots the correction product contains only reciprocal f factors, with one for each pair. Each removed original root also contributes one factor β joining it to the row argument at the step that removes it.

Proof. For this proof extend the vacuum base to added arguments by

$$\mathcal{A}_Y(p) = \begin{cases} A_Y(p), & p \in Y, \\ a(p)F_Y(p), & p \notin Y. \end{cases}$$

The coefficient of an upper removal is $\mathcal{A}_Y(p)\beta(v' - p)U_Y(T, p)$. The corresponding lower coefficient is $\mathcal{A}_Y(p)\chi_Y(p)\beta(p - v')L_Y(T, p)$: on a root the equality of the two vacuum bases is precisely (2.12) at $t = 0$, including its self factor -1 ; off a root their ratio is $q_Y(p)$. For two removals the extra factor is $f(q - p)$, exactly as in (2.25). Thus the coefficient of a path before taking its scalar product is the product of its corrections γ_j times one $\mathcal{A}_Y(p)$ for every removed element.

Every original root is either retained or removed once. Similarly, each new spectral argument is either retained in V or removed once, possibly in the very action that introduced it. Consequently the product of vacuum bases along the path is

$$\left(\prod_{p \in H} A_Y(p) \right) \left(\prod_{v' \in \{v'_1, \dots, v'_m\} \setminus V} a(v')F_Y(v') \right).$$

Multiply by (2.32). The deleted-root bases cancel, and the surviving arguments supply the missing vacuum bases. Dividing by $\prod_j \tau_0(v'_j; Y)$ leaves exactly $\prod_j (1 + q_Y(v'_j))^{-1}$ and $(-s_\lambda)^{-k}$. Equation (2.33) supplies the determinant and weights.

The Hilbert dual contributes the same $(-1)^n$ to the scalar product and norm, so it cancels by Lemma 2.5. Each physical entry, after division by $(-1)^n$, contributes $(-1)^{n_{j-1}-n}$ from $\mathcal{Q}_L$ on its input sector; adjacent $\mathcal{P}_L$ factors cancel. This proves the explicit parity sign. Different final spin sectors are orthogonal, proving the zero assertion.

Finally consider a pair of deleted original roots. If removed at different steps, the earlier root occurs in the denominator of (2.39) at the later removal. If removed together, the two denominators contribute $f(q-p)^{-2}$ and the extra factor $f(q-p)$ leaves one reciprocal. There are no other factors involving two original roots. The factor β at each original removal is explicit in (2.41). This proves the last claims. $\square$

For real u , $|f(u)|^2 = 1 + \sin^2 \lambda / \sinh^2 u \geq 1$. Thus the reciprocal factors between deleted roots cause no collision growth, and for a bounded generic added argument the corresponding β factor decays like $e^{-|p|}$ as its real root p escapes. These bounds control the pair factors and the individual removal factors. Simultaneous escape of several roots still requires a determinant bound, which will be proved in Section 3.

2.6 Two-twist overlaps

The normalization argument in Section 5 will also use a different consequence of the same scalar-product formula: the overlap of the unit vectors associated with two twists. Unlike a fixed row word, this overlap retains determinants of the full root-set size. We record its exact pair product here, before taking any thermodynamic limit.

To factor (2.37), denote the twists of Y, Z by t_Y, t_Z , respectively. Put

$$\Delta(Y) = \prod_{j < k} [y_k - y_j], \quad \mathcal{H}(Y, U) = \prod_{j, k} [y_j - U_k + i\lambda], \quad c_n = (2\pi i)^n (-1)^{n(n-1)/2}.$$

The trigonometric Cauchy identity rewrites (2.30) as

$$S(Y, U) = c_n \left(\prod_{y \in Y} l(y) \right) \left(\prod_{u \in U} a(u) \right) \frac{\mathcal{H}(Y, U)}{\Delta(Y)\Delta(U)} \det M^{t_Y}(Y, U). \quad (2.43)$$

Consequently all one-set factors cancel separately, leaving

$$|\langle \Omega_Y, \Omega_Z \rangle|^2 = \frac{\mathcal{H}(Y, Z)\mathcal{H}(Z, Y)}{\mathcal{H}(Y, Y)\mathcal{H}(Z, Z)} \frac{\det M^{t_Y}(Y, Z) \det M^{t_Z}(Z, Y)}{\det G_Y \det G_Z}. \quad (2.44)$$

This is a finite identity, including the full shifted pair product. In particular, along a finite regular twist branch $Y(t)$, take $Y = Y(0)$ and $Z = Y(t)$. The overlap is nonzero for t near zero, since its value at zero is one. Its local logarithm and derivatives there are therefore well defined. Neither changing the phases of the unit vectors nor changing the logarithm branch changes the quadratic coefficient. No neighborhood uniform in L is asserted here.

We now return to the fixed-word formula (2.42). Its scalar coefficients are explicit, but passing to the plane requires uniform control of the root measures, the Gaudin inverses, and determinants with several roots in the sparse ends. Section 3 supplies those estimates.

3 Thermodynamic row kernels

The finite formulas of Section 2 reduce a row word to sums over finitely many selected Bethe roots. To pass to a plane kernel we must control those sums even when several selected roots approach the

sparse ends of the root configuration. We first construct the real roots, then prove the inverse and determinant estimates that justify this passage. All constants in this Section may depend on the fixed parameter $0 < \lambda < \pi$. They need not remain bounded as $\lambda \downarrow 0$ or $\lambda \uparrow \pi$.

3.1 The three limiting procedures

We use the notation s, K, r, G, ξ of Section 2, and put

$$h = \int_{\mathbb{R}} |K(u)| du = |1 - 2\eta| < 1, \quad R_0 = K * (I + K*)^{-1}.$$

Here $K*$ denotes convolution on the line; the last expression means the convolution kernel of the indicated operator. Since $h < 1$, its Neumann series converges on bounded functions. There is also a number $a > 0$ such that the same series converges with the weight $e^{a|u|}$: choose a so that $\int e^{a|u|} |K(u)| du < 1$. In particular R_0 and each of its fixed derivatives are exponentially integrable.

The following limits have different purposes.

1. In the *homogeneous plane limit*, all $b_j = 0$ and L tends to infinity through even integers. If $\lambda > \pi/2$ we use $n = L/2$. If $0 < \lambda \leq \pi/2$, we use $n = L/2 - \lfloor L^\epsilon \rfloor$, where $\epsilon > 0$ is sufficiently small and fixed. Its permitted size is specified in the proof of Lemma 3.3.
2. In the *array limit*, which we use only for $\lambda > 2\pi/3$, put

$$b_j = \left(j - \frac{L-1}{2} \right) \frac{d}{2}, \quad B = \frac{Ld}{2}, \quad n = L/2.$$

First $B \rightarrow \infty$ through the allowed even widths, and then $d \downarrow 0$. A common real center a_B of the row arguments may vary with B , provided $|a_B| \leq (1-\gamma)B/2$ for a fixed $\gamma > 0$. Differences of arguments range over a fixed compact set.

3. In the *regular-filling limit*, also used subsequently only for $\lambda > 2\pi/3$, the chain is homogeneous. First $L \rightarrow \infty$ with $n/L \rightarrow \rho < 1/2$, and then $\rho \uparrow 1/2$. We assert comparison with the plane state in this procedure for equal-time local matrices.

For example, uniform array convergence to a function F means

$$\lim_{d \downarrow 0} \limsup_{B \rightarrow \infty} \sup_{\substack{|a_B| \leq (1-\gamma)B/2 \\ \mathbf{x} \in \mathcal{K}}} |F_{d,B}(a_B + \mathbf{x}) - F(\mathbf{x})| = 0 \quad (3.1)$$

for every fixed compact parameter set $\mathcal{K}$ under discussion. It is not necessary to assert a limit for every fixed d . A uniform bound in this procedure means that there are $C, d_0 > 0$ such that for every $0 < d < d_0$ there is $B_0(d)$ for which the bound holds whenever $B \geq B_0(d)$. The corresponding convention for regular filling is $\lim_{\rho \uparrow 1/2} \limsup_{L \rightarrow \infty}$. Word lengths, matrices, derivative orders, and compact parameter sets are fixed before these limits. No estimate below is asserted uniformly in an unbounded word length.

3.2 Real roots, counting cells, and the sparse ends

There are two constructions of roots. At exact half filling with $K < 0$, the last few roots remain sparse and need individual estimates. At regular filling, and in the homogeneous near-half-filled sequence used when $K \geq 0$, even the terminal cells become dense. The next lemma supplies both constructions and the quadrature measure that will enter the finite-word formula.

Lemma 3.1 (Construction and density of the roots). *The zero-twist consecutive-root equations of Lemma 2.2 have distinct real solutions with the following properties.*

1. *If $\lambda > \pi/2$, the solutions at half filling exist for every finite homogeneous or real-array chain. Their counting derivative $\rho_L = \xi'$ is positive. If*

$$p_L(u) = \sum_{j=0}^{L-1} r(u - b_j),$$

then $\rho_L - p_L$ and their integrated discrepancy are uniformly bounded on the entire line.

2. *In the homogeneous chain, let*

$$r_Q = (I + K_Q)^{-1} s, \quad H_Q = (I + K_Q)^{-1} 1,$$

where K_Q integrates over $[-Q, Q]$. For all $Q \geq 0$,

$$cr(u) \leq r_Q(u) \leq Cr(u), \quad c \leq H_Q(u) \leq C \quad (|u| \leq Q). \quad (3.2)$$

The mass $\int_{-Q}^Q r_Q$ increases to $1/2$, and

$$\frac{1}{2} - \int_{-Q}^Q r_Q \asymp e^{-Q/\eta}.$$

Choose $Q = Q_L$ by $\int_{-Q}^Q r_Q = n/L$. If Q stays bounded with a positive limiting filling, or if $Lr(Q)$ tends to infinity at least as a fixed positive power of L , the roots exist for all sufficiently large L . If y_j^0 are the mid-quantiles of Lr_Q , then $Lr_Q(y_j^0)|y_j - y_j^0| \leq C$ uniformly in j , and

$$\xi'(u) = Lr_Q(u) + O(1) \quad (3.3)$$

on the interval, including a bounded counting extension at its endpoints.

3. *In all three limiting procedures the measures*

$$\nu_L = \sum_j \ell_j \delta_{y_j}, \quad \ell_j = \xi'(y_j)^{-1},$$

have uniformly bounded mass on intervals of length one. In a region whose slightly enlarged neighborhood has density at least M , their error against bounded Lipschitz tests on a fixed bounded interval is at most

$$\frac{C}{M} (\|\phi\|_\infty + \text{Lip}(\phi)). \quad (3.4)$$

Thus ν_L converges locally to Lebesgue measure in the dense portions of the limits.

At zero twist the homogeneous vectors constructed from these roots are the sector Perron vectors. The Gaudin matrices are nonsingular, so each constructed finite root list has a real-analytic twist branch near zero.

Proof. We first construct the exactly half-filled lists and control their sparse ends. We then compare the finite-interval density with the full-line density and use its quantiles to construct the dense lists. Both constructions will give the same local quadrature estimate.

Exactly half-filled roots. Suppose $K < 0$ and write $k = -K > 0$. The Jacobian of the counting equations is

$$G_{ii} = \sum_j s(y_i - b_j) + \sum_{j \neq i} k(y_i - y_j), \quad G_{ij} = -k(y_i - y_j) \quad (i \neq j).$$

Its quadratic form is the sum of a positive diagonal form and $\sum_{i < j} k(y_i - y_j)|u_i - u_j|^2$. It is strictly positive, so the equations are strictly convex gradient equations. Start at $\lambda = \pi/2$, where $K = 0$ and each counting level has one real preimage, and continue in a compact parameter interval. There can be no coincident roots because their prescribed counting levels are distinct.

There can be no escaping roots either. If k_0 roots tend to $+\infty$, sum their equations. The interaction primitives within that set cancel by oddness. Dividing by k_0 , the limiting excess of the left side over the prescribed levels is

$$(1 - \eta)(L/2 - n + k_0) > 0.$$

The same argument applies at $-\infty$. All site parameters here are finite in each continuation problem. Nonsingularity and this compactness continue the solution to the required parameter.

At half filling the total mass of $\rho_L = \xi'$ is n . The two outer counting half-levels occur at infinity. Consequently the signed measure and its primitive

$$e_L = \sum_j \delta_{y_j} - \rho_L(u) du, \quad F_L(u) = e_L((-\infty, u])$$

satisfy $|F_L| \leq 1/2$ and $F_L(\pm\infty) = 0$. The density equation and convolution inversion give

$$\rho_L - p_L = -R_0 * e_L = -R'_0 * F_L. \quad (3.5)$$

Integrating once also bounds the difference of the tail integrals by $\|R_0\|_1/2$. This proves the first assertion. Moreover $|\rho'_L| \leq C\rho_L$: both s and k are positive kernels with bounded logarithmic derivatives.

The sparse terminal cells. The preceding discrepancy estimate identifies the dense bulk, but a uniform inverse must also control the last few roots. We therefore estimate their densities and gaps directly. Number the roots from the right as $x_1 > x_2 > \dots$ and write $T(u) = \int_u^\infty \rho_L(v) dv$. Then $T(x_j) = j - 1/2$. For either $q = s$ or $q = k$, put $\bar{q}(u) = \int_u^\infty q(v) dv$. The displayed hyperbolic formulas for the kernels imply

$$\begin{aligned} q(u) &\leq C\bar{q}(u) & (u \in \mathbb{R}), \\ c\bar{q}(u) &\leq q(u), \quad \bar{q}(u+t) \leq Ce^{-2t}\bar{q}(u) & (u, t \geq 0). \end{aligned} \quad (3.6)$$

At a root x_j lying beyond every bare center, its own kernel and the $j - 1$ farther-out kernels contribute at most $h/2 + h(j - 1) = hT(x_j)$ to this tail. The remaining centers are inward. Their tail is at least $(1 - h)T(x_j)$, so

$$c(j - \tfrac{1}{2}) \leq \rho_L(x_j) \leq C(j - \tfrac{1}{2}). \quad (3.7)$$

This bounds each fixed-index reciprocal density away from both zero and infinity.

Terminal gaps cannot become long. If x_{j+1} also lies beyond the bare centers and $\Delta = x_j - x_{j+1}$, compare the inward tail at x_j with its tail at x_{j+1} using (3.6). It gives

$$(1 - h)(j - \tfrac{1}{2}) \leq Ce^{-2\Delta}(j + \tfrac{1}{2}), \quad \Delta \leq \frac{1}{2} \log \frac{3C}{1 - h}.$$

Conversely $1 = \int_{x_{j+1}}^{x_j} \rho_L$ and the upper tail comparison give $\Delta \geq c/(j+1)$. Thus each fixed-index gap has positive finite subsequential limits.

The explicit tail of p_L , together with its bounded integrated error, locates any sufficiently large fixed edge index at $e_L^{\text{edge}} + O(1)$, where

$$e_L^{\text{edge}} = \eta \log L \quad \text{or} \quad e_{d,B}^{\text{edge}} = B/2 + \eta \log(1/d) \quad (3.8)$$

in the homogeneous and array cases, respectively. For example, choose an index larger than the integrated error bound; the p_L tail is comparable to $\exp(-(u - e_L^{\text{edge}})/\eta)$ near the displayed edge. That root lies beyond the bare centers in the limiting procedures. The gap estimate then places the last root no more than a bounded distance beyond that edge. The left end is identical. For each fixed density threshold M , the portions with density below M start within a distance C_M of the nominal edges and contain only a bounded number of root indices. Both the number and the distance may depend on M .

These statements have their asserted array uniformity. In a purported failure sequence we may take $d \downarrow 0$ and B arbitrarily large depending on d . Every fixed edge index is then eventually beyond all bare centers, and (3.7) applies to every such index. One should not instead take a fixed- d edge limit and assert that its reciprocal weights tend to zero inward: the subsequent $d \downarrow 0$ limit is essential for that assertion.

Finite-interval densities. We turn to the dense construction. Its reference configuration will consist of quantiles of Lr_Q , so we first need positivity and a relative comparison with r , uniformly as Q grows. The inverse of $I + K_Q$ exists on bounded functions because its perturbation has norm at most $h < 1$. Comparison with the full-line equation gives

$$r_Q - r = (I + K_Q)^{-1} K_{\text{out}} r, \quad (3.9)$$

where K_{out} integrates over $\mathbb{R} \setminus [-Q, Q]$. When $K < 0$, write the right side as the negative of the exit kernel $(I - k_Q)^{-1} k_{\text{out}}$ applied to r . That positive kernel has mass at most h : the constant h is a supersolution of its mass equation, since $h - k_Q h - k_{\text{out}} 1 = (1 - h)k_Q 1 \geq 0$. For exterior v and interior u , we have $r(v) \leq r(Q) \leq r(u)$. Applying the exit-kernel mass bound to this inequality gives $(1 - h)r \leq r_Q \leq r$ on the interval. The positive Neumann series also gives $1 \leq H_Q \leq (1 - h)^{-1}$ in this case.

For $0 < \lambda < \pi/2$, a direct bound on the alternating Neumann series for $I + K_Q$ would lose positivity when h is large. Instead the full-line dressed kernel is positive. To see this explicitly, put

$$\vartheta = \frac{\pi\lambda}{\pi - \lambda}, \quad J_\lambda(u) = \frac{\sin \vartheta}{(\pi - \lambda)(\cosh(2\pi u/(\pi - \lambda)) - \cos \vartheta)}.$$

Here $0 < \vartheta < \pi$, so $J_\lambda \geq 0$. Rescaling the bare-kernel transform in (2.20) gives

$$\hat{J}_\lambda(p) = \frac{\sinh((\pi - 2\lambda)p/2)}{\sinh((\pi - \lambda)p/2)}.$$

Together with $\hat{r}(p) = (2 \cosh(\lambda p/2))^{-1}$, this identifies $R_0 = J_\lambda * r$. In particular

$$R_0 \geq 0, \quad \int_{\mathbb{R}} R_0 = \frac{h}{1 + h} =: \tau < \frac{1}{2}. \quad (3.10)$$

This argument uses integrable kernels, so Fourier uniqueness identifies the actual convolution kernel, not just its multiplier.

Let $I_Q = [-Q, Q]$ and $O_Q = \mathbb{R} \setminus I_Q$. The positive operator $u \mapsto R_0 * (\mathbf{1}_{O_Q} u)$ has sup norm at most τ . Its Neumann series defines bounded nonnegative functions on the entire line by

$$\begin{aligned}\bar{r}_Q &= r + R_0 * (\mathbf{1}_{O_Q} \bar{r}_Q), \\ \bar{H}_Q &= \frac{1}{1+h} + R_0 * (\mathbf{1}_{O_Q} \bar{H}_Q).\end{aligned}\tag{3.11}$$

Multiplying by $I + K*$ and using $(I + K*)R_0* = K*$ gives

$$\bar{r}_Q + K * (\mathbf{1}_{I_Q} \bar{r}_Q) = s, \quad \bar{H}_Q + K * (\mathbf{1}_{I_Q} \bar{H}_Q) = 1.$$

Their restrictions are therefore r_Q, H_Q , by uniqueness for $I + K_Q$. Since r decreases on $[0, \infty)$, the first equation in (3.11), restricted to O_Q , gives

$$\|\bar{r}_Q\|_{L^\infty(O_Q)} \leq r(Q) + \tau \|\bar{r}_Q\|_{L^\infty(O_Q)}, \quad \|\bar{r}_Q\|_{L^\infty(O_Q)} \leq \frac{r(Q)}{1-\tau}.$$

Applying the same equation inside I_Q and the second equation on the full line now yields

$$\begin{aligned}r(u) &\leq r_Q(u) \leq r(u) + \frac{\tau r(Q)}{1-\tau} \leq \frac{r(u)}{1-\tau}, \\ \frac{1}{1+h} &\leq H_Q(u) \leq 1 \quad (|u| \leq Q).\end{aligned}\tag{3.12}$$

At $\lambda = \pi/2$, $K = R_0 = 0$, $r_Q = r = s$, and $H_Q = 1$. We have thus proved (3.2) for every fixed $0 < \lambda < \pi$.

These arguments also give the relative derivative bounds needed for quadrature. In the positive-kernel case, differentiate the convolution in (3.11); each fixed derivative of R_0 is integrable, and the exterior source is bounded by $r(Q)/(1-\tau)$. Thus every fixed derivative of $r_Q - r$ is $O(r(Q))$ on I_Q . For $K < 0$, put $d_Q = r_Q - r$ and differentiate $d_Q = K_{\text{out}} r - K_Q d_Q$ in its output variable; the same bound follows from $\|d_Q\|_\infty \leq h r(Q)$ and the integrability of all fixed derivatives of K . Since $|r^{(j)}| \leq C_j r$ and $r(u) \geq r(Q)$ on the interval, in both cases $|r_Q^{(j)}(u)| \leq C_j r_Q(u)$. All these constants can be chosen uniformly when λ ranges over a compact subinterval of $(0, \pi)$. At an endpoint we extend r_Q by $s - K_Q r_Q$ outside the interval. For $K \geq 0$ this is the positive function $\bar{r}_Q$ just constructed; for $K < 0$ it is the positive sum $s + k_Q r_Q$. The same derivative bounds hold on an extension of bounded counting length at either endpoint, since its physical width is $O((Lr(Q))^{-1})$ in the dense limits used below.

Differentiation with respect to the interval endpoint, followed by symmetry of the kernel, gives

$$\frac{d}{dQ} \int_{-Q}^Q r_Q(u) du = 2r_Q(Q)H_Q(Q) > 0.\tag{3.13}$$

Indeed $(I + K_Q)\partial_Q r_Q = -r_Q(Q)(K(\cdot - Q) + K(\cdot + Q))$; pair this with H_Q . The just proved bound $|r_Q - r| \leq Cr(Q)$ shows that the integrated difference on $[-Q, Q]$ is at most $2CQr(Q)$ and tends to zero. Thus the mass tends to $\int_{\mathbb{R}} r = 1/2$. By (3.2), the right side of (3.13) is bounded above and below by positive constants times $r(Q)$. Integrating (3.13) from Q to infinity proves the asserted mass-deficit estimate.

From dense quantiles to exact roots. The interval density now has a strictly increasing mass and uniform relative derivative bounds. Choose Q to have the desired mass n/L , and let y_j^0 be the mid-quantiles of Lr_Q . The continuum counting function has derivative Lr_Q on the interval and, by oddness, gives the required levels at these quantiles. Midpoint quadrature against a function

of bounded variation bounds the discrete residual of each root equation by a constant. The same holds for every fixed derivative of the counting function: apply that quadrature to translates of the interaction kernel or its derivatives.

Put $m_L = Lr(Q)$, and use displacement coordinates $\zeta_j = Lr_Q(y_j^0)(y_j - y_j^0)$. At the quantiles the Jacobian in these coordinates is

$$I + \left(\frac{K(y_i^0 - y_j^0)}{Lr_Q(y_j^0)} \right)_{ij} + O(m_L^{-1})$$

in the sup row norm. The middle matrix has absolute row norm at most $h + o(1)$ by quadrature and the relative derivative bounds just proved. Its inverse is uniformly bounded. On any fixed-radius ball in the ζ coordinates, the physical displacements are $O(m_L^{-1})$. Neighbor sums of kernel derivatives, weighted by $1/(Lr_Q)$, are bounded. Comparing the bare-minus-root density to Lr_Q before displacing the roots also bounds the change of the divided diagonal entries by $O(m_L^{-1})$. Thus the Jacobian varies by $o(1)$ on that ball.

For clarity, Newton's map here may be taken with the inverse Jacobian frozen at the quantiles. Choose the ball radius larger than twice that inverse bound times the residual bound. For large L its derivative has norm less than $1/2$, so it maps the ball into itself and is a contraction. The fixed point gives the exact roots and (3.3). Positivity of the resulting counting derivative forces their ordering. The same estimates hold uniformly on every fixed compact continuation interval joining $\lambda > 0$ to $\pi/2$: throughout that interval h is bounded strictly below one, and the density and derivative comparisons above are uniform. For the near-half-filled sequence, the mass-deficit comparison also gives $Lr(Q) \asymp L^\epsilon$ uniformly there. The Jacobian is nonsingular; equivalently its symmetrization has positive eigenvalues, since the absolute row bound for the perturbation of the identity is below one.

Quadrature and the Perron vector. Both root constructions are now available, with positive counting density and a nonsingular Gaudin matrix. Their common local quadrature estimate follows from the counting cells. In a cell of density at least M , its length Δ_j is $O(M^{-1})$. The relative derivative bound gives $|\ell_j - \Delta_j| \leq C\Delta_j^2$. Integrate a Lipschitz test cell by cell and include the two partial boundary cells to obtain (3.4). Uniform local mass follows in the dense region from the same argument and at the ends from (3.7) and the gap bounds. In regular filling every cell is dense, including the bounded counting extension of the interval.

The Gaudin determinant and Lemma 2.5 make each Bethe vector nonzero. In the homogeneous chain the logarithmic row derivative at the swap has, up to an additive scalar, local link matrix equal to the swap times $-\eta R'(0)$. Its hopping coefficient on an opposite-spin pair is $-\eta/\sin \lambda < 0$. Within each nontrivial particle sector the hopping graph is connected, so the sector minimum is simple and has a positive vector. At $\lambda = \pi/2$ this is the free nearest-neighbor exclusion Hamiltonian. In ordered particle coordinates its eigenvectors are antisymmetrized plane waves with circle sign $(-1)^{n-1}$, and the minimum fills the consecutive momenta about zero. These are precisely the consecutive Bethe roots: their physical shift phase is $e^{ip} = -F_{\{y\}}(i\lambda/2)$, and differentiation of the eigenvalue gives $-2\pi\eta s(y) = -\cos p$ at the free point. The sign gauge thus centers the occupied momenta at zero.

The real-root branch remains this simple minimum by continuation. The physical transfer matrix is primitive in the sector: its diagonal is positive, and successive allowed auxiliary swaps move any particle to a neighboring hole, connecting the sector. It commutes with the link Hamiltonian. The positive simple minimum is consequently its Perron vector. Taking the long cylinder limit gives its normalized vacuum expectations. Finally nonsingularity of G gives the finite analytic twist branch by the implicit function theorem. This last statement does not require a twist neighborhood uniform in L . $\square$

3.3 Uniform inverses on pointwise function spaces

Lebesgue L^∞ equivalence classes cannot be used when an operator evaluates a function at the roots. We use $C_b(\mathbb{R})$ with its pointwise sup norm, or the space of bounded continuously differentiable functions with norm $\|u\|_\infty + \|u'\|_\infty$. The root-node spaces carry their ordinary finite-dimensional sup norms.

Lemma 3.2 (Gaudin and mixed-measure inverse bounds). *In the limits specified above,*

$$\|(I + K\ell)^{-1}\|_{\infty \rightarrow \infty} \leq C, \quad (K\ell)_{ij} = K(y_i - y_j)\ell_j. \quad (3.14)$$

There is also a small $a > 0$ for which the inverse has uniformly bounded row norm after conjugation by either e^{ay_i} or e^{-ay_i} .

Let χ_-, χ_0, χ_+ be a nonnegative partition of unity whose two transitions are translates of fixed smooth profiles with bounded first derivatives. The middle function has compact support, the transitions lie in regions of a fixed sufficiently high density, and the two transition regions recede from one another. The left function is one to the left of its transition and zero to its right; the right function has the opposite convention at the right transition. Put

$$D_L = \nu_L - du, \quad D^\bullet = \chi_\bullet D_L.$$

The mixed operator

$$I + R_0 * (D^0 + D^+) \quad (3.15)$$

has a uniform lower bound in the pointwise sup norm. The analogous assertion holds with the two ends interchanged.

Proof. For regular filling, local quadrature up to the endpoints bounds the absolute row norm of $K\ell$ by $h + o(1) < 1$. This proves (3.14) there. For $K < 0$ at half filling, finite strict diagonal dominance does not give a uniform inverse: its bare-source deficit may shrink at the last roots. We first isolate those roots from the contracting bulk, then use their reciprocal densities to obtain a decaying solution of a one-sided averaging equation.

Truncate the exponentially small kernel tail at a fixed distance and then choose a high density threshold. Local mass and (3.4) show that every row sufficiently far inside the dense portion has absolute row sum at most a fixed $q < 1$. The edge-index bounds confine all other rows to a fixed number of indices at either end.

If the inverse bound failed, there would be vectors W_L with $\|W_L\|_\infty = 1$ and $(I - k\ell)W_L \rightarrow 0$ in sup norm. A near-unit coordinate must lie among those fixed edge indices. Retain one end, translate its terminal root to zero, and extract all fixed-index gaps and weights. At the right end write the limiting points as $x_1 = 0 > x_2 > \dots$; the left end is treated by reflection. The bounds of Lemma 3.1 make the limiting weights positive and finite; at every fixed index they satisfy $c/j \leq \ell_j \leq C/j$. The limiting points have no finite accumulation, since otherwise this lower bound would contradict the uniform local mass bound. In the array case this extraction is along $d \downarrow 0$ with B sufficiently large, exactly as in (3.1).

Put $v_j = \ell_j W_j$ in the resulting one-sided limit. It is nonzero, and $v_j \rightarrow 0$ as $j \rightarrow \infty$. Uniform local mass and exponential tails justify passing the equation to this limit. It reads

$$(\rho_j - k(0))v_j = \sum_{i \neq j} k(x_j - x_i)v_i.$$

The left coefficient is positive: finite row dominance bounds it below by $k(x_j - x_i) > 0$ for any other retained index i , whose limiting gap is finite. After division by this coefficient, the equation is an

averaging equation with strictly positive off-diagonal weights whose sum is at most one. The latter follows from finite Gaudin row dominance and Fatou's lemma; a strictly positive limiting deficit is not required. The absolute maximum of v is attained, and some other index has smaller modulus. Its positive weight strictly decreases this maximum, a contradiction.

We spell out the mixed case. The measure

$$\tilde{\nu} = du + D^0 + D^+ = \chi_- du + (1 - \chi_-)\nu_L$$

is nonnegative. Exponential decay and bounded local variation justify Fubini, and $R_0 + K * R_0 = K$ gives the exact identity

$$(I + K*)(I + R_0 * (D^0 + D^+))u = u + K * (\tilde{\nu}u). \quad (3.16)$$

The line convolution $I + K*$ is boundedly invertible because $h < 1$.

For $K < 0$, suppose the undressed operator had normalized approximate null functions u_L . They satisfy $u_L = k * (\tilde{\nu}u_L) + o(1)$ uniformly. The quantity $\sup_x \int k(x - y)\tilde{\nu}(dy)$ is uniformly bounded. It follows that the supremum of $|u_L|$ on $\text{supp } \tilde{\nu}$ is bounded below. Divide by that supremum. The support supremum is now one and the full supremum remains bounded. On the support, every row except a fixed number of right-edge atoms has sum at most $q < 1$: the replaced side is Lebesgue, the blend is dense, and the remaining noncontracting part is the original discrete right end.

Thus a near-unit support value survives at one of those atoms. The modified left side is arbitrarily far away from each retained right-edge index, so its contribution disappears in the extracted equation. The preceding harmonic-maximum argument applies. There is no additional approximate mode between the atoms: $k * (\tilde{\nu}u_L)$ has a uniformly bounded derivative, so the approximate equation makes u_L equicontinuous up to a vanishing sup error. In the regular-filling case the entire mixed measure instead has absolute kernel row norm $h + o(1)$, by quadrature, and contracts directly. Equation (3.16) proves the claimed lower bound.

Finally exponential conjugation changes the discrete kernel row norm by at most

$$\sup_i \sum_j |K(y_i - y_j)|\ell_j |e^{a(y_i - y_j)} - 1| \leq C|a|.$$

This follows by summing over unit intervals and using local mass and exponential kernel decay. A sufficiently small a therefore preserves the inverse bound by the inverse identity. The same perturbation estimate preserves the mixed lower bounds. The weighted middle contraction is proved explicitly in Lemma 3.4. $\square$

3.4 An analytic normalization for all marked rows

The inverse estimates control the scalar-product columns. To continue the resulting kernels from one convenient spectral region to the whole strip, we also need a common operator bound. Write

$$\mathcal{S}_\lambda = \{x \in \mathbb{C} : |\text{Im } x| < \lambda/2\}.$$

This is the angle strip $|\text{Re } z| < v$ under $x = \eta z$.

Lemma 3.3 (Harmonic row normalization). *There is a real harmonic function U on $\mathcal{S}_\lambda$ such that, for every finite real-inhomogeneity chain,*

$$\|T^D(x)\| \leq 2\|D\| \exp \left(\sum_j U(x - b_j) \right). \quad (3.17)$$

Choose a nonvanishing analytic function $\mathcal{N}_L$ whose modulus is the exponential on the right. In the homogeneous plane and array limits, the untwisted vacuum eigenvalue τ_L satisfies

$$|\tau_L(x)/\mathcal{N}_L(x)| \longrightarrow 1 \quad (3.18)$$

locally uniformly in the strip, with the moving-bulk convention (3.1) in the array case. Consequently, for each compact $\mathcal{K} \subset \mathcal{S}_\lambda$,

$$\limsup \sup_{x \in \mathcal{K}} \|A_L^D(x)\| \leq 2\|D\|. \quad (3.19)$$

In particular the vacuum denominator has no zeros on such a compact for all sufficiently large systems in the indicated sense.

Proof. On $\text{Im } x = \lambda/2$, local inversion says that each site matrix divided by its equal-spin entry is unitary. Taking the auxiliary trace costs at most $2\|D\|$, and gives the boundary estimate with $|a(x)|$. Crossing gives the reflected lower boundary estimate. For $0 < \text{Im } x < \lambda/2$ define

$$U(x) = \log \left| \frac{[x + i\lambda/2]}{[i\lambda]} \right| + \int_{\mathbb{R}} r(u) \log \left| \frac{[u - x + i\lambda]}{[u - x]} \right| du. \quad (3.20)$$

The integrand has no singularities in that open half-strip and decays exponentially at both real ends. Its boundary value on $\text{Im } x = \lambda/2$ is zero, because the two sinh moduli coincide. Reflect (3.20) conjugately to the lower half-strip. The two values agree on the real line. Differentiating the logarithmic sinh kernels at that line, with the denominator term interpreted by its Poisson-kernel limit, gives the upper normal derivative

$$\pi(s - K * r - r) = 0.$$

Thus the two harmonic functions glue harmonically. Their growth at the real ends is at most linear.

For unit vectors u, v , apply the strip maximum principle to $\log |\langle u, T^D(x)v \rangle| - \sum_j U(x - b_j)$. The boundary bound is $\log(2\|D\|)$ and the growth is at most linear. One may justify the unbounded-strip principle by subtracting $\varepsilon \cosh(a \text{Re } x) \cos(a \text{Im } x)$, with $0 < a < \pi/\lambda$, applying the principle in finite rectangles, and then sending ε to zero. Taking the supremum over u, v proves (3.17). The strip is simply connected, so a harmonic conjugate gives the nonvanishing $\mathcal{N}_L$.

It remains to compare it with the actual eigenvalue. It suffices first to work in a fixed small open neighborhood inside the upper swap point $i\lambda/2$. There $q_Y(x) \rightarrow 0$. Homogeneously, for each fixed $0 < \lambda < \pi$ choose a sufficiently small disc about $i\lambda/2$. The root ratio $[x - y + i\lambda]/[x - y - i\lambda]$ is bounded by a constant M_λ for every real y on that disc: the imaginary part of its denominator stays away from $\pi\mathbb{Z}$, and the ratio has finite limits at both real ends. The one-site ratio $[x - i\lambda/2]/[x + i\lambda/2]$ tends uniformly to zero as the disc shrinks. Shrink it until its supremum times $\max(1, \sqrt{M_\lambda})$ is less than some $q_* < 1$. Then $n \leq L/2$ gives $|q_Y(x)| \leq q_*^L$ on the disc, and hence on its intersection with the strict strip. The disc and q_* may depend on the fixed λ . For the array, $\lambda > 2\pi/3$ and the neighborhood can be chosen so that each site and root ratio has modulus at most one. Indeed the difference of the squared sinh moduli is governed, respectively, by $\sin(2 \text{Im } x) \sin \lambda > 0$ and $\sin(2 \text{Im } x) \sin(2\lambda) < 0$. There are order $1/d$ sites at bounded distance from the moving bulk center, with site ratios uniformly below one. Hence $q_Y(x) = O(e^{-c/d})$ after $B \rightarrow \infty$, uniformly in that bulk.

Put

$$H_x(u) = \log \left| \frac{[u - x + i\lambda]}{[u - x]} \right|, \quad \tilde{H}_x = (I + K*)^{-1} H_x.$$

These are smooth tests in u on the chosen neighborhood. The exponentially weighted Neumann series gives uniform exponential decay for $\tilde{H}_x$ and all its fixed derivatives, also after real recentering.

The density equation gives the exact identity

$$\sum_{y \in Y} H_x(y) - \int_{\mathbb{R}} p_L(u) H_x(u) du = \int_{\mathbb{R}} \tilde{H}_x(u) e_L(du). \quad (3.21)$$

At half filling, integrate by parts against the bounded counting sawtooth F_L . In counting coordinates its mean over every complete cell is zero. On a dense fixed window, the relative derivative bound for ρ_L makes the physical cell averages converge to that mean. Thus F_L tends weakly to zero there. The exponential tail of the test and the uniform bound $|F_L| \leq 1/2$ discard the complement of an increasing window. This proves that (3.21) tends to zero, including the array limit after bulk recentering.

For the homogeneous near-half-filled sequence, use the true outer counting half-levels as cell endpoints. They are a bounded counting distance from $\pm Q$ by (3.3), and the positive comparison extends to them. The cellwise primitive is zero at the endpoints and is bounded by $1/2$ inside. The same midpoint argument therefore works within these cells. Outside them, the root measure vanishes and

$$|\xi'(u)| \leq CLr(Q) = O(L^\epsilon).$$

To see this uniform exterior bound, replace the root sum of any translate of K by its Lr_Q integral. Midpoint quadrature and bounded counting displacement cost $O(1)$ by variation and reciprocal-density summation. The remaining exterior expression $L(s - K_Q r_Q)$ is $O(Lr(Q))$, by the full-line equation and (3.9). Its pairing with the dressed test outside the cells is consequently $O(L^\epsilon e^{-aQ})$ for a fixed $a > 0$. Since $Q = \eta(1 - \epsilon) \log L + O(1)$, it tends to zero when

$$0 < \epsilon < \frac{a\eta}{1 + a\eta}. \quad (3.22)$$

Decreasing a if necessary gives one such choice on the compact continuation interval used to construct the roots.

The first-term eigenvalue formula, $q_Y \rightarrow 0$, and (3.21) now prove (3.18) on this open neighborhood. On the whole strip, $\tau_L/\mathcal{N}_L$ is bounded by two. Every subsequential holomorphic limit therefore has modulus one on the neighborhood and is a unimodular constant by the maximum modulus principle. Normal-family compactness gives local uniform modulus convergence throughout the strip. The same argument works for any purported failing moving-bulk array sequence. Dividing (3.17) by $|\tau_L|$ proves the last assertions. $\square$

3.5 Joint tails from finite residue conditions

Fix a word length m and generic arguments $v'_1, \dots, v'_m$ in a small complex neighborhood of $i\lambda/2$. Generic here means that the finite difference factors in Corollary 2.6 have no poles or forbidden coincidences. This is a nonempty open condition. We may recenter all real coordinates in the array case. The columns in (2.33), at zero twist, are denoted by $W_{v'}$. Their source columns are

$$m_{v'}(u) = \frac{(1 + q_Y(v')) \coth(v' - u) - \coth(v' - u - i\lambda) - q_Y(v') \coth(v' - u + i\lambda)}{-2\pi i}. \quad (3.23)$$

Extend each finite node solution to the line by $W = m - K * (\nu_L W)$. Lemma 3.2, local mass, and the kernel derivative bounds give a uniform bound for W and each fixed derivative on the line. Undoing this convolution equation gives

$$W = W^0 - R_0 * (D_L W), \quad W^0 = (I + K*)^{-1} m. \quad (3.24)$$

Write $E = \mathbb{C}^m$ for the coefficient space and $W(c) = \sum_{j=1}^m c_j W_{v'_j}$ for $c \in E$, with the same notation for linear combinations of the free columns $W_{v'_j}^0$. All these statements are uniform for $\|c\|_2 \leq 1$.

We need decay for a whole minor of these columns. To see how much, fix a finite action path and write $H = (p_1, \dots, p_k)$ for its deleted original roots and V for its surviving row arguments. In (2.42), leave the determinant $\det(W_{v'}(p))_{H,V}$ aside for the moment and extract the quadrature weights $\prod_{p \in H} \ell_p$. The hyperbolic Cauchy determinant in (2.42) contributes at most

$$C \exp \left(k \sum_i |p_i| - \sum_{i < j} |p_i - p_j| \right)$$

after the root-pair cancellations. To check those cancellations also near collisions, for each pair of deleted original roots Corollary 2.6 supplies a reciprocal f factor. Combined with its Cauchy denominator, its modulus is bounded by

$$|[p_i - p_j \pm i\lambda]|^{-1} \leq C_\lambda e^{-|p_i - p_j|}.$$

All other fixed-argument factors are bounded on the generic parameter set. There is at least one removal factor $\beta(v' - p_i)$ or $\beta(p_i - v')$ for each original root deleted, and it is at most $Ce^{-|p_i|}$. The factors $q_Y(v')$ are bounded and the row denominators $1 + q_Y(v')$ are bounded away from zero in the spectral region used here. Consequently, apart from the dressed determinant and the quadrature weights, the summand has modulus at most

$$C \exp \left((k-1) \sum_i |p_i| - \sum_{i < j} |p_i - p_j| \right).$$

With $k_+(t) = \#\{i : p_i > t\}$ and $k_-(t) = \#\{i : -p_i > t\}$, its exponent is exactly

$$\int_0^\infty (k_+(t)(k_+(t) - 1) + k_-(t)(k_-(t) - 1)) dt. \quad (3.25)$$

One verifies this identity first on a region with the signs and ordering fixed, or by writing every distance as an integral of indicator functions.

Thus r roots escaping together at one end cost $r(r-1)$ in the exponential rate. A bound on each column by the same decaying function would not control arbitrarily large fixed minors. We will instead find nested subspaces of column combinations with successively faster decay. For the j th direction it suffices to obtain a rate $\beta_j > 2j-2$, because the first r such rates then sum to more than $r(r-1)$. The number of slow directions is determined by the poles of the full-line dressed equation.

The pole spaces needed for this equation are particularly small. Let (d_j) be the increasing list, with multiplicities, formed from

$$\left\{ \frac{2k-1}{\eta} : k \geq 1 \right\} \cup \left\{ \frac{2k}{1-\eta} : k \geq 1 \right\}. \quad (3.26)$$

At coincident entries, retain both copies; the list bounds the maximum possible pole orders. For every j ,

$$d_j \geq 2j-1. \quad (3.27)$$

Indeed, if n_1, n_2 count the entries of the two progressions strictly below $T = 2j-1$, then $2n_1-1 < \eta T$ and $2n_2 < (1-\eta)T$. Adding gives $n_1 + n_2 < j$.

For completeness, the Fourier calculation behind the list uses

$$1 + \widehat{K}(p) = \frac{2 \sinh((\pi - \lambda)p/2) \cosh(\lambda p/2)}{\sinh(\pi p/2)}.$$

In particular

$$\widehat{R}_0(p) = \frac{\sinh((\pi - 2\lambda)p/2)}{2 \sinh((\pi - \lambda)p/2) \cosh(\lambda p/2)}.$$

The Fourier transform of each coth difference in (3.23) has denominator $\sinh(\pi p/2)$, as follows by shifting the integration contour by $i\pi$ and accounting for the coth residues. This denominator cancels upon dressing. The remaining possible poles for both W^0 and R_0 are exactly those bounded by (3.26); the apparent singularity at zero is removable. On any horizontal Fourier contour avoiding these poles, the transforms decay exponentially in the real direction. The generic arguments range in a compact analytic neighborhood, so this decay is uniform. Multiplication by any fixed power of the Fourier variable preserves integrability.

Thus, for $\beta > 0$ not in the list, shifting the contour and subtracting the crossed residues gives

$$R_{0,\beta} = R_0 - P_\beta, \quad W_\beta^0(c) = W^0(c) - p_\beta(c),$$

where the global exponential polynomials $P_\beta, p_\beta(c)$ belong to

$$\mathcal{E}_\beta = \text{span}\{u^q e^{-du} : d < \beta, \ 0 \leq q < m_d\}. \quad (3.28)$$

Here m_d is the multiplicity in the list, and $\dim \mathcal{E}_\beta \leq r_\beta := \#\{j : d_j < \beta\}$. Using two nearby pole-free contours gives, for some $\delta > 0$,

$$e^{\beta u} |R_{0,\beta}^{(q)}(u)| \leq C_{\beta,q} e^{-\delta|u|} \quad (q = 0, 1, 2), \quad (3.29)$$

and the corresponding bounds for the free columns. Any other fixed derivative order is available in the same way. The residue polynomials are subtracted on the whole line, without cutoffs; this preserves their finite-dimensional translation span.

Lemma 3.4 (Finite-chain weighted flags). *Let $E = \mathbb{C}^m$ be the coefficient space of the fixed column list. For each admissible $\beta > 0$ there is a subspace $F_{\beta,L} \subset E$ of codimension at most r_β such that*

$$|W(c)(u)| \leq C_\beta e^{-\beta u} \|c\|_2 \quad \text{at every positive root } u, \quad c \in F_{\beta,L}. \quad (3.30)$$

For a fixed finite list of increasing shifts, choose one discrepancy partition for the whole list. The resulting subspaces may be chosen nested. There are analogous flags at the negative end. The constants are uniform in the three limiting procedures, with regular filling understood as $Q \rightarrow \infty$ after $L \rightarrow \infty$. Restriction to any column coordinate subspace preserves the codimension bound and all unit-vector estimates.

Proof. Separate the two sparse ends. Choose the fixed-profile partition in Lemma 3.2 so that

$$\text{supp } D^- \subset (-\infty, a_L], \quad \text{supp } D^+ \subset [b_L, \infty), \quad a_L < 0 < b_L.$$

The word center is deep between these boundaries. The middle support is compact and dense. At a fixed density threshold, Lemma 3.1 gives

$$\max Y \leq b_L + C_0, \quad \min Y \geq a_L - C_0, \quad (3.31)$$

with a threshold-dependent constant uniform in the limits. For regular filling take the transitions just inside $[-Q, Q]$. For the array both the last root and the transition contain the same $\eta \log(1/d)$ shift in (3.8). All three signed pieces have uniformly bounded variation mass on intervals of length one.

Use the pointwise Banach space

$$X_\beta = \left\{ u \in C^1(\mathbb{R}) : \|u\|_\beta := \sup_x e^{\beta x} (|u(x)| + |u'(x)|) < \infty \right\}.$$

The quadrature error on each unit-scale test is as small as desired on the middle support and its taper. To bound $R_{0,\beta} * (D^0 u)$, insert a smooth unit-scale partition in the integration variable and apply (3.4). For the output and its first derivative, the test derivatives involve $R_{0,\beta}, R'_{0,\beta}, R''_{0,\beta}$ and u, u' . The envelopes (3.29) sum after weighting to give

$$\|R_{0,\beta} * (D^0 u)\|_\beta \leq C_\beta \varepsilon_{\text{quad}} \|u\|_\beta.$$

Choose the density threshold so this norm is below 1/2 for the fixed finite list of shifts, including zero.

We first remove the right-end discrepancy from the column equation, keeping its true left-end input. Let

$$W^p(c) = (I + R_0 * D^0)^{-1} (W^0(c) - R_0 * (D^- W(c))),$$

using the shift-zero contraction on X_0 . Our task is to characterize the combinations c for which this bounded column also belongs to X_β . The omitted right end will then be restored with the mixed-operator lower bound.

Solve after subtracting the slow residues. For $c \in E$ solve in X_β

$$(I + R_{0,\beta} * D^0) U_\beta(c) = W_\beta^0(c) - R_{0,\beta} * (D^- W(c)). \quad (3.32)$$

The true-column bound and local variation imply

$$\|R_{0,\beta} * (D^- W(c))\|_\beta \leq C_\beta e^{\beta a_L} \|c\|_2.$$

For example multiply the convolution integrand by $e^{\beta(x-y)} e^{\beta y}$, use $y \leq a_L$, and sum the exponentially decaying weighted kernel envelopes over unit intervals. The same argument applies to the first derivative. Consequently

$$\|U_\beta(c)\|_\beta \leq C_\beta \|c\|_2. \quad (3.33)$$

This estimate holds for every coefficient vector before imposing any residue condition.

Impose only the residue conditions. Define the linear map

$$\Lambda_\beta(c) = P_\beta * (D^0 U_\beta(c) + D^- W(c)) - p_\beta(c). \quad (3.34)$$

Its values lie in $\mathcal{E}_\beta$. Indeed convolution of an exponential polynomial with a source preserves that span: expand each $(x - y)^q$. The middle source is compactly supported, and the left moments converge because $d > 0$ and the left variation mass is locally bounded. Rank-nullity therefore gives $\text{codim ker } \Lambda_\beta \leq r_\beta$. No inverse or condition-number bound for Λ_β is used.

Restoring the residues in (3.32) gives

$$(I + R_0 * D^0) U_\beta(c) = W^0(c) - R_0 * (D^- W(c)) + \Lambda_\beta(c).$$

For $c \in \ker \Lambda_\beta$, this identity first proves $U_\beta(c) \in X_0$: in the rearranged equation the middle source $D^0 U_\beta$ is a finite compactly supported measure, so its convolution and first derivative are bounded at

both ends. The other two terms are already bounded. This membership assertion for a fixed chain does not yet require a uniform estimate for that compact source. Only now apply the shift-zero inverse to identify $U_\beta(c) = W^p(c)$ and obtain its uniform unweighted bound.

Conversely, if $W^p(c) \in X_\beta$, rewrite its equation with the shifted kernels. The defect is an element of $\mathcal{E}_\beta$ which also belongs to X_β . It is zero: every nonzero exponential polynomial in (3.28) violates the β -weighted bound at $+\infty$. Weighted uniqueness then gives $W^p(c) = U_\beta(c)$ and $\Lambda_\beta(c) = 0$. Hence

$$F_{\beta,L} := \ker \Lambda_\beta = \{c : W^p(c) \in X_\beta\}. \quad (3.35)$$

The spaces are nested, because $W^p(c)$ is already unweighted bounded at $-\infty$ and a larger positive weight at $+\infty$ implies every smaller one. Estimates (3.33) hold uniformly for every unit vector in these subspaces.

Restore the right end at the actual nodes. For such a vector, subtract the true and middle equations:

$$(I + R_0 * (D^0 + D^+))(W(c) - W^p(c)) = -R_0 * (D^+ W^p(c)).$$

The right side has sup norm at most $C_\beta e^{-\beta b_L}$. The infinite Lebesgue portion of D^+ causes no extra factor: sum its uniformly bounded local variation against $e^{-\beta(b_L+j)}$ over $j \geq 0$. The mixed lower bound therefore gives the same estimate for $W - W^p$. At every actual positive root, (3.31) converts it to $C_\beta e^{-\beta u}$, which proves (3.30). Reflecting proves the negative-end assertion. Finally restriction of (3.34) to a coordinate subspace still has rank at most r_β , and a unit vector there is a unit vector in E . All the asserted restriction properties follow. $\square$

The flag estimates now supply the decay rates required by (3.25).

Lemma 3.5 (Joint tail bound). *For a fixed generic row word and any one of its nonzero finite action paths, let $H = (p_1, \dots, p_k)$ be the deleted original roots and V its surviving row arguments. After extracting the quadrature weights $\prod_{i=1}^k \ell_{p_i}$ from (2.42), the remaining summand is bounded in modulus by*

$$C \exp \left(-\varepsilon \sum_{i=1}^k |p_i| \right) \quad (3.36)$$

for some $\varepsilon > 0$. The constant and exponent may depend on the fixed word and generic compact parameter set, but are uniform in the limiting procedures. In particular, the sum over paths with $\max_i |p_i| > R$ is $O(e^{-\varepsilon' R})$ uniformly, for some $\varepsilon' > 0$.

Proof. Choose pole-free numbers $\beta_j \in (2j-2, 2j-1)$ for $1 \leq j \leq k$. By (3.27), the corresponding flag has codimension at most $j-1$. For a positive-row minor with r columns, order its rows as $p_1 \geq \dots \geq p_r \geq 0$. Choose an orthonormal basis of its coefficient space backwards: the j th basis vector belongs to $F_{\beta_j, L}$. This is possible because these spaces are nested and have dimension at least $r-j+1$. A unitary basis change does not alter the determinant modulus. Applying (3.30) to its columns and using the rearrangement inequality in the determinant expansion gives

$$|\det W| \leq C \exp \left(-\sum_{j=1}^r \beta_j p_j \right) = C \exp \left(-\int_0^\infty \sum_{j=1}^{k_+(t)} \beta_j dt \right), \quad (3.37)$$

where $k_+(t) = \#\{i : p_i > t\}$. For a determinant with both signs, first expand by its positive and negative rows. Every Laplace minor has a column coordinate subspace, on which the same codimension and unit-vector estimates hold. Choose its flag basis separately. Thus no simultaneous

adaptation of the two tail flags is required. The finite Laplace sum gives the product of the positive bound and its negative counterpart.

Set $\varepsilon = \min_{j \leq k} (\beta_j - (2j - 2)) > 0$. For every $r \leq k$, $\sum_{j=1}^r \beta_j \geq r(r - 1) + \varepsilon r$. Combining (3.37) at both ends with (3.25) proves (3.36). The case $k = 0$ is the empty determinant and needs no tail estimate. Uniform local quadrature mass now bounds the sums of the majorant by convergent geometric series on unit boxes. Discarding boxes with some coordinate beyond R proves the last assertion, with a harmless smaller exponent. $\square$

3.6 Limits of fixed row words

The joint tail bound lets us take the thermodynamic limit first on a bounded rapidity window, and then remove that window. The harmonic row bound extends the resulting convergence to the whole strict strip.

Proposition 3.6 (Limits of fixed row words). *For each fixed integer $m \geq 0$ and fixed 2×2 matrices $D_1, \dots, D_m$, the homogeneous plane coefficients*

$$\langle \Omega_L, A_L^{D_m}(x_m) \cdots A_L^{D_1}(x_1) \Omega_L \rangle \quad (3.38)$$

converge locally uniformly on $\mathcal{S}_\lambda^m$ to a holomorphic function $\mathcal{K}_{D_m, \dots, D_1}(x_m, \dots, x_1)$. It is invariant under every common complex translation for which the translated arguments remain in the strip. For $\lambda > 2\pi/3$, the same functions are the array limits in (3.1).

If v_L is any fixed finite linear combination of row words applied to Ω_L , then

$$\lim \|A_L^D(x)v_L\|^2 \leq 4\|D\|^2 \lim \|v_L\|^2.$$

The two limits are determined by the word coefficients and the adjoint rule. This is the bound used to construct bounded row actions in Section 4. For $\lambda > 2\pi/3$, in the regular-filling procedure the expectations of every fixed equal-time local matrix product converge to those of the homogeneous plane construction.

Proof. First keep all arguments in the generic neighborhood used above. The finite row expansion has finitely many action patterns, each with at most m deleted original roots. Words with a net change of particle number have coefficient zero. On each bounded rapidity window, the line extensions of the dressed columns and their derivatives are uniformly bounded. Local quadrature and their convolution equation therefore give, along any subsequence, a smooth local limit satisfying

$$W + K * W = m_{v'}^{(0)}, \quad m_{v'}^{(0)}(u) = \frac{\coth(v' - u) - \coth(v' - u - i\lambda)}{-2\pi i}.$$

The tails in this equation are negligible by exponential kernel decay and bounded local mass. Since $h < 1$, the bounded solution is unique. The elementary identity $m_{v'}^{(0)}(u) = s(u - v' + i\lambda/2)$ shows that dressing replaces s by r . Equivalently its Fourier transform is $e^{-ip(v' - i\lambda/2)} / (2 \cosh(\lambda p/2))$. This gives

$$W_{v'}(u) \longrightarrow \frac{1}{2\lambda \cosh((u - v' + i\lambda/2)/\eta)}. \quad (3.39)$$

For the array, the same conclusion holds along every bulk-recentered sequence in its prescribed double-limit sense.

Apply (2.42) on a bounded rapidity window. Its weights are exactly the product quadrature measure. Local quadrature, (3.39), and $q_Y(v') \rightarrow 0$ turn its sum into the corresponding line integral.

The apparent near-collision singularities have already been canceled by the root-pair factors; deleting diagonal tuples has no effect in the limit, since the maximal atom on such a window tends to zero. Lemma 3.5 then permits the window to grow to the whole line. This proves convergence of every path and hence of the word. It also proves independence of the choice between homogeneous and array limits. All surviving ingredients depend only on rapidity differences. Changing all variables of integration by a common real amount therefore proves real-translation invariance in the x coordinate.

Lemma 3.3 bounds all the finite words locally uniformly on the full strip. Normal-family compactness and the identity theorem extend the convergence from the generic open set to $\mathcal{S}_\lambda^m$, with uniqueness of every subsequential limit. The derivative in the common translation direction vanishes on that open set and hence everywhere. Integrating it along translations staying in the strip gives the stated complex-translation invariance. For any finite linear combination of word vectors, apply the finite operator inequality (3.19) to its Gram norm. Passing to the limit gives the bound $2\|D\|$ for the induced row action and also shows that it respects zero Gram vectors.

For the remaining equal-time assertion, first fix $Q < \infty$ and send $L \rightarrow \infty$ with filling $\int_{-Q}^Q r_Q$. The same finite path calculation is valid, now with dense nodes on $[-Q, Q]$ and the interval dressing operator. Then send $Q \rightarrow \infty$. The inverse bounds, local convergence, and joint tail bound are uniform in this second limit, so the same full-line integrals result.

This calculation can be made on small contours surrounding $i\lambda/2$, using nested distinct radii to keep all arguments generic. For the regular-filling assertion we only use $\lambda > 2\pi/3$; choose the contour radii so small that all the fixed shifted poles stay outside them. On these contours the homogeneous q_Y still tends to zero exponentially, and the eigenvalue first term is analytic and nonzero. Thus each complete finite normalized word is holomorphic on the product of the enclosed discs for sufficiently large L . The same determinant bounds apply on their boundary contours, including the portions just outside the strict strip. Convergence on these contours is uniform: first take the fixed- Q chain limit on a bounded rapidity window, then use the uniform joint tail bound when taking $Q \rightarrow \infty$ and removing that window. The iterated Cauchy formula therefore gives the coincident endpoint values.

At the swap, a marked row implements an arbitrary matrix on one site together with the cyclic shift. Multiplying marked swap rows conjugates the successive site matrices by that shift; their product reads any finite set of sites, with identity marks at unused sites. The residual shift fixes the homogeneous Perron vector, so its vacuum expectation is exactly the desired local-matrix expectation. The half-filled homogeneous calculation has the same endpoint contours and integrals. This proves equality of the equal-time local states in the stated order of limits, without asserting an unrestricted row-strip limit at a fixed nonzero magnetization. $\square$

3.7 Clustering of diagonal row words

Proposition 3.7 (Separated diagonal words). *Partition a fixed finite list of diagonal marked rows into groups. Within each group fix its internal arguments in the strict angle strip. Translate the arguments of group g by a common imaginary amount it_g , and let every $|t_g - t_{g'}|$ tend to infinity. Then the plane word kernel factors into the product of the individual group kernels. Convergence is locally uniform in the internal strict arguments. The order within each group is retained.*

Proof. We first use generic arguments near the upper boundary in the x coordinate. Their real parts may be separated by arbitrary translations. The integral formula of Proposition 3.6 applies there by analytic continuation of its difference integrals; keep the imaginary parts in a fixed small generic box and avoid the finitely many coincidence poles during continuation.

For a path with deleted roots H and surviving arguments V , write $N_H(t)$ and $N_V(t)$ for the numbers of their real parts to the left of t . These two lists have equal length. The quotient of

the continuum determinant (3.39) and the original hyperbolic Cauchy determinant, including the root-pair cancellations, is bounded by

$$C \exp \left(-(\eta^{-1} - 1) \int_{\mathbb{R}} (N_H(t) - N_V(t))^2 dt \right). \quad (3.40)$$

Here the constant is uniform as the fixed internal lists separate. To verify the exponent, use

$$\cosh((p - v' + i\lambda/2)/\eta) = i \sinh((p - v')/\eta)$$

and apply the sinh Cauchy identity at scales $1/\eta$ and 1. The exponent in their ratio is $-(\eta^{-1} - 1)$ times

$$\sum_{p \in H, v' \in V} |p - \operatorname{Re} v'| - \sum_{p < p'} |p - p'| - \sum_{v' < v''} |\operatorname{Re} v' - \operatorname{Re} v''|.$$

This expression is exactly the integral in (3.40). The bounded imaginary offsets and fixed within-group generic factors change only the constant. The reciprocal f factors remove the near-root-collision denominators as in Lemma 3.5.

The integral in (3.40) controls ordered matching distance. Indeed $N_H - N_V$ is integer-valued, so its square is at least its absolute value, and

$$\int_{\mathbb{R}} |N_H - N_V| dt = \sum_j |p_j - \operatorname{Re} v'_j|$$

for the two ordered lists. Since $\eta < 1$, the estimate localizes all integration variables integrably around the surviving arguments, uniformly in their separations.

For diagonal words every action has one removal. Each removal also has its factor β decaying exponentially in the distance between the adding argument and the removed argument or root. If a path couples two different groups, either this factor or (3.40) makes its integral vanish as the gap grows. More explicitly, first restrict every matched root to a fixed radius of its matching argument using the uniform exponential bound. On this restricted region any removal between distinct groups has distance tending to infinity. Discard these paths and then increase the matching radius. The surviving paths have their additions and removals inside individual groups; in each group the numbers of deleted original roots and surviving added arguments are equal.

On a bounded matching region the two determinants are block diagonal in the limit, so their quotient tends to the product of the group quotients. Cross-group correction products tend to one. In fact $f(u) \rightarrow e^{\pm i\lambda}$ as $u \rightarrow \pm\infty$, and at every removal the numbers of extra surviving arguments and missing original roots in another group agree. Their limiting phases cancel. The constants $(-i\lambda)^{-1}$ occur once per deleted column and multiply in the same way. Termwise limits, the integrable matching bound, and the finite sum over paths therefore give exactly the product of the separate group answers.

If different groups were interlaced in the original word, they may first be gathered while retaining each internal order. The normalized intertwiner in RTT tends exponentially to an invertible diagonal matrix at large real separation. Its conjugation of a pair of diagonal marks therefore differs from the unchanged pair by an exponentially small matrix. The bounded limiting rows make the corresponding exchange error tend to zero.

Finally the row norm bound makes these translated kernels a normal family as functions of the internal strict arguments. Every subsequential holomorphic limit equals the product on the generic open box just treated. The identity theorem determines it throughout the product strip, proving the proposition and its local uniformity. $\square$

The construction now supplies all row-kernel inputs for the angular Hilbert space: holomorphy, the common-translation symmetry, bounded marked actions, and diagonal clustering. The physical identification and endpoint continuation are treated next.

4 The plane representation and its angular spectrum

The word kernels constructed in Section 3 now become operators on a Hilbert space. We identify their boundary values with local spins and spatial translations, and their physical-angle coefficients with the prescribed plane law. The resulting spectral description will provide both the normalization calculation and the separation into two chiral limits. Throughout this section

$$0 < \lambda < \pi, \quad \eta = \lambda/\pi, \quad v = \pi/2, \quad w = e^{iz}.$$

All constants may depend on the fixed parameter. The arguments below use the ordinary homogeneous-row Hilbert norm. No metric associated with a diagonal cut, and no positive-association statement for $c < 1$, is used.

4.1 From word kernels to operators

Let $\mathcal{W}$ be the complex vector space spanned by formal strict-row words

$$A^{D_m}(z_m) \cdots A^{D_1}(z_1)\Omega, \quad |\operatorname{Re} z_j| < v,$$

including the empty word Ω . Give two words the limiting scalar product obtained by reflecting the first word with the adjoint rule of Lemma 2.1. Proposition 3.6 proves existence of all these simultaneous limits. The form is positive, since every finite Gram matrix is a limit of positive finite-chain Gram matrices. We quotient by its null space and complete; the resulting Hilbert space is denoted by $\mathcal{H}$.

Proposition 4.1 (The angular representation). *On $\mathcal{H}$ the rows act as bounded holomorphic operator families, with*

$$\|A^D(z)\| \leq 2\|D\|, \quad A^D(z)^* = A^{X\overline{D}X}(-\bar{z}).$$

They satisfy the finite RTT exchange identities wherever their auxiliary intertwiner is invertible. There is a strongly continuous unitary group U_b , $b \in \mathbb{R}$, fixing Ω , such that

$$U_b A^D(z) U_b^* = A^D(z + ib). \quad (4.1)$$

Words with real arguments in any nonempty open subinterval of $(-v, v)$ span a dense subspace. The family $P = A^I$ consists of commuting normal operators and has a simultaneous representation by almost surely holomorphic scalar functions f on the strip, with

$$f(-\bar{z}) = \overline{f(z)}, \quad |f(z)| \leq 2. \quad (4.2)$$

In the half-plane coordinate this reality condition reads $f(\bar{w}) = \overline{f(w)}$.

Proof. The uniform normalized row bounds of Lemma 3.3, applied before taking each Gram limit, show that left multiplication by a row preserves null vectors and has the asserted bound. Thus it extends to the completion. Adjoint and RTT identities pass to all word pairings and then to bounded operators. Matrix coefficients on words are holomorphic by Proposition 3.6; local boundedness extends this to all matrix coefficients and hence to operator holomorphy.

Translate all arguments of a word by ib . In its Gram product the adjoint arguments are translated by the same amount, because $-\overline{z + ib} = -\bar{z} + ib$. Common-translation invariance of the word kernels therefore makes this map an isometry. Translation by $-ib$ is its inverse. Continuity of the kernels gives strong continuity on words, and the isometry bound extends it to $\mathcal{H}$. This proves (4.1). Notice that positive dilation of w corresponds to one of the two signs of this unitary parameter.

If a vector is orthogonal to every word whose real arguments lie in a fixed open interval, successive applications of the identity theorem to the word arguments show that it is orthogonal to all strict words. This proves density. Commutation of the untwisted rows and their adjoint rule put them in one abelian von Neumann algebra. Generate that algebra using a countable dense set of arguments. A multiplication representation, together with the uniformly bounded Taylor series on a countable cover of the strip, gives scalar analytic representatives on one common set of full measure. This also proves (4.2). The spectral measure class is invariant under imaginary translation by (4.1). $\square$

4.2 Finite division and the two boundary lines

Boundary rows identify the local spins and the site shift. To construct them on $\mathcal{H}$, we first divide the finite marked row by its unmarked counterpart near a swap point. The resulting uniform quotient bound continues the vacuum vectors; RTT then continues dense word vectors on the same disc. Passing their Taylor series to spectral fibers will give scalar continuation with a common radius.

Lemma 4.2 (Boundary division and scalar continuation). *There is $r_\lambda > 0$, independent of the even chain length, for which the two homogeneous quotients*

$$T^D(z)T^I(z)^{-1}, \quad T^I(z)^{-1}T^D(z)$$

are holomorphic and bounded by $C_\lambda \|D\|$ on discs of radius r_λ about $z = v$ and $z = -v$. The limiting rows have strong boundary values

$$A^D(v) = SD_0, \quad A^D(-v) = (XD^t X)_0 S^{-1}, \quad (4.3)$$

where S is the unitary site shift and $S\Omega = \Omega$. Almost every spectral function extends analytically and without zeros to bands of a common positive width about $\operatorname{Re} z = \pm v$, has modulus one on these lines, and satisfies $|f| \leq 1$ between them. Consequently it is inner in the right half-plane, and every zero ζ satisfies

$$\operatorname{Re} \zeta \geq c_0 |\zeta| \quad (4.4)$$

for a fixed $c_0 > 0$. At each specified strict argument the spectral set on which f vanishes has measure zero.

Proof. Write $u = \eta(v - z)$, so $u = 0$ at the upper endpoint, and put $b(u) = \sin u / \sin \lambda$ and $r(u) = 1 - b(u)^2$. Define the reversed unmarked row by

$$\tilde{T}^I(-u) = \operatorname{tr}_{a'}(R_{a',L-1}(-u) \cdots R_{a',0}(-u)).$$

Multiply $T^D(u)$ by this row. Transposition in a' reverses the second auxiliary product; factors at different quantum sites then commute past one another. The product is therefore the two-auxiliary partial trace of

$$(D \otimes I)G_0(u) \cdots G_{L-1}(u), \quad G_j(u) = R_{aj}(u)R_{a'j}(-u)^{t_{a'}}.$$

The auxiliary line spanned by $\Phi = 2^{-1/2}(|++\rangle + |--\rangle)$ is invariant. Inversion of R makes the action on this line equal to $r(u)I$. At $u = 0$ the swap identity makes the range of $G_j(0)$ lie in

$\mathbb{C}\Phi \otimes \mathbb{C}^2$. Thus, in the same auxiliary decomposition for every site,

$$r(u)^{-1}G_j(u) = \begin{pmatrix} I & B_j(u) \\ 0 & C_j(u) \end{pmatrix}, \quad \|B_j(u)\| \leq C_\lambda, \quad \|C_j(u)\| \leq C_\lambda|u|. \quad (4.5)$$

These bounds follow directly from the entries of R , since $\sin \lambda \neq 0$. For example, on a sufficiently small disc of radius proportional to $\sin \lambda$ one has $|r| \geq 3/4$ and $\|C_j\| \leq 1/4$. The upper-right block of the product in (4.5) is bounded by a geometric series, while its lower-right block has norm at most 4^{-L} . Taking the unmarked auxiliary trace therefore gives

$$r(u)^{-L}T^I(u)\tilde{T}^I(-u) = I + F_L(u), \quad \|F_L(u)\| \leq 3 \cdot 4^{-L}.$$

The off-diagonal triangular block has zero auxiliary trace. The marked partial trace is uniformly bounded by $C_\lambda\|D\|$. The displayed identity is invertible, so both finite square factors are invertible and

$$T^D(u)T^I(u)^{-1} = (r(u)^{-L}T^D(u)\tilde{T}^I(-u))(I + F_L(u))^{-1}$$

has the required uniform bound. Reversing the local product gives the same argument with an invariant covector and proves the other quotient. Crossing transfers it to the lower endpoint. None of these steps uses the sign of $\cos \lambda$.

The common finite vacuum remains an eigenvector of T^I in these discs by analytic continuation. Its eigenvalue is nonzero there because the row is invertible. Thus

$$A_L^D(z)\Omega_L = (T^D(z)T^I(z)^{-1})\Omega_L$$

is uniformly bounded and holomorphic on a common endpoint disc. Cauchy's estimates control every fixed Taylor coefficient and the norm of each Taylor remainder on a smaller disc, uniformly in L .

We first use these bounds to construct continued vacuum vectors inside $\mathcal{H}$. Take a subsequential Gram enlargement containing the vectors obtained by applying finite words of strict rows, endpoint rows and shifts to the vacuum and to these Taylor coefficients. Use countable dense strict arguments and a basis of marks. The coefficient bounds define a holomorphic vector series on the common disc in this enlargement. On its intersection with the strict strip, uniform Taylor remainder bounds identify that series with the already constructed strict vacuum vector. Its orthogonal projection onto $\mathcal{H}^\perp$ is consequently zero on this open intersection, and hence on the entire disc by the identity theorem. The continued vacuum vectors therefore belong to $\mathcal{H}$. Their pairings with strict words also show uniqueness, independent of the enlargement. At the center, the finite swap and crossing identities give the endpoint vectors in (4.3); the Cauchy bounds approximate them in norm by nearby strict vacuum vectors.

Choose a closed interval $I \subset (-v, v)$ with nonempty interior such that every RTT intertwiner between either endpoint and an angle in I is invertible. Such an interval exists: the excluded angles form a finite set. More explicitly, the possible zeros of the intertwiner determinant occur at $s \in \pi\mathbb{Z} \pm \lambda$, where $s = \eta(z - \alpha)$. When $\lambda > \pi/2$ the possible interior exceptions at the endpoints are $\alpha = \pm(v - (\pi - \lambda)/\eta)$; avoiding these points suffices. There is then a common endpoint disc on which all these inverses are bounded. Commute an endpoint-near row through a fixed word with angles in I . RTT expresses the result as a finite sum of bounded strict rows acting on continued marked vacuum vectors. The number and bounds of the terms may depend on the word, but the radius of analyticity does not. The same finite-chain estimates give norm convergence to the endpoint action. Density of the I -words and the uniform interior row bound extend this to strong convergence on all of $\mathcal{H}$. In particular both S and S^{-1} preserve $\mathcal{H}$; their restrictions are unitary inverses. Successive

strong approximations give the same Gram limits for products containing several endpoint rows. Define $D_0 = S^{-1}A^D(v)$ and $D_j = S^{-j}D_0S^j$. The finite site-matrix identities pass to these products: at each site this is a unital star representation of the 2×2 matrix algebra, and matrices at distinct sites commute. Products of these matrices are the local spin operators used below.

To pass from this vector continuation to scalar continuation with one radius, choose countably many dense I -words h_j and a finite measure equivalent to the scalar spectral measure. On a fixed endpoint disc write the continued vector $P(z)h_j$ as $\sum_{k \geq 0} a_{jk}(z - v)^k$. For $r' < R$ smaller than its common radius, Cauchy's estimate gives

$$\|a_{jk}\|_{L^2} \leq M_{j,R}R^{-k}, \quad \sum_{k \geq 0} (r')^k \|a_{jk}\|_{L^2} < \infty.$$

The series of fiber norms is therefore summable almost everywhere. Intersecting over the countable words and radii produces pointwise holomorphic vector series on the same disc. On the interior part each series equals $f(z)h_j$. Its continuation is consequently proportional to h_j . At almost every spectral point at least one h_j is nonzero, and the resulting scalar ratios agree by the identity theorem. This constructs the scalar continuation on a common disc.

Equation (4.3) gives unitary values at both endpoints. Covariance and strong convergence give unitary boundary values at $\pm v + ib$. Applying this first on a countable dense set of b 's and then using continuity shows that $|f| = 1$ on each continued boundary diameter. Schwarz reflection gives $f(z)f(2v - \bar{z}) = 1$ there, and the analogous formula at $-v$. Thus the continued functions are nonzero. Rational imaginary translations cover the boundary lines by discs of one radius, with a common null set. The bounded strip maximum principle now yields $|f| \leq 1$. Under $w = e^{iz}$ the excluded boundary bands give (4.4). Finally each scalar function has isolated zeros. Fubini on an imaginary-translation orbit gives a null zero set at almost every point of that orbit, and invariance of the spectral measure class transfers the assertion to every specified point. $\square$

The scalar continuation has a consequence on the entire Hilbert space $\mathcal{H}$, before any restriction on the marks. Canonical inner factorization, in the form recalled in [1, pp. 1367–1368, Equations (1)–(3)] and transferred from the disk by the Cayley map, gives

$$f(w) = \gamma e^{-aw-b/w} \prod_j \beta_j(w), \quad a, b \geq 0, \quad |\gamma| = 1, \quad (4.6)$$

where β_j is a constant-phase multiple of $(w - \zeta_j)/(w + \bar{\zeta}_j)$ and $\sum_j \operatorname{Re} \zeta_j / (1 + |\zeta_j|^2) < \infty$. Continuation through every nonzero finite point of the imaginary axis excludes singular measure there. The only possible remaining boundary singularities are at 0 and ∞ , giving the factors $e^{-b/w}$ and e^{-aw} . The zero cone (4.4) holds on this full space as well. We will use this general form for the full-space energy bounds. Diagonal words permit a further conclusion: below we show that their spectral functions have no singular factors and only finitely many zeros.

4.3 Identification with the physical plane

We next fix the probabilistic meaning of this representation. The physical row angle is $z = 0$, or $w = 1$, and the physical weights are $cR(\lambda/2)$, with $a = b = 1$. Multiplying every vertex weight by c does not affect any normalized finite-volume law.

Proposition 4.3 (Physical law, both axes, and reflection). *The row representation gives the local probabilities of the plane law $\mathbb{P}_c$ in Theorem 1.1. In particular its prescribed iterated balanced-torus limit exists when $c < 1$. Quantization with time pointing east and quantization with time pointing*

north describe this same law. For an ordinary observable supported on one side of either coordinate cut, the squared norm of its prepared state is its expectation paired with its complex-conjugate reflected observable, with the corresponding spin reflection. Products of neutral height observables may be prepared using paths confined to that side.

Proof. At fixed even transverse length, primitivity of the physical transfer in the chosen sector identifies the long-torus limit with its normalized Perron vector. A finite collection of arrows is read by physical row steps on a seam and by spins on finitely many equal-time sections. Repeated seam marks are combined, and the spins are expressed by (4.3). Its joint probabilities are therefore finite linear combinations of strict and endpoint word coefficients. Their limits exist by Proposition 3.6 and Lemma 4.2. When $c < 1$ the root construction used the exactly balanced sector, so these are precisely the two limits, in their prescribed order, defining $\mathbb{P}_c$.

The two axes when $c < 1$. Here the exactly balanced roots already identify the cylinder limit. To compare its two axial quantizations, fix a rectangle. Orient its input path upward along the left side and then eastward along the top side, and express all spins relative to those directed steps. Let K be the tensor of its unmarked physical tiles, viewed as a matrix from the input sides to the other two sides. It is a product of physical $R(\lambda/2)$ matrices and permutations of tensor factors. The local eigenvalues before the harmless factor c are $1/c, 1/c, 1/c + 1, 1/c - 1$. Thus K is invertible for $c < 1$.

For a finite surrounding cylinder, cut on all four sides and contract the exterior with its normalized vacuum data. This gives a linear functional L of the inside matrix, with $L(K) = 1$. The normalization includes the exterior transfer eigenvalues appropriate to that cylinder. Putting elementary matrices $D_1, \dots, D_q$ on the input path tests the functional on $K(D_1 \otimes \dots \otimes D_q)$: the row index of each mark is inside and its column index outside. Since tensor products of elementary matrices span the input matrix algebra and K is fixed and invertible, the limiting marked path values determine every entry of the limiting exterior functional. They also prove that those entries have limits.

In the eastward representation the successive path angles are $v, 0$; in the northward representation, with space pointing west, they are $0, -v$. To compare them, first replace the first pair by $(v - \varepsilon, \varepsilon)$ and translate both angles by $-v$. The resulting pair is $(-\varepsilon, -v + \varepsilon)$, still strict. Common-translation invariance identifies the marked word values. Strong boundary convergence then permits $\varepsilon \downarrow 0$. The spin and matrix-index conventions agree: at a reversed collapsed step the mark is XD^tX , exactly as in (4.3). Crossing only relabels those spins; the interior physical tensor remains K . Thus the two normalized limiting functionals agree on KD for every elementary tensor D , and hence on every inside matrix. All local arrow probabilities agree. This argument compares normalized exterior functionals; it requires no equality between eigenvalues of cylinders of different circumferences.

Identification when $c \geq 1$. In this range we must identify the state also when the root construction uses sectors tending to half filling. Let W_c be the physical vertex weight and, for a stationary probability measure ρ on allowed arrow configurations, put

$$\mathcal{F}_c(\rho) = -h(\rho) - \mathbb{E}_\rho \log W_c, \quad (4.7)$$

where h is specific entropy per vertex. The equivalent gradient formulation fixes the height parity and forgets an additive constant; these changes have zero entropy cost per unit area. Spatial translation acts on gradients: after translating an anchored height representative, one subtracts its new reference value. Thus fixing the reference parity does not remove any physical translation from the gradient action. The domain and reference-height convention are those of [26, Section 2.3 and Appendix B]; the entropy–energy decomposition is also explicit in [21, Sections 2.3.2–2.3.4]. Since the alphabet is finite and $0 < c < \infty$, the entropy and local energy in (4.7) are finite. Hard constraints are included by restricting to allowed configurations.

Write $U(0)$ for the homogeneous harmonic normalization in Lemma 3.3, and set $p_* = U(0) + \log c$. The same boundary-unitarity and strip argument used there applies to a row with specified auxiliary input and output spins: it is a matrix-unit auxiliary mark. A rectangle of width a and height b , with any boundary data, therefore has partition sum at most

$$\exp\{abp_* + C(a + b)\}.$$

Indeed each open row has operator norm at most $2e^{ap_*}$, and summing the boundary spins costs only an exponential in its perimeter. The finite entropy variational inequality on the rectangle then gives $h(\rho) + \mathbb{E}_\rho \log W_c \leq p_*$ for every stationary allowed state.

For the near-balanced tori, choose their longitudinal lengths sufficiently large to realize the Perron asymptotic. Their constrained log partition functions per vertex tend to p_* by Lemma 3.3. Tile each torus by fixed-size squares. Entropy subadditivity, with a negligible uncovered fraction, bounds its entropy per vertex above by the entropy of a fixed-square marginal divided by the square area, plus a vanishing boundary error. Pass to the local limit at this fixed square size, then let the square size grow. The exact finite identity “entropy plus mean log weight equals log partition function” proves the reverse inequality $h(\rho) + \mathbb{E}_\rho \log W_c \geq p_*$. Hence the limiting state attains the global minimum of (4.7). Its expected spatial tilt is zero by the sector sizes. The other expected tilt is zero as well: at the physical angle the current mark is skew-adjoint, whereas its finite physical expectation is real. This observation specifies the tilt; it is not being used as a uniqueness argument.

The equilibrium characterization following equation (6) in [7], for $1 \leq c \leq 2$, identifies the finite-specific-free-energy minimizer with the balanced plane law. We spell out the variational-domain reduction needed to apply this characterization, since a zero expected tilt alone would not suffice. For a fixed spatial translation lattice $\mathcal{L}$, let $\mathcal{P}_\mathcal{L}$ denote invariant laws of allowed gradients and write

$$\sigma_\mathcal{L}(a) = \inf\{\mathcal{F}_c(\rho) : \rho \in \mathcal{P}_\mathcal{L}, S(\rho) = a\}.$$

Here $S(\rho)$ is the expected physical slope. Both the preceding perimeter bound and the tiling proof apply on this domain, using the average energy over a fundamental cell if necessary. We keep the same $\mathcal{L}$ when forming the infimum and the ergodic decomposition. The surface tension is unchanged on passing from full translations to a finite-index sublattice: average any sublattice-invariant law over the finitely many translation cosets. This preserves its slope and its entropy–energy density and makes it fully invariant; the reverse inequality between the infima follows from inclusion of the domains. Thus one may use the full physical gradient translation action. If a periodic height-potential representation uses a parity-preserving sublattice, the argument is performed throughout on that sublattice and is then projected back to arrows.

Specific free energy is affine under this ergodic decomposition [26, Chapter 3]. Thus every component of a globally minimizing law is itself a global minimizer: each component has free energy at least the global infimum, and their average equals it. For $c \geq 1$, strict convexity of the six-vertex surface tension on the interior of its slope domain is [21, Theorem 13.16 of the 2020 preprint]. Its stated condition is $c^2 \geq \max(1, 1)$, and the accompanying simply-attractive-potential description supplies the height-model setting. Arrow reversal makes $\sigma_\mathcal{L}$ even. The origin is an interior slope: height functions formed as sums of two one-dimensional unit-increment paths give allowed stationary gradient laws at every slope in $[-1, 1]^2$, all of finite specific free energy. Hence strict convexity implies $\sigma_\mathcal{L}(a) > \sigma_\mathcal{L}(0)$ for every nonzero interior a . A boundary slope cannot be another minimizer: otherwise convexity along the segment from 0 to that slope would contradict this strict inequality at an interior point. Every minimizing ergodic component therefore has slope zero and is a fixed-slope minimizer. The variational principle makes it a minimal gradient phase.

At this last step we use the zero-slope uniqueness supplied by the combined equilibrium characterization in [7], with its Sheffield and delocalization inputs and its range $1 \leq c \leq 2$. This is not an

assertion of uniqueness for every Gibbs state, nor an extension to $c > 2$. The domain reduction above explains why the possibly nonergodic auxiliary state is covered, even when a formulation of the underlying height result is stated for minimal phases. The gradient-to-arrow correspondence is one-to-one after forgetting the additive height constant, so their common law is $\mathbb{P}_c$. The same argument applies to either axial quantization, and therefore identifies the two orientations as well.

Reflection across a coordinate cut. In either homogeneous axial quantization, transposing the finite row product at a coordinate cut reflects its tiles and changes the spin mark by the stated adjoint rule. The finite Gram square is therefore the expectation of the observable and its conjugate reflected copy. Pass to the local and Gram limits to obtain the assertion. Neutral height differences can be read along paths on the prescribed side of the cut, so the same identity applies to their finite products. $\square$

4.4 Finite inner functions and a measure on dilation orbits

We now restrict when indicated to the closed diagonal-word subspace $\mathcal{D} \subset \mathcal{H}$: it is generated by words all of whose marks are diagonal. It is invariant under P, J and their adjoints, where $J = A^Z$ and $Z = \text{diag}(1, -1)$. Endpoint diagonal rows show that it contains the states needed for ordinary height insertions. Write $P(w)$ and $J(w)$ in half-plane coordinates from now on.

Lemma 4.4 (Inner spectra and the current measure). *On $\mathcal{D}$, almost every spectral function is either 1 or a finite Blaschke product in the right half-plane, with*

$$f(0) = f(\infty) = 1, \quad f(\bar{w}) = \overline{f(w)}. \quad (4.8)$$

Its zeros obey (4.4). In particular its degree is even; write it as $2n_f$ when it is nonconstant. Every zero produces a fixed-width, fixed-depth dip of $|f(e^s)|$ near its logarithmic radius.

There is a possibly infinite positive measure μ on the nonconstant functions and a measurable spectral vector $\mathbf{h}$ such that

$$J(w)\Omega = (1 - f(w))\mathbf{h}, \quad \mu(df) = \|\mathbf{h}(f)\|^2 \sigma(df). \quad (4.9)$$

Here σ is any scalar measure used for the direct-integral representation; the resulting measure μ does not depend on that choice. It is invariant under dilation of the argument. On the cross-section where the smallest zero radius is one, it has the form

$$\mu = ds \nu(dF), \quad f(w) = F(e^s w), \quad \kappa := \nu(1) < \infty. \quad (4.10)$$

If $\mathcal{L}(F)$ is the difference between its largest and smallest logarithmic zero radii, then

$$\nu\{\mathcal{L} > t\} \leq C e^{-2\eta t} \quad (t \geq 1). \quad (4.11)$$

For a fixed diagonal word h ,

$$\|(1 - P(e^s))h\| \leq C_h e^{-\eta|s|}. \quad (4.12)$$

For products of fixed words placed at scales r_i , the corresponding bound outside their fixed internal argument ranges is $C \sum_i e^{-\eta|s - \log r_i|}$, uniformly when these ranges separate.

Proof. As an auxiliary argument is dilated far from all arguments in a fixed word, the scalar-normalized RTT intertwiner tends to an invertible diagonal matrix, with error $O(e^{-\eta|s|})$. Its limiting diagonal matrix commutes with the diagonal joint auxiliary marks. Move $P(e^s)$ through the word to the vacuum, where it equals the identity. The exchange errors are finite sums of bounded marked-row

products, proving (4.12). Moving through several separated groups gives the stated sum, since the number of factors is fixed. These are operator-norm bounds on each finite exchange error, not just bounds on its vacuum expectation.

On a countable total set in $\mathcal{D}$, the spectral integral of $|1 - f(e^s)|^2$ over s is consequently finite. Apply the full-space factorization (4.6). If $a > 0$ or $b > 0$, then $|f(e^s)|$ tends to zero at one end, contrary to the integrability just proved. For a zero ζ with $r = |\zeta|$, the identity

$$\left| \frac{t - \zeta}{t + \bar{\zeta}} \right|^2 = 1 - \frac{4t \operatorname{Re} \zeta}{|t + \bar{\zeta}|^2}, \quad t > 0,$$

and (4.4) imply $1 - |f(e^s)| \geq \varepsilon_0$ whenever $|s - \log r| \leq \delta_0$, with fixed positive ε_0, δ_0 . Infinitely many zeros would escape to one end and supply infinitely many disjoint such intervals, again contradicting integrability. The resulting finite product has limits at both ends, and integrability forces both limits to be one. Reality was already established. Nonreal zeros occur in conjugate pairs, and the endpoint quotient is $(-1)^{\deg f}$, so the degree is even. This proves (4.8) and the dip assertion.

Differentiate commutation of equal diagonal-twist transfers at twist zero and apply it to Ω . Since $P(w)\Omega = \Omega$, it gives

$$(1 - P(w'))J(w)\Omega = (1 - P(w))J(w')\Omega. \quad (4.13)$$

There is no component of $J(w)\Omega$ on $f = 1$. Indeed, let Q be the projection to that spectral set, which contains Ω . The endpoint identities and their dilates glue $QJ(w)Q$ to the left half-plane by $QJ(-w)Q = -QJ(w)Q$, because $P(w) = I$ on Q . It is bounded on the punctured plane, so the singularities at zero and infinity are removable. Liouville's theorem and oddness make it zero. On every nonconstant fiber $|f(1)| < 1$, so one may set $h = J(1)\Omega/(1 - f(1))$. Equation (4.13) gives (4.9), first at a countable set of arguments and then everywhere, including endpoints by strong convergence. Covariance, with $U_b\Omega = \Omega$, makes this quotient transform without a scalar multiplier. Its squared fiber norm therefore gives a dilation-invariant measure.

Every nonconstant finite product has a unique smallest zero radius. Rescale it to one to obtain a measurable cross-section. Translation invariance in its logarithmic dilation coordinate gives (4.10), initially as a sigma-finite disintegration. Each orbit has a dip of fixed width at its first zero, where $|1 - f(1)| \geq \varepsilon_0$. Since

$$\int |1 - f(1)|^2 d\mu = \|J(1)\Omega\|^2 < \infty,$$

this disintegration is locally finite and $\kappa < \infty$.

Applying (4.12) to $J(1)\Omega$ gives

$$I(l) := \int |1 - f(1)|^2 |1 - f(e^l)|^2 d\mu \leq Ce^{-2\eta l} \quad (l \geq 1). \quad (4.14)$$

On an orbit of width $\mathcal{L}$, integrate first over the dip window of its smallest zero and then over l placing the second argument in the dip window of its largest zero. Both intervals have a fixed width and both factors a fixed lower bound. Thus

$$c\nu\{\mathcal{L} > t\} \leq \int_{t-2\delta_0}^{\infty} I(l) dl,$$

after changing the constant for bounded t . This proves (4.11). $\square$

4.5 Low spectral bands and the two leading current modes

We now return to the full space $\mathcal{H}$. The low-energy estimates will be used with arbitrary local spin states, so the finite-product conclusion on $\mathcal{D}$ is not enough for this part of the argument. The full-space factorization (4.6) will instead control the zeros and the possible singular factors. Our aim is to isolate the two leading angular modes of a current after time damping. Define

$$H_0 = -\log |P(1)|, \quad Q_u = \mathbf{1}_{[0,u]}(H_0), \quad Q_u^\sigma = Q_u \mathbf{1}_{\{\text{sgn } P(1)=\sigma\}}, \quad \sigma = \pm 1.$$

The logarithm is defined by spectral calculus; the fixed-argument no-zero assertion makes its value finite almost everywhere. Put

$$N(w) = \frac{1}{2} \{J(w)P(w) + P(w)J(w)\}. \quad (4.15)$$

This current averages the two positions of a mark among two consecutive steps. Its two leading Laurent modes will become the two continuum currents. The factor $1/2$ will be retained throughout.

Lemma 4.5 (Low-band current estimates). *There are constants $A, a, C, u_0 > 0$ such that, for $0 < u < u_0$, $Q_u J(w) Q_u$ extends to a bounded holomorphic function on the full annulus $Au < |w| < a/u$. If*

$$Q_u J(w) Q_u = \sum_{j \in \mathbb{Z}} w^j J_{u,j}, \quad (4.16)$$

then $\|J_{u,j}\| \leq C(Cu)^{|j|}$, with the exponent zero for $j = 0$. Set $\theta = \eta/2 > 0$. Uniformly on the closed right unit semicircle,

$$\|Q_u^\sigma J(w) Q_u^\sigma\| \leq Cu, \quad \|Q_u^\sigma J(w) Q_u^{-\sigma}\| \leq Cu^\theta, \quad (4.17)$$

$$Q_u^\sigma N(w) Q_u^\sigma = \sigma Q_u^\sigma (w J_{u,1} + w^{-1} J_{u,-1}) Q_u^\sigma + O(u^2), \quad (4.18)$$

$$\|Q_u^\sigma N(w) Q_u^{-\sigma}\| \leq Cu^{1+\theta}, \quad \|Q_u N(w) Q_u\| \leq Cu. \quad (4.19)$$

The estimates remain valid for unequal input and output cutoffs, using their maximum for u . With m ordinary time rows on each side, N costs $O(m^{-1})$ and a sign-opposite undoubled J costs $O(m^{-\theta})$. The leading-mode remainder is $O(m^{-1-\theta})$ when, on each dyadic input/output block, the leading expression is taken at the larger cutoff; the precise block sum is defined in the proof.

Finally, if $S = P(i)$ and K is the principal argument of S^2 , then

$$S^2 = e^{iK}, \quad |K| \leq CH_0. \quad (4.20)$$

Proof. The zero budget and analytic annulus. Evaluate (4.6) at 1. On the band $H_0 \leq u$ it gives

$$a + b + \frac{1}{2} \sum_j \log \left(1 + \frac{4 \operatorname{Re} \zeta_j}{|1 - \zeta_j|^2} \right) \leq u.$$

The cone condition implies the zero budget

$$a + b + \sum_j \min(|\zeta_j|, |\zeta_j|^{-1}) \leq Cu. \quad (4.21)$$

Indeed $\operatorname{Re} \zeta_j \geq c_0 |\zeta_j|$ and $|1 - \zeta_j| \leq 1 + |\zeta_j|$. Each zero is therefore either of radius at most Cu or of reciprocal radius at most Cu . Comparing each factor at w and at 1 shows, on $Au < |w| < a/u$, $\operatorname{Re} w \geq 0$, that

$$\left| \log \frac{f(w)}{f(1)} \right| \leq Cu(|w| + |w|^{-1}).$$

Choose A large and a small once and for all. The function and its inverse are then bounded on this half-annulus. More precisely,

$$P(w)Q_u^\sigma = \sigma Q_u^\sigma + O(u(|w| + |w|^{-1})), \quad (4.22)$$

and the same estimate holds for the inverse restricted to the band.

The endpoint formulas give

$$J(i) = SZ_0, \quad J(-i) = -Z_0 S^{-1}, \quad J(-i) = -P(i)^{-1}J(i)P(i)^{-1}.$$

The other endpoint gives the analogous identity on the other ray. Dilate these uncompressed identities using (4.1), and only then compress. This is important because Q_u need not commute with the dilation group. The formula

$$J_u(-w) = -P(w)^{-1}J_u(w)P(w)^{-1}, \quad J_u(w) = Q_u J(w) Q_u, \quad (4.23)$$

defines the left half of the annulus; both inverses here and below are only on $Q_u \mathcal{H}$. Boundedness and strong boundary convergence make the two definitions agree on both imaginary rays. Scalar Morera applied to every matrix coefficient gives holomorphic gluing. Thus $\|J_u(w)\| \leq C$ on the full annulus. Cauchy's formula on circles near its inner and outer boundaries proves (4.16) and its coefficient bounds. In particular, uniformly on $|w| = 1$,

$$\frac{1}{2}(J_u(w) - J_u(-w)) = wJ_{u,1} + w^{-1}J_{u,-1} + O(u^3) = O(u). \quad (4.24)$$

The nonconstant even modes are $O(u^2)$, but this observation alone does not control the constant coefficient.

Sign blocks and the leading modes. For $X_{\sigma\tau}(w) = Q_u^\sigma J(w) Q_u^\tau$, (4.23) and (4.22) give

$$X_{\sigma\tau}(-w) = -\sigma\tau X_{\sigma\tau}(w) + O(u)\|X_{\sigma\tau}(w)\|.$$

On an equal-sign block its even part is therefore bounded by $Cu\|X_{\sigma\sigma}(w)\|$. Its odd part is $O(u)$ by (4.24). Absorbing gives $\|X_{\sigma\sigma}\| \leq Cu$ and then an $O(u^2)$ bound for its whole even part, including the constant coefficient. This proves the first estimate in (4.17).

For opposite signs use an additional RTT comparison at $r = \sqrt{u}$. The normalized intertwiner for arguments r, w is $R_\infty + O(r^\eta)$ with bounded inverse, uniformly for w on the closed right unit semicircle; R_∞ is diagonal and invertible. It commutes with the joint mark $I \otimes Z$. Conjugating this mark and expanding the $O(r^\eta)$ error in tensor matrix units, each producing a bounded marked-row product, proves

$$\|[P(r), J(w)]\| \leq Cr^\eta.$$

Strong approximation includes the endpoints. On Q_u^σ , $P(r) = \sigma I + O(\sqrt{u})$ by (4.22). Compression of the commutator yields

$$(2 - C\sqrt{u})\|X_{\sigma,-\sigma}(w)\| \leq Cu^{\eta/2},$$

which proves the second estimate in (4.17). Inserting $P(w) = \sigma I + O(u)$ in (4.15) now proves (4.18)–(4.19): equal signs add, whereas opposite signs cancel. In the latter case the odd part of X is itself $O(u^{1+\theta})$ by the gluing identity. Thus the leading expression can equivalently be written

$$\frac{1}{2}\{\text{sgn}(P(1))Q_u, wJ_{u,1} + w^{-1}J_{u,-1}\} + O(u^{1+\theta})$$

on the whole band. Higher modes need not vanish at a fixed cutoff; the displayed error is what removes them after rescaling.

Time damping and shift phase. For unequal bands embed both in their larger band. Here is the precise meaning of the damped leading-mode remainder. For large m , put $t_q = 2^q/m$ and let q_m be the largest integer with $t_{q_m} < u_0$. Let $B_0 = \mathbf{1}_{[0,t_0]}(H_0)$, $B_q = \mathbf{1}_{(t_{q-1}, t_q]}(H_0)$ for $1 \leq q \leq q_m$, and let $B_* = I - \sum_{q=0}^{q_m} B_q$. Write $\Sigma = \text{sgn } P(1)$ and define

$$L_u(w) = \frac{1}{2} \{ \Sigma Q_u, w J_{u,1} + w^{-1} J_{u,-1} \}, \quad u_{qr} = \max(t_q, t_r).$$

On the pair of low bands B_q, B_r the preceding estimates give

$$\|B_q(N(w) - L_{u_{qr}}(w))B_r\| \leq C u_{qr}^{1+\theta}.$$

Thus the leading operator after damping is the finite block sum

$$L_m^{\text{damp}}(w) = \sum_{q,r=0}^{q_m} P(1)^m B_q L_{u_{qr}}(w) B_r P(1)^m.$$

Each block uses its own larger cutoff; no cutoff-independent Laurent coefficient is asserted. On B_q with $q \geq 1$, time damping contributes at most $e^{-mt_q/2}$. For every $\beta > 0$ the corresponding double sum is bounded by

$$C_\beta m^{-\beta} \sum_{q,r \geq 0} \max(2^q, 2^r)^\beta e^{-c(2^q+2^r)} \leq C'_\beta m^{-\beta},$$

where the lowest bands are absorbed into the constant. Taking $\beta = 1 + \theta$ gives

$$\|P(1)^m N(w) P(1)^m - L_m^{\text{damp}}(w)\| \leq C m^{-1-\theta}.$$

Indeed any term involving B_* is exponentially small: its spectrum lies above $t_{q_m} \geq u_0/2$, and N is globally bounded. The same block sum with $\beta = 1$ bounds the damped N itself, and with $\beta = \theta$ bounds a sign-opposite J . These arguments also work when the damping rows range in a fixed compact strict-angle interval. Schwarz–Pick compares their modulus loss $1 - |f(w)|$ uniformly with $1 - |f(1)|$. On low bands this is comparable to H_0 ; on the remaining spectrum it is bounded below by a positive constant.

On a sufficiently small band, (4.22) at $w = i$ gives $|\arg f(i)^2| \leq C H_0(f)$, by applying the estimate with cutoffs decreasing to the fiber value. On its complement the principal argument is at most π , which is also bounded by a fixed multiple of H_0 . Spectral calculus proves (4.20). $\square$

4.6 The endpoint Hamiltonian and the sum rule

The contraction $P(1)$ measures time damping. We also need the generator obtained by differentiating the unitary endpoint, because it is local on spin states. These are different operators: we keep the notation H_0 for the former and E for the latter.

Lemma 4.6 (The local endpoint generator). *The nonnegative self-adjoint spectral operator*

$$E = S^{-1} \partial_z P(z) \big|_{z=v}, \quad E(f) = - \partial_s \arg f(i e^s) \big|_{s=0}, \quad (4.25)$$

agrees on local-spin vectors with the vacuum-subtracted nearest-neighbor Hamiltonian whose link is

$$h_{j,j+1} = -\eta \text{swap}_{j,j+1} R'_{j,j+1}(0) = \begin{pmatrix} \eta \cot \lambda & 0 & 0 & 0 \\ 0 & 0 & -\eta/\sin \lambda & 0 \\ 0 & -\eta/\sin \lambda & 0 & 0 \\ 0 & 0 & 0 & \eta \cot \lambda \end{pmatrix}_{j,j+1}. \quad (4.26)$$

In particular $[h_{j,j+1}, Z_j + Z_{j+1}] = 0$ and $E\Omega = 0$. The closure of the local-spin subspace reduces E , and its real-time evolution is the local Hamiltonian evolution. For all sufficiently small u ,

$$\mathbf{1}_{\{E \leq u\}} \leq \mathbf{1}_{\{H_0 \leq Cu\}}. \quad (4.27)$$

With $Z_1 = S^{-1}Z_0S$, the exact adjacent-pair identity is

$$N(i) = \frac{1}{2}S^2(Z_0 + Z_1). \quad (4.28)$$

Proof. Differentiate the inner factors along the imaginary boundary. The singular factors in (4.6) contribute $a + b$, and a zero ζ contributes

$$\frac{2 \operatorname{Re} \zeta}{|i - \zeta|^2}. \quad (4.29)$$

All contributions are nonnegative. Scalar continuation makes the derivative finite almost everywhere and justifies the sum, or one may first differentiate finite partial products and use positivity. Since $\partial_z = iw\partial_w$, this is exactly the spectral derivative in (4.25). The cone condition bounds (4.29) below by a constant times $\min(|\zeta|, |\zeta|^{-1})$. Thus small E implies the small/large zero budget in (4.21). On that budget the negative logarithm of each factor at 1 is bounded above by the same quantity. This proves (4.27).

At finite length, differentiate one factor in the swap row and divide by its shift. The inserted link is $-\eta \operatorname{swap} R'(0)$, proving (4.26). Dividing by the vacuum eigenvalue subtracts its vacuum energy. For a fixed local matrix O , the finite action on $O\Omega_L$ is therefore

$$\left[\sum_j h_{j,j+1}, O \right] \Omega_L,$$

where only links meeting the support of O occur. To justify passing this derivative to the limit, write the local matrix as a linear combination of products of one-site matrices, inserting identity marks at gaps. Put $B_L(D) = A_L^D(v) = S_L D_0$. The finite identity

$$B_L(D_m) \cdots B_L(D_1) = S_L^m (D_m)_{m-1} \cdots (D_1)_0$$

expresses $O\Omega_L$, after a fixed translation if needed, as a finite sum of upper-endpoint words with a single power of S_L outside the word. First commute the unmarked row $P_L(z)$ through that outer shift. Then move it through the upper-endpoint rows by RTT. The intertwiner at coincident upper-endpoint arguments is the invertible permutation matrix, so all these exchanges are analytic on a fixed small disc about v . Changed auxiliary marks still belong to the same finite-dimensional matrix algebra. Each resulting term is a bounded endpoint word applied to a continued marked vacuum vector, which has the uniform Taylor bounds from the finite division estimate. This proves wordwise vector Taylor bounds for $P_L(z)O\Omega_L$ on that disc. The same Gram and uniform-remainder argument as in Lemma 4.2 identifies the continued vector in $\mathcal{H}$. Cauchy's formula therefore passes its derivative, in norm and in all Gram pairings, to the spectral derivative. Consequently

$$E(O\Omega) = \left[\sum_j h_{j,j+1}, O \right] \Omega. \quad (4.30)$$

Iterating is legitimate because the commutator remains local. If O is supported on an interval of length ℓ , each new commutator enlarges the support by at most two. Its norm is bounded by $2\|h\|$ times the number of interacting links. Hence

$$\|E^k O\Omega\| \leq C_O C_O^k k!.$$

Local vectors are analytic for the symmetric operator specified by (4.30). Analytic-vector uniqueness makes its closure self-adjoint on their closed span. Since E extends this operator, the span reduces E and the two self-adjoint operators agree there. This also identifies real-time evolution. Finally $J(i) = SZ_0$ and $P(i) = S$ in (4.15) give (4.28). $\square$

Lemma 4.7 (Energy and winding sum rule). *Let p denote the phase of the unit shift $S = P(i)$, modulo 2π , and let sym average a measure at p and $-p$. Set*

$$D_0 = -\langle \Omega, [Z_0, [Z_0, h_{0,1}]] \Omega \rangle.$$

Away from $p = 0$, the projection of the current measure obeys

$$2 \text{sym}((p)_*(E\mu)) = D_0 \frac{dp}{2\pi}. \quad (4.31)$$

On each orbit of degree $2n_F$, the phase winds monotonically through n_F complete turns. In particular

$$\int n_F d\nu(F) < \infty. \quad (4.32)$$

Proof. Since $Z_0\Omega = S^{-1}J(i)\Omega$, the spectral measure of the local density vector is

$$\|Z_0\Omega\|^2(df) = |1 - f(i)|^2 d\mu(f). \quad (4.33)$$

Set $c(j) = \langle \Omega, [Z_0, [E, Z_j]] \Omega \rangle$ and take its spatial Fourier transform. By $E\Omega = 0$, the transform is twice the symmetrized energy-weighted spectral measure of $Z_0\Omega$. On the other hand (4.30) and conservation of $Z_0 + Z_1$ leave only the three coefficients at separations 0, 1, -1 , namely $2D_0, -D_0, -D_0$. Their transform is $|1 - e^{ip}|^2 D_0$. Thus

$$2 \text{sym}((p)_*(E|1 - f(i)|^2 \mu)) = |1 - e^{ip}|^2 D_0 \frac{dp}{2\pi}.$$

Testing on compact subsets of the circle without zero permits division by the squared factor and proves (4.31).

On a nonconstant finite-product orbit, direct differentiation gives $-dp/ds = E(F(e^s)) > 0$. The boundary phase change on the positive imaginary ray is one half of the full boundary change: this follows either by pairing conjugate zeros in (4.8) or by integrating (4.29) with its dilated argument. It is $2\pi n_F$. At each nonzero phase there are consequently n_F crossings, counted with multiplicity, and none has zero derivative. The change of variables from s to phase gives

$$(p)_*(E ds \nu) = \left(\int n_F d\nu(F) \right) dp$$

away from zero, first with truncated orbit measures and then by monotone convergence. Equation (4.31) implies finiteness; in these conventions it also gives $4\pi \int n_F d\nu = D_0$. The endpoints of an orbit and its finitely many zero-phase crossings have zero ds measure, so no missing phase atom is involved in this computation. $\square$

4.7 The height variogram

The measure constructed above already determines height covariances. The following bound will also supply the logarithmic majorants needed for nonsmooth tests and regularity.

Proposition 4.8 (Exact variogram and logarithmic growth). *After an axial symmetry, any nonzero lattice displacement may be written (k, j) with $k \geq 1$ and $|j| \leq k$. Its height variance is*

$$\begin{aligned} \mathbb{E}_c(h(k, j) - h(0, 0))^2 &= 2 \operatorname{Re} \int (1 - f(1))^k f(\pm i)^{|j|} d\mu(f) \\ &= 4\kappa \log k + O(1), \end{aligned} \quad (4.34)$$

where the sign of i is chosen according to the vertical direction. The error is uniform over these displacements. In particular the variance is bounded by $C(1 + \log(1 + |k| + |j|))$.

Proof. First consider a directed path of physical or endpoint steps, and write their spectral multipliers as $t_1, \dots, t_m$. Reflection changes the sign of a current. Equations (4.9) and (4.3) therefore express the covariance of two distinct increments, in their path order, as the negative real part of the integral of

$$(1 - t_a)t_{a+1} \cdots t_{b-1}(1 - t_b), \quad a < b,$$

with the conjugate term supplying the real covariance. A single increment has variance one. At an endpoint this equals $2 \operatorname{Re} \int (1 - f(i)) d\mu$, because $|f(i)| = 1$ and (4.33) integrates to $\|Z_0 \Omega\|^2 = 1$. The same identity holds at the physical step $w = 1$: on each orbit move the ray from $i\mathbb{R}_+$ to $\mathbb{R}_+$ by Cauchy's theorem for $(1 - F(w)) dw/w$. Its small and large circular arcs vanish, since $F(0) = F(\infty) = 1$. The integrated identity is justified by the bounds below.

The elementary telescoping identity

$$1 - \prod_{a=1}^m t_a = \sum_{a=1}^m (1 - t_a) - \sum_{a < b} (1 - t_a)t_{a+1} \cdots t_{b-1}(1 - t_b)$$

therefore gives the first equality of (4.34), taking k horizontal and $|j|$ vertical steps. It is equally valid by truncating the orbit measure first and then passing to the absolutely convergent integrals below.

Here are bounds that both justify this passage and give the asymptotic. On an orbit choose one zero radius as origin in logarithmic coordinates and let $\mathcal{L}$ be the range of the log radii. Beyond either extreme by distance d , the factor formulas and the endpoint normalizations give

$$|1 - F(e^s)| + |1 - F(\pm i e^s)| \leq C n_F e^{-d}, \quad (4.35)$$

or the trivial constant bound when it is smaller. The factors on the imaginary ray have modulus one; those on the positive ray have modulus at most one. Hence the integrand in (4.34) is absolutely integrable on each orbit. Outside the zero range enlarged by $\log k + \log(1 + n_F)$, the inequality $|1 - a^k b^{|j|}| \leq k|1 - a| + |j||1 - b|$, with $|j| \leq k$, and (4.35) give a bounded integral. The remaining extra interval lengths cost at most $C(\mathcal{L} + \log(1 + n_F) + 1)$.

For the main interval, use any single zero. Its cone bound gives

$$|F(e^s)|^k \leq \exp\{-cke^{-|s-s_0|}\} \quad \text{for } |s - s_0| \leq \log k,$$

where s_0 is its logarithmic radius (with the sign dictated by the chosen orbit coordinate). The integral of the right side over this interval is bounded uniformly in k . Multiplication by $F(\pm i e^s)^{|j|}$ does not change that bound. Consequently the integral of the real part of $1 - F(e^s)^k F(\pm i e^s)^{|j|}$ is

$$2 \log k + O(1 + \mathcal{L} + \log(1 + n_F)).$$

The remainder is integrable against ν by (4.11), (4.32), and $\kappa < \infty$. Multiplication by the factor 2 in the variogram proves (4.34). The same majorants justify the orbitwise contour move and the telescoping passage above. Axial symmetry, already proved for the physical law, supplies the final uniform displacement bound. $\square$

We have thus obtained a physical representation with a finite measure ν of dilation orbits, an as yet unevaluated constant κ , and uniform low-energy current estimates. Section 5 determines κ for $c < 1$; for $1 \leq c < 2$, the final proof uses the conditional amplitude theorem after convergence has been proved. Section 6 uses the same estimates, including their higher-mode errors, to identify all limiting current correlations.

5 Determining the covariance below $c = 1$

Throughout this section $0 < c < 1$, so $2\pi/3 < \lambda < \pi$. Our objective is to determine the orbit mass κ defined in Section 4. Two different finite systems give opposite bounds. A homogeneous chain at fixed particle density gives a lower bound through its response to a seam twist. A dense array of real inhomogeneities gives an upper bound through the change of its vacuum vector under the same twist. Neither comparison uses positive association.

Proposition 5.1 (The covariance constant below one). *With the row and height conventions of Sections 2 and 4,*

$$\kappa = \frac{1}{2\pi(\pi - \lambda)}. \quad (5.1)$$

Consequently a limiting covariance $-2\kappa \log |x - y|$ has squared multiplier $4\pi\kappa = 2/(\pi - \lambda) = \sigma(c)^2$ relative to (1.1).

5.1 Homogeneous stiffness at fixed density

For $Q < \infty$, let K_Q denote convolution with K restricted to $[-Q, Q]$, and write

$$r_Q = (I + K_Q)^{-1}s, \quad H_Q = (I + K_Q)^{-1}\mathbf{1}.$$

These are the density and dressed constant of Lemma 3.1. Choose the homogeneous finite chain with particle number $n/L \rightarrow \int_{-Q}^Q r_Q(u) du$. The first limit in this subsection is $L \rightarrow \infty$ at this fixed Q ; only afterwards does $Q \rightarrow \infty$. After both limits, the equal-time local state is the balanced plane state, by Proposition 3.6 and Proposition 4.3. At fixed Q , the filling remains strictly below one half.

Insert the auxiliary seam e^{itZ} , and let $\mathcal{H}_L(t)$ be the endpoint Hamiltonian before subtracting its ground energy. Denote its continued sector ground energy by $e_L(t)$, its unit ground vector at zero twist by Ω_L , and

$$E_{\text{exc},L} = \mathcal{H}_L(0) - e_L(0)$$

on that magnetization sector. The untwisted sector minimum is simple. All inverse-energy expressions below are restricted to $\Omega_L^\perp$ in that sector.

At the swap endpoint, the auxiliary spin is carried from site to site. The seam therefore twists the closing link. Conjugating by site rotations whose angles increase in increments t/L distributes this twist over all links, since each link conserves the sum of its two spins. Let b_j and d_j be respectively the first and second derivatives of the local link under the single-link twist, with a common choice of orientation. Put $\mathcal{J}_L = \sum_j b_j$ and $D_{0,L} = L^{-1} \sum_j d_j$. Ordinary second-order perturbation theory then gives

$$Le_L''(0) = \langle D_{0,L} \rangle - \frac{2}{L} \left\| E_{\text{exc},L}^{-1/2} (I - |\Omega_L\rangle\langle\Omega_L|) \mathcal{J}_L \Omega_L \right\|^2. \quad (5.2)$$

This boundary-twist curvature identity has a standard spin-chain formulation in Shastry and Sutherland [25, pp. 243–244], who trace the method to Kohn. Indeed, after distributing the seam, $\mathcal{H}'_L(0) = L^{-1} \mathcal{J}_L$ and $\mathcal{H}''_L(0) = L^{-2} \sum_j d_j$. The sign chosen for t has no effect on (5.2). For either

orientation, the second link derivative is $d_j = -[Z_j, [Z_j, h_{j,j+1}]]$. Thus the completed local-state limit of $\langle D_{0,L} \rangle$ is precisely the constant D_0 in Lemma 4.7.

Lemma 5.2 (Homogeneous stiffness). *The iterated homogeneous stiffness satisfies*

$$\lim_{Q \rightarrow \infty} \lim_{L \rightarrow \infty} L e_L''(0) = \frac{2}{\pi - \lambda}. \quad (5.3)$$

Proof. The root-dependent part of the endpoint energy is $\sum_j e(y_j)$, where

$$e(u) = -2\pi\eta s(u).$$

Differentiating the counting equations at zero twist gives

$$G y' = \frac{1}{\pi}, \quad (G y'')_j = -L s'(y_j) (y'_j)^2 + \sum_k K'(y_j - y_k) (y'_j - y'_k)^2.$$

All derivatives here are taken on the regular real-root branch at fixed finite L . Lemma 3.2 and the fixed- Q quadratures imply, at the nodes,

$$L y'_j \longrightarrow W(y_j), \quad W = \frac{H_Q}{\pi r_Q}.$$

Set $\varepsilon_Q = (I + K_Q)^{-1} e'$. The symmetry of G gives $LG^{-1} e' \rightarrow \varepsilon_Q / r_Q$. Substituting the equation for y'' into the second energy derivative and expanding its square therefore gives

$$\begin{aligned} \lim_{L \rightarrow \infty} L e_L''(0) &= \int_{-Q}^Q \left\{ (r_Q \varepsilon'_Q - \varepsilon_Q r'_Q) W^2 + 2 \varepsilon_Q W (r_Q W)' \right\} du \\ &= 2(\varepsilon_Q r_Q W^2)(Q). \end{aligned} \quad (5.4)$$

Here is the cancellation in the first equality. The direct squared displacement term is $\int r_Q e'' W^2$. The term $-s' + K'_Q r_Q$ in the implicit derivative is $-r'_Q$, so it contributes $-\int \varepsilon_Q r'_Q W^2$. Transposing the other K'_Q term replaces e'' by ε'_Q . The remaining cross term uses $(I + K_Q)(r_Q W) = 1/\pi$. The integrand is now the derivative of $\varepsilon_Q r_Q W^2$. The density and W are even and ε_Q is odd, proving the boundary expression.

We next evaluate its boundary factors, rather than differentiating a bulk asymptotic. Write

$$r(u) = \frac{1}{2\lambda \cosh(u/\eta)}.$$

If g solves the full-line equation with the relevant source, its restriction is corrected to the interval solution by

$$g \longmapsto g + (I + K_Q)^{-1} K_{\text{out}} g,$$

where $K_{\text{out}} g$ integrates over $\mathbb{R} \setminus [-Q, Q]$. Apply this map to r and to $-2\pi\eta r'$. On the outer tail near Q their ratio tends uniformly to 2π . The contribution of the opposite tail is negligible relative to $r(Q)$: insert a sufficiently small exponential weight in the convergent Neumann series for K_Q , and use the separation of the two ends. The same comparison gives $r_Q(Q) \asymp r(Q)$, hence

$$\frac{\varepsilon_Q(Q)}{r_Q(Q)} \longrightarrow 2\pi.$$

After translating the endpoint Q to zero, the Neumann series for H_Q converges term by term to its half-line series and is dominated by the geometric series with ratio $\|K\|_1 < 1$. Thus $H_Q(Q)$ has a limit. Symmetry of the resolvent, or differentiation of its series, gives the exact identity

$$\frac{d}{dQ} \int_{-Q}^Q H_Q(u) du = 2H_Q(Q)^2. \quad (5.5)$$

The bulk series also gives

$$\frac{1}{2Q} \int_{-Q}^Q H_Q(u) du \longrightarrow \frac{1}{1 + \widehat{K}(0)}.$$

Since the right side of (5.5) has a limit, averaging that identity identifies

$$H_Q(Q)^2 \longrightarrow \frac{1}{1 + \widehat{K}(0)} = \frac{\pi}{2(\pi - \lambda)}.$$

Finally (5.4) equals $2\varepsilon_Q(Q)H_Q(Q)^2/[\pi^2 r_Q(Q)]$. The two boundary limits yield (5.3). $\square$

5.2 An infrared comparison with the plane

The inverse in (5.2) is not a local observable. We therefore regularize it before passing to the plane. For $u > 0$ put

$$g_u(E) = \frac{1 - e^{-uE^2}}{E}, \quad g_u(0) = 0.$$

On nonnegative energies $0 \leq g_u(E) \leq E^{-1}$. Extend g_u as an odd function to the real line when using its Fourier representation.

Lemma 5.3 (Infrared comparison). *With κ as in Section 4,*

$$\kappa \geq \frac{1}{2\pi(\pi - \lambda)}. \quad (5.6)$$

Proof. Replace the inverse-energy quadratic form in (5.2) by the smaller form with $g_u(E_{\text{exc},L})$, keeping u fixed. Translation invariance rewrites its factor $2/L$ as the sum over link separations of twice a symmetrized single-source pairing.

An explicit real-time representation is

$$g_u(E) = \int_0^\infty \text{erfc}\left(\frac{t}{2\sqrt{u}}\right) \sin(tE) dt, \quad \text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-s^2} ds.$$

Integration by parts gives this identity. Its sine kernel is bounded at zero, integrable, and has a Gaussian tail at large real times. Oddness expresses the symmetrized pairing through expectations of real-time commutators of the local currents. The endpoint-Hamiltonian identification in Lemma 4.6 makes this the same real-time evolution as that generated by the finite-range link Hamiltonian. For link separation r , split the time integral at a fixed small multiple of r . In the shorter-time part the finite-range Lieb–Robinson estimate gives exponential decay in r ; in the longer-time part the filter tail is summable without using locality [23]. The estimates are uniform in the circle distance. For bounded times the local evolutions are norm approximated on a bounded neighborhood, so local convergence of the ground states gives convergence of each such commutator.

Let β_b be the spectral measure of the plane vector $b_0\Omega$ for the commuting angular rows; the endpoint energy E and shift phase p are functions on this spectrum. The preceding summability shows that

$$2 \text{sym}((p)_*(g_u(E)\beta_b)) = \mathcal{S}_u(p) \frac{dp}{2\pi}$$

has a continuous density. The finite regularized term converges to $\mathcal{S}_u(0)$, first as $L \rightarrow \infty$ and then as $Q \rightarrow \infty$. Throughout this passage u is fixed. Consequently (5.2) and Lemma 5.2 imply

$$\frac{2}{\pi - \lambda} \leq D_0 - \mathcal{S}_u(0). \quad (5.7)$$

The current and local spin preserve the magnetization sector, so the nonnegativity used in this comparison is sector nonnegativity. The real-time commutator estimate itself needs no ground-state hypothesis.

At nonzero momentum, the continuity equation

$$i[E, Z_j] = b_j - b_{j-1}$$

identifies the current spectral density. On the vacuum, $\|Z_0\Omega\|^2(df) = |1 - f(i)|^2\mu(df)$. The continuity equation implies, away from $p = 0$, that the energy-weighted current measure satisfies $g_u(E)\beta_b = E(1 - e^{-uE^2})\mu$. Indeed the spin measure carries $|1 - f(i)|^2$, which cancels the squared shift difference in the continuity equation. Subtract this identity from the sum rule in Lemma 4.7. On the punctured phase circle this gives

$$(D_0 - \mathcal{S}_u(p)) \frac{dp}{2\pi} = 2 \text{sym}((p)_*(Ee^{-uE^2}\mu)). \quad (5.8)$$

The equality extends to the limit $p \rightarrow 0$ because the left side is continuous. Indeed, the summable commutator bounds show that the fixed- u susceptibility measure has an absolutely convergent spatial Fourier series, hence no atom at phase zero. The regulator annihilates zero energy, and the $f = 1$ current component was removed in the construction of μ . Thus the off-zero division has omitted no additional contribution to this continuous density.

On a dilation orbit write $p(s) = \arg f(ie^s)$. Its absolute derivative is E , and its winding number is n_f when the Blaschke degree is $2n_f$. For small nonzero p of either sign, changing variables from s to p in (5.8) leaves n_f crossings, weighted by e^{-uE^2} , and the density convention contributes 4π . As $p \rightarrow 0$, precisely one crossing escapes to the corresponding orbit end, where $E \rightarrow 0$; the other crossings have positive limiting energy. The sum is bounded by n_f , and $\int n_f d\nu < \infty$ by the sum rule. Dominated convergence, first at $p \rightarrow 0$ and then at $u \rightarrow \infty$, therefore gives

$$\lim_{u \rightarrow \infty} (D_0 - \mathcal{S}_u(0)) = 4\pi\kappa.$$

Combining this with (5.7) proves (5.6). □

5.3 The exact overlap to be estimated

We now use the other finite system: real site parameters on an array of spacing $d/2$, of length $B = Ld/2$, at half filling. Let $\Omega_{L,t}$ be the unit real-root vector with seam twist e^{itZ} , continued near zero, and put

$$I_L = \left\| (I - |\Omega_L\rangle\langle\Omega_L|) \partial_t \Omega_{L,t} \Big|_{t=0} \right\|^2. \quad (5.9)$$

These staggered vectors need no Perron interpretation.

Let A and C be the root lists at zero and at t . Use $[x] = \sinh x$, and let M_{AC} be the reduced scalar-product matrix of Section 2, with on-shell roots A and column arguments C . In M_{CA} , use the twisted on-shell roots C and their twisted Bethe ratio $e^{-2it}q_C$. At $t = 0$ both coincident matrices are

the appropriate Gaudin matrices. Lemmas 2.4 and 2.5, followed by the hyperbolic Cauchy identity, give

$$|\langle \Omega_{L,0}, \Omega_{L,t} \rangle|^2 = \prod_{j,k} \frac{[A_j - C_k + i\lambda][C_j - A_k + i\lambda]}{[A_j - A_k + i\lambda][C_j - C_k + i\lambda]} \frac{\det M_{AC} \det M_{CA}}{\det G_A \det G_C}. \quad (5.10)$$

This is the identity (2.44): its one-set factors cancel, including those converting the algebraic duals into Hilbert duals. It therefore measures the overlap of unit vectors.

For each finite chain, the analytic implicit-function theorem applies to the counting equations because G is nonsingular. The roots remain distinct, the vector remains nonzero and its squared normalized overlap is nonzero on a sufficiently small neighborhood of zero. The local logarithm with value zero at $t = 0$ is therefore defined. Differentiating a unit vector shows

$$\log |\langle \Omega_{L,0}, \Omega_{L,t} \rangle|^2 = -I_L t^2 + O_L(t^3). \quad (5.11)$$

Only its quadratic coefficient is used. There is no need for a complex neighborhood whose radius is uniform in L .

Lemma 5.4 (Staggered overlap asymptotic). *In the order $B \rightarrow \infty$, then $d \downarrow 0$,*

$$\frac{I_L}{B} \longrightarrow \frac{1}{2\lambda(\pi - \lambda)}. \quad (5.12)$$

The limit is understood in the double-limit convention (3.1).

We first use this asymptotic to bound κ . Its proof occupies Sections 5.5–5.6 below.

5.4 Packing angular windows

We connect the staggered susceptibility to the plane by detecting separated angular ranges in the spectrum of its derivative vector. Let ζ_L be the spectral measure of the orthogonal derivative vector in (5.9), for the simultaneous self-adjoint rows $P_L(e^s)$, divided by their designated untwisted vacuum eigenvalues. Its total mass is I_L . Denote its spectral coordinates by f_s . The differentiated eigen-equation expresses multiplication of that vector by $1 - f_s$ as the marked row vector, with its vacuum component removed. Consequently convergence of all finite Gram words gives

$$(1 - f_s)^2 \zeta_L \longrightarrow (1 - f(e^s))^2 \mu \quad (5.13)$$

against bounded continuous functions of finitely many coordinates in a fixed window. The convergence is uniform when the window center stays in a fixed-fraction array interior.

We need a finite-array bound for two dips before summing over windows. Write ψ_L for the orthogonal derivative vector and $P_s = P_L(e^s)$, $J_s = J_L(e^s)$. The differentiated eigen-equation is

$$(1 - P_{s'})\psi_L = iJ_{s'}\Omega_L - a_{s'}\Omega_L$$

for a scalar $a_{s'}$. Since $P_s\Omega_L = \Omega_L$,

$$(1 - P_s)(1 - P_{s'})\psi_L = -i[P_s, J_{s'}]\Omega_L.$$

The spectral theorem therefore gives the exact identity

$$\int (1 - f_s)^2 (1 - f_{s'})^2 d\zeta_L = \|[P_s, J_{s'}]\Omega_L\|^2.$$

Lemma 3.3 bounds every singly marked row, uniformly at all centers in a fixed-fraction array interior, once the array is sufficiently long and its spacing sufficiently small. In finite RTT the normalized intertwiner at separation $s - s'$ is an invertible diagonal matrix plus $O(e^{-\eta|s-s'|})$; the diagonal matrix commutes with the mark $I \otimes Z$. Expanding the conjugation error in auxiliary matrix units expresses the commutator as a fixed finite sum of two-row products. The single-row bounds consequently imply

$$\int (1 - f_s)^2 (1 - f_{s'})^2 d\zeta_L \leq C e^{-2\eta|s-s'|}. \quad (5.14)$$

The constant is independent of the array length, small spacing and both bulk centers. In particular there is no additive convergence error whose sum over a growing number of pairs could accumulate.

Lemma 5.5 (Packing angular windows). *The plane orbit mass satisfies*

$$\kappa \leq \frac{\eta}{2\lambda(\pi - \lambda)}. \quad (5.15)$$

Proof. Choose finitely many real samples in a window, and let $Q(f) = \sum_j (1 - f_{s_j})^2$. Any bounded continuous detector F supported away from the all-ones point can be written

$$F = \sum_j \frac{F}{Q} (1 - f_{s_j})^2,$$

with each quotient extended continuously by zero near that point. Thus (5.13) gives convergence for such detectors without assuming convergence of the full unweighted mass. Similarly, products of two detectors are bounded by finite sums of two-dip factors. The constants depend on the fixed window and mesh of samples, not on the array center in the permitted bulk.

The cone condition and the dip estimate for inner functions provide a fixed dip width and depth around each zero scale, independent of the degree. Choose a fixed fine sampling mesh and a continuous detector $0 \leq F_T \leq 1$ in a window of length T , equal to one when a prescribed interior dip is detected and zero near all ones. Each dilation orbit contributes an interval of centers of length at least $T - C$. Hence

$$\int F_T d\mu \geq (T - C)\kappa.$$

Take the detector to be a continuous cutoff of $\max_j |1 - f_{s_j}|$ at a fixed positive dip threshold a . Then $F_T \leq C_a \sum_j (1 - f_{s_j})^2$, with C_a independent of T . Equation (5.14) bounds the product for two windows separated by a gap g by $CT^2 e^{-2\eta g}$. For N regularly spaced windows, their pair integrals sum to at most

$$\frac{CNT^2 e^{-2\eta g}}{1 - e^{-2\eta(T+g)}}.$$

Pack windows of length T and gaps g into a bulk interval of logarithmic length $(1 - \gamma)B/\eta$. Their number N_B satisfies

$$\frac{N_B}{B} \rightarrow \frac{1 - \gamma}{\eta(T + g)}.$$

For numbers $a_j \in [0, 1]$, $\sum_j a_j \leq 1 + \sum_{j < k} a_j a_k$. Integrating this inequality against ζ_L , using bulk-uniform detector convergence and the finite two-dip estimate, gives

$$\frac{1 - \gamma}{\eta(T + g)} (T - C)\kappa \leq \lim_{d \downarrow 0} \limsup_{B \rightarrow \infty} \frac{I_L}{B} + \frac{CT^2 e^{-2\eta g}}{T + g}.$$

For fixed T, g, γ , all detectors use only a fixed finite number of samples, so this application does not ask for convergence of a growing word. The uniform estimates allow the inequalities to be

summed over their translated windows. The single-window convergence error, after division by B , is $(N_B/B)\varepsilon_{d,B,T}$, where $\lim_{d \downarrow 0} \limsup_{B \rightarrow \infty} \varepsilon_{d,B,T} = 0$. Shortening the bulk interval at its ends by $O(T+g)$ keeps every sample in the permitted interior and does not change the limiting count.

Use Lemma 5.4, choose $g = A \log T$ with A sufficiently large, let $T \rightarrow \infty$, and then let $\gamma \downarrow 0$. This proves (5.15). $\square$

Proof of Proposition 5.1. Since $\eta = \lambda/\pi$, the upper bound (5.15) equals the lower bound (5.6). Thus (5.1) holds. The covariance conversion follows directly from (1.1). $\square$

5.5 Counting-coordinate accuracy and the finite trace limit

It remains to prove the overlap asymptotic used above. This subsection and Section 5.6 together prove Lemma 5.4. We first compare the finite quadratic logarithmic traces, divided by the physical length B , with their bulk convolution traces; the next subsection evaluates the resulting symbols. All Taylor coefficients are taken at finite L before $B \rightarrow \infty$ at fixed d , and then $d \downarrow 0$, with $B = Ld/2$.

We first justify the precision needed to differentiate (5.10). The leading root density alone would not suffice. In a dense interior interval, Lemmas 3.1 and 3.2 give

$$\xi'(u) = d^{-1} + O(1), \quad |\xi^{(m)}(u)| \leq C_m d^{-1} \quad (m \geq 1),$$

with fixed derivative order. In the coordinate $z = \xi(u)$, roots have unit midpoint spacing and derivatives of the inverse coordinate satisfy

$$\left| \frac{d^m u}{dz^m} \right| \leq C_m d^m.$$

For a smooth exponentially localized test F , localized away from the ends, this gives

$$\int \left| \frac{d^m}{dz^m} F(u(z)) \right| dz \leq C_m d^{m-1}. \quad (5.16)$$

Cutoffs can be put arbitrarily many decay lengths from the test center; their errors tend to zero as $B \rightarrow \infty$ at fixed d .

Midpoint summation minus integration can be integrated by parts against bounded periodic primitives of the unit-spaced counting discrepancy. Using (5.16) to arbitrarily high fixed order gives an error $O_N(d^N)$ for every fixed N , besides the vanishing end errors. Apply this to the dressed density equation and all its fixed derivatives. The uniform site array has the same quadrature accuracy, so convolution inversion improves the density estimate to

$$\xi'(u) = d^{-1} + O_N(d^N) \quad (5.17)$$

with the same accuracy for its fixed derivatives in the interior.

Differentiating the Gaudin equation for the twist gives the constant interior displacement

$$y_j(t) - y_j(0) = dz_* + O_N(d^N), \quad z_* = \frac{t}{2(\pi - \lambda)},$$

to each fixed Taylor order at zero. For the first derivative this is

$$\frac{d}{\pi(1 + \widehat{K}(0))} = \frac{d}{2(\pi - \lambda)}.$$

For higher derivatives, the differentiated site-array source is arbitrarily small by midpoint quadrature, and difference terms vanish on a constant displacement. Induction with the Gaudin inverse proves the assertion. The statement concerns spacings and root differences, not an arbitrarily chosen absolute mesh origin. Errors over several spacings can grow polynomially with index separation; the kernels against which they are summed decrease exponentially.

We next establish the finite matrix estimates needed for the trace comparison, including the sparse ends. Write

$$x_j = y_j(0), \quad \rho(u) = \xi'_0(u), \quad \rho_j = \rho(x_j), \quad \ell_j = \rho_j^{-1}, \quad D_0 = \text{diag}(\ell_j), \quad \mathcal{A} = G_0 D_0 = I + K\ell.$$

For each matrix $M(t)$ among $M_{AC}(t), M_{CA}(t), G_C(t), G_A$, use $N(t) = M(t)D_0$, keeping D_0 fixed at zero twist. Then $N(0) = \mathcal{A}$, and $\log \det N - \log \det M$ is constant. Equivalently, $N(0)^{-1}N^{(r)}(0) = D_0^{-1}G_0^{-1}M^{(r)}(0)D_0$, so the finite trace identity is

$$\left. \frac{d^2}{dt^2} \log \det M(t) \right|_{t=0} = \text{Tr}(\mathcal{A}^{-1}N''(0) - \mathcal{A}^{-1}N'(0)\mathcal{A}^{-1}N'(0)). \quad (5.18)$$

In particular, no derivative of ℓ_j occurs, and an edge weight has not been replaced by d .

Here and below constants are uniform for sufficiently small d and $B \geq B_0(d)$, as in Section 3. Let $p(u) = \sum_{\alpha} r(u - b_{\alpha})$. The sawtooth estimate (3.5) gives $\rho = p + O(1)$, with bounded integrated error. The $d/2$ -spaced array and exponential decay of r give $p(u) \leq C/d$ on the line. Between the outer bare sites, at least c/d sites lie within distance one of u , once $B \geq 2$; hence $\rho(u) \geq c/d$ there for small d . At an exterior root, (3.7) instead gives $\rho_j \geq c$. Consecutive counting levels differ by one, so

$$\begin{aligned} 0 < \rho(u) \leq C/d, \quad cd \leq \ell_j \leq C, \quad x_{j+1} - x_j \geq cd, \\ \sup_u \sum_j \ell_j e^{-\gamma|u-x_j|} \leq C_{\gamma} \quad (\gamma > 0). \end{aligned} \quad (5.19)$$

The last estimate is the local quadrature-mass bound of Lemma 3.1, summed over unit intervals. Since $K < 0$, ρ is a sum of positive kernels s and $k = -K$; their fixed derivatives are bounded by constant multiples of themselves. Thus $|\rho^{(m)}(u)| \leq C_m \rho(u)$.

Put $v_j = y'_j(0)$, $w_j = y''_j(0)$ on the twisted branch, and $\sigma(u) = \sum_{\alpha} s(u - b_{\alpha})$. Twice differentiating the finite counting equations gives

$$G_0 v = \pi^{-1} \mathbf{1}, \quad G_0 w = -H, \quad H_i = \sigma'(x_i) v_i^2 - \sum_j K'(x_i - x_j) (v_i - v_j)^2.$$

Since $G_0^{-1} = D_0 A^{-1}$, Lemma 3.2 first gives $|v_j| \leq C \ell_j$. Using $\sigma(x_i) + \sum_j k(x_i - x_j) = \rho_i$, we then obtain

$$|H_i| \leq C \left(\ell_i^2 \rho_i + \sum_j k(x_i - x_j) \ell_j^2 \right) \leq C.$$

Here $\rho_i \ell_i = 1$, $\ell_i \leq C$, and the last bound in (5.19) were used. Consequently

$$|v_j| + |w_j| \leq C \ell_j. \quad (5.20)$$

For the moving counting density $\rho_t = \xi'_{C(t)}$, derivatives at fixed u are

$$\dot{\rho}(u) = \sum_j K'(u - x_j) v_j, \quad \ddot{\rho}(u) = \sum_j \{K'(u - x_j) w_j - K''(u - x_j) v_j^2\}.$$

These functions and their required fixed spatial derivatives are $O(1)$, by (5.19) and (5.20). This includes all terminal indices.

We remove the coincident-column singularity before differentiation. For either overlap orientation, let $Y(t)$ be its on-shell list, $U(t)$ its column list, and $Q(t, u) = e^{-2it_Y} q_{Y(t)}(u)$, where $t_Y = 0$ for M_{AC} and $t_Y = t$ for M_{CA} . Thus $Q(t, Y_j(t)) = -1$. Set

$$h_j = U_j - Y_j, \quad L_j(t) = \int_0^1 \rho_{Y(t)}(Y_j + \theta h_j) d\theta, \quad \mathcal{E}(z) = \frac{e^z - 1}{z}, \quad \mathcal{E}(0) = 1.$$

The identity $q'/q = 2\pi i \rho_Y$ gives $Q(t, U_j) = -e^{2\pi i h_j L_j}$. With

$$H_\lambda(u) = \frac{\coth u - \coth(u + i\lambda)}{-2\pi i}, \quad \beta_j(t) = 1 - e^{2\pi i h_j L_j},$$

the finite scalar-product matrix is exactly $M_{ij} = K(u_{ij}) + \beta_j H_\lambda(u_{ij})$, where $u_{ij} = U_j - Y_i$. Its constant $\coth$ tails have canceled. On the diagonal this becomes the analytic identity

$$M_{jj} = K(h_j) + L_j \mathcal{E}(2\pi i h_j L_j) [h_j \coth h_j - h_j \coth(h_j + i\lambda)]. \quad (5.21)$$

At zero twist, $h_j = 0$, $L_j = \rho_j$, and $|h'_j| + |h''_j| \leq C\ell_j$. Differentiating the integral defining L_j , using the preceding density bounds, gives $|L'_j| + |L''_j| \leq C$: every spatial density derivative is multiplied by a velocity or acceleration $O(\ell_j)$, and the fixed-position time derivatives are $O(1)$. For $z_j = 2\pi i h_j L_j$, this implies

$$z'_j(0) = 2\pi i \rho_j h'_j(0) = O(1), \quad z''_j(0) = 2\pi i (\rho_j h''_j + 2h'_j L'_j)|_{t=0} = O(1).$$

The bracket in (5.21) is analytic with value one at zero; it and $\mathcal{E}(z_j)$ have bounded first two Taylor derivatives. The product rule and $\ell_j \rho_j = 1$ therefore give $\ell_j (|M'_{jj}(0)| + |M''_{jj}(0)|) \leq C$. No reciprocal displacement or reciprocal root velocity is introduced.

For $i \neq j$, ordinary differentiation at $u = x_j - x_i$ gives

$$\begin{aligned} M'_{ij} &= K'(u) u'_{ij} + \beta'_j H_\lambda(u), \\ M''_{ij} &= K''(u) (u'_{ij})^2 + K'(u) u''_{ij} + \beta''_j H_\lambda(u) + 2\beta'_j H'_\lambda(u) u'_{ij}. \end{aligned}$$

Here $\beta_j(0) = 0$, $|\beta'_j(0)| + |\beta''_j(0)| \leq C$, and u'_{ij}, u''_{ij} are bounded by (5.20). For some fixed $\gamma > 0$,

$$|K^{(r)}(u)| \leq C_r e^{-\gamma|u|}, \quad |H_\lambda(u)| \leq C(1 + |u|^{-1}) e^{-\gamma|u|}, \quad |H'_\lambda(u)| \leq C(1 + |u|^{-2}) e^{-\gamma|u|}.$$

Choose small $\alpha > 0$ and write $\|T\|_{\alpha, x} = \sup_i \sum_j e^{\alpha|x_i - x_j|} |T_{ij}|$. The minimum gap bounds the two pole factors by C/d and C/d^2 . Multiplication by the column weight ℓ_j , local quadrature mass, and the diagonal estimate prove

$$\|M'(0)D_0\|_{\alpha, x} \leq C/d, \quad \|M''(0)D_0\|_{\alpha, x} \leq C/d^2 \quad (M = M_{AC}, M_{CA}). \quad (5.22)$$

For the Gaudin denominator, put $r_i(t) = \rho_t(y_i(t))$. The formulas

$$r'_i = \dot{\rho}_i + \rho'_i v_i, \quad r''_i = \ddot{\rho}_i + 2\dot{\rho}'_i v_i + \rho''_i v_i^2 + \rho'_i w_i$$

give bounded first two derivatives. Differentiating $G_{ij}(t) = r_i(t)\delta_{ij} + K(y_i(t) - y_j(t))$ and summing with the same column weights thus gives $\|G'_C(0)D_0\|_{\alpha, x} + \|G''_C(0)D_0\|_{\alpha, x} \leq C$. Lemma 3.2, with both signs of its exponential weight, gives $\|\mathcal{A}^{-1}\|_{\alpha, x} \leq C$. Weighted absolute row norms are

submultiplicative by the triangle inequality. Hence the matrix inside the trace in (5.18), and each of its diagonal entries, are bounded by C/d^2 , uniformly in B .

The discarded count includes all exterior roots. Indeed, with $T(u) = \int_u^\infty \rho(v) dv$, the half-integer tail levels and integrated sawtooth estimate give

$$\#\{j : x_j \geq x\} \leq T(x) + 1 \leq \sum_\alpha \bar{r}(x - b_\alpha) + C, \quad \bar{r}(v) = \int_v^\infty r(u) du.$$

For $x = B/2 - R$, at most $C(R/d + 1)$ bare sites lie to its right, each contributing at most $1/2$. The other sites contribute at most $C \sum_{m \geq 0} e^{-md/(2\eta)} \leq C/d$. Reflection therefore proves

$$\#\{j : |x_j| \geq B/2 - R\} \leq C(R + 1)/d. \quad (5.23)$$

Deleting these diagonal entries costs at most $C(R + 1)/(d^3 B)$ after division by B . This tends to zero at fixed d if $R = R(B) \rightarrow \infty$ and $R(B)/B \rightarrow 0$; a fixed number of nested strips has the same property.

For the inverse comparison we use one common weighted row algebra, including at the sparse ends:

$$\|T\|_{a,d} = \sup_i \sum_j e^{ad|i-j|} |T_{ij}|, \quad T_n = I + (dK(d(i-j)))_{1 \leq i,j \leq n}. \quad (5.24)$$

Since $|x_i - x_j| \geq cd|i-j|$, the actual-matrix bounds above hold in this norm for sufficiently small fixed $a > 0$. Choose a also so that $\int e^{a|u|} |K(u)| du < 1$. For small d , $\|T_n - I\|_{a,d} \leq q < 1$; the finite Neumann series gives $\|T_n^{-1}\|_{a,d} \leq (1 - q)^{-1}$, uniformly in n . Thus the exact identity

$$\mathcal{A}^{-1} - T_n^{-1} = \mathcal{A}^{-1}(T_n - \mathcal{A})T_n^{-1}$$

holds in the same weighted algebra for both inverses. Two nested physical cutoffs separated by R inside the bare array have at least $cR/d - O(1)$ intervening root indices, since $\rho \geq c/d$ there. Any index path from a retained inner diagonal to the outer strip therefore gains a factor $Ce^{-c'R}$.

In the interior, (5.17) and the twist-jet estimates give $\ell_j = d + O_N(d^N)$ and arbitrarily accurate comparisons of root differences with $d(j - i)$. Apply these to the normalized Taylor matrices, using (5.21) on the diagonal. Fixed derivatives at distinct-root poles cost only powers of $1/d$; polynomial growth with index separation is absorbed by exponential decay. Choosing a higher initial accuracy order therefore makes the interior row errors of the Taylor matrices and of $\mathcal{A} - T_n$ $O_N(d^N)$ for every fixed N , besides errors vanishing at fixed d as the strips recede. Their global row bounds have only polynomial costs in $1/d$. The inverse identity and telescoping the fixed products in (5.18), with nested strips, give a retained diagonal error at most

$$P(d^{-1})[O_N(d^N) + o_{B \rightarrow \infty, d}(1) + C_d e^{-c'R(B)}]$$

for a fixed polynomial P . No intermediate row sum costs a factor growing with B . Since exactly $n = B/d$ indices occur, the trace per physical length costs only one additional factor $1/d$. Send $B \rightarrow \infty$ first, and choose $N > \deg P + 1$ before letting $d \downarrow 0$.

Finally, the weighted Neumann series compares T_n^{-1} with the full-lattice convolution inverse: every omitted path from an interior row crosses the matrix boundary, and so has the same exponentially small bound. The same argument applies to the finite products. The corresponding convolution trace integrand has diagonal coefficient $d(2\pi)^{-1} \int_{-\pi/d}^{\pi/d} \hat{F}(p) dp$. Multiplication by $n/B = 1/d$ gives precisely the Fourier measure $dp/(2\pi)$ per unit physical length. No infinite-volume trace-class assertion is involved.

5.6 Evaluation of the symbol

Proof of Lemma 5.4. In the product part of (5.10), the quadratic coefficient is a sum of $(\log[x+i\lambda])''$ times the two first displacements. Its integral on the real line is 2, from the two limits of $\coth(x+i\lambda)$. The constant accelerations cancel between cross and within-set terms. With root density $1/d$ and displacement dz_* , the coefficient per unit length is $2z_*^2$, with a vanishing two-limit error.

For the determinants, the bulk sequence of dM_{AC} , at row-column difference $y = d(k + z_*)$, is d times

$$\frac{(1 - e^{2\pi iz_*}) \coth y - \coth(y - i\lambda) + e^{2\pi iz_*} \coth(y + i\lambda)}{-2\pi i}. \quad (5.25)$$

For M_{CA} replace z_* by $-z_*$; for the two Gaudin denominators use $z_* = 0$. The coincident pole is continued by its removable limit. The preceding trace comparison reduces the computation to the quadratic coefficient of these bulk symbols.

Use Fourier exponent $-ipd(k + z_*)$. The simple pole contributes the shifted-sampling identity

$$\sum_{k \in \mathbb{Z}} \frac{e^{-ipd(k+z_*)}}{k + z_*} = \pi(\cot(\pi z_*) - i \operatorname{sgn} p), \quad 0 < |p| < \pi/d,$$

with symmetric summation. Subtract $1/y$ from the singular cotangent before treating the remaining part. Its derivative is integrable and analytic in every strip $|\operatorname{Im} y| < a < \min(\lambda, \pi - \lambda)$. Contour shift therefore makes every nonzero alias exponentially small in $1/d$, including its fixed Taylor coefficients. This remains uniform at the band ends since a nonzero alias has absolute frequency at least π/d ; the value at $p = 0$ is irrelevant to the integral.

Write $a_0 = e^{-\lambda p}$ and $b_0 = e^{-(\pi-\lambda)p}$. The continuous transform together with the sampled pole gives

$$\frac{(1 - e^{2\pi iz_*})(1 + a_0 b_0)/2 - b_0 + e^{2\pi iz_*} a_0}{1 - a_0 b_0} + \frac{1 + e^{2\pi iz_*}}{2} = \frac{(1 - b_0)(1 + e^{2\pi iz_*} a_0)}{1 - a_0 b_0}.$$

Dividing by the value at zero twist, the relative symbol is

$$\frac{1 + e^{2\pi iz_*} a_0}{1 + a_0}.$$

The offset Fourier phases cancel between the two orientations. Their combined quadratic logarithmic coefficient is

$$-4\pi^2 z_*^2 \frac{a_0}{(1 + a_0)^2}.$$

Since

$$\int_{\mathbb{R}} \frac{e^{-\lambda p}}{(1 + e^{-\lambda p})^2} \frac{dp}{2\pi} = \frac{1}{2\pi\lambda},$$

the determinant contribution per length is $-2\pi z_*^2/\lambda$. For each finite L , let $[t^2]$ denote the Taylor coefficient at $t = 0$ of the local logarithm fixed above. Adding the product contribution proves the explicit coefficient limit

$$\lim_{d \downarrow 0} \limsup_{\substack{B \rightarrow \infty \\ B = Ld/2}} \left| \frac{1}{B} [t^2] \log |\langle \Omega_{L,0}, \Omega_{L,t} \rangle|^2 + \frac{1}{2\lambda(\pi - \lambda)} \right| = 0.$$

The Taylor coefficient is taken before either array limit; no common analytic neighborhood or uniform finite- t expansion is asserted. By (5.11) that coefficient is $-I_L$, which proves the lemma. $\square$

6 Chiral currents and Gaussian correlations

The goal is to identify the limiting correlations of rescaled centered height differences. We first express these finite lattice observables as marked-row products. The angular spectrum then separates each product into a holomorphic and an antiholomorphic part; a second separation of scales identifies the collision coefficient that determines their Gaussian correlations.

We work at any fixed $0 < c < 2$, equivalently $0 < \lambda < \pi$, and use the diagonal-word Hilbert space $\mathcal{D}$ of Section 4. Its vacuum is Ω . The commuting rows $P(w)$ have spectral functions f , the marked row is $J(w)$, and

$$H_0 = -\log |P(1)|, \quad S = P(i), \quad N(w) = \frac{J(w)P(w) + P(w)J(w)}{2}.$$

Throughout this section $z = x + iy$ denotes a physical point; w remains the angular row parameter. At a mesh face define the centered height differences

$$\begin{aligned} D_{x,\delta} h_\delta(z) &= \frac{1}{2} (h_\delta(z + \delta) - h_\delta(z - \delta)), \\ D_{y,\delta} h_\delta(z) &= \frac{1}{2} (h_\delta(z + i\delta) - h_\delta(z - i\delta)), \\ D_\delta^+ &= \frac{1}{2} (D_{x,\delta} - iD_{y,\delta}), \quad D_\delta^- = \frac{1}{2} (D_{x,\delta} + iD_{y,\delta}). \end{aligned}$$

The mesh inverse δ^{-1} is not included in these definitions. One may evaluate at any point using the stipulated face convention. All arguments below allow the chosen lattice faces to move with the mesh.

Take points whose real coordinates are pairwise distinct. In eastward quantization, a centered horizontal or vertical difference is the mean of two consecutive height steps. The path convention and $N = (JP + PJ)/2$ identify its insertion as $-N(1)$ or $-N(i)$, respectively. To read a product, join these two-step segments in increasing time order, using positive time steps and signed transverse steps. Between their endpoints the propagators are $P(1)^{m_\delta} S^{l_\delta}$. For successive points $z_i = x_i + iy_i$ and $z_j = x_j + iy_j$ in that order, $\delta m_\delta \rightarrow x_j - x_i > 0$ and $\delta l_\delta \rightarrow y_j - y_i$. The endpoint offsets change m_δ, l_δ by at most a fixed number of sites and do not change these limits.

Thus the concrete task is to determine the scaling limit of marked-row products with positive time gaps. We construct the two limiting current forms first, then prove convergence of these products and identify their correlations.

On $\mathcal{D}$, Lemma 4.4 gives finite Blaschke products with $f(0) = f(\infty) = 1$ and real symmetry, including the constant product $f = 1$. Their zeros ζ lie in the fixed cone $\operatorname{Re} \zeta \geq c_0 |\zeta|$. The current vacuum identity and its measure are

$$J(w)\Omega = (1 - f(w))\mathbf{h}, \quad \mu(df) = \|\mathbf{h}(f)\|^2 \sigma(df), \quad \mu = ds\nu, \quad \nu(1) = \kappa < \infty. \quad (6.1)$$

Here σ is a scalar spectral measure and s is logarithmic dilation along a nonconstant spectral orbit. The symbol $\mathbf{h}$ is a spectral field; it need not be a vector of finite norm. Every use of it below includes an integrable multiplier.

6.1 The two chiral forms

Define nonnegative spectral multiplication operators on $\mathcal{D}$ by

$$a_+(f) = -f'(0) = \sum_{\zeta} 2 \operatorname{Re}(\zeta^{-1}), \quad a_-(f) = -\frac{d}{dv} f(v^{-1}) \Big|_{v=0} = \sum_{\zeta} 2 \operatorname{Re} \zeta. \quad (6.2)$$

Both values are zero on $f = 1$, and both are strictly positive on every nonconstant fiber. Write $E_{\pm}(R) = \mathbf{1}_{a_{\pm} \leq R}$ and $\mathcal{D}_{\pm}^{\text{fin}} = \bigcup_{R>0} E_{\pm}(R)\mathcal{D}$. These are dense linear subspaces. A current form on this domain means a sesquilinear form whose restriction to each bounded spectral band is a bounded operator; no unbounded-operator closure is implicit in this usage.

Lemma 6.1. *There are consistent skew-adjoint forms $j_{\pm}$ on $\mathcal{D}_{\pm}^{\text{fin}}$ such that, for every $R > 0$,*

$$\|E_{\pm}(R)j_{\pm}E_{\pm}(R)\| \leq CR, \quad E_{\pm}(R)j_{\pm}\Omega = E_{\pm}(R)a_{\pm}\mathbf{h}. \quad (6.3)$$

Their vacuum means vanish. For $s > 0$, the damped forms extend to bounded operators and satisfy

$$\|e^{-sa_{\pm}}j_{\pm}e^{-sa_{\pm}}\| \leq C/s. \quad (6.4)$$

For every nonnegative measurable function g on $(0, \infty)$,

$$\int g(a_{\pm}(f)) d\mu(f) = \kappa \int_0^{\infty} g(t) \frac{dt}{t}. \quad (6.5)$$

Proof. We give the construction at zero. The construction at infinity uses the coordinate $v = 1/w$ and the same estimates. The reciprocal-radius budget in (6.2), together with the cone condition, gives, for a sufficiently small constant $c_* > 0$, uniformly on the band $E_+(R)$

$$P(w) = I + O(R|w|), \quad \|P(w)^{-1}\| \leq C, \quad \operatorname{Re} w \geq 0, \quad |w| < c_*/R. \quad (6.6)$$

For example, normalize each Blaschke factor to one at zero. Its difference from one is at most $C|w| \operatorname{Re}(\zeta^{-1})$ in this disc, and the sum of these bounds is at most $CR|w|$. Taking c_* small bounds the product and its inverse. All inverses in this proof are restricted to the stated band.

On $E_+(R)\mathcal{D}$ write $P_R(w)$ for the restriction of $P(w)$. Compress J by $E_+(R)$ and continue it to the other half-disc by the reversal identity

$$J_R(-w) = -P_R(w)^{-1}J_R(w)P_R(w)^{-1}, \quad J_R = E_+(R)JE_+(R). \quad (6.7)$$

For this boundary gluing, first dilate the uncompressed endpoint reversal identity and then compress by $E_+(R)$, which commutes with every $P(w)$. Thus the argument of Lemma 4.5 applies without moving a cutoff through a boost. The resulting function is bounded and holomorphic on the punctured disc. Its center is removable: the negative Laurent coefficients vanish by the Cauchy bounds. At the center (6.7) says $J_R(0) = -J_R(0)$, so that constant coefficient is zero. Define $E_+(R)j_+E_+(R)$ to be the linear Taylor coefficient. The Cauchy estimate on a circle of radius proportional to R^{-1} proves the bound CR .

Compression from a larger band to a smaller band commutes with taking this coefficient. This proves consistency and defines the form on the union of bands. For positive real w , $J(w)^* = -J(w)$, so the coefficient is skew-adjoint as a form. The same argument gives j_- .

Dilation multiplies a_+ by the dilation factor and a_- by its reciprocal. On each nonconstant orbit either substitution changes Haar measure ds to dt/t . Equation (6.1) therefore gives (6.5). In particular, $\int_{a_{\pm} \leq R} a_{\pm}^2 d\mu = \kappa R^2/2$. Dividing the vacuum identity by w , or by v at infinity, and taking the band Taylor coefficient now gives the vacuum assertion in (6.3). Its inner product with Ω is zero: Ω lies in the zero-energy band, and its compressed current mean is bounded by CR for every $R > 0$.

For the damping assertion, divide the spectrum of $a_{\pm}$ into $[0, s^{-1}]$ and $(2^{q-1}s^{-1}, 2^qs^{-1}]$, $q \geq 1$. The form bound controls a block between levels q, r by $Cs^{-1} \max(2^q, 2^r)$. Damping on both sides supplies $e^{-2^{q-1}}e^{-2^{r-1}}$ when the respective index is positive. The sum of these block norms is at most C/s , which proves both existence of the bounded extension and (6.4). The same argument shows that tails of these damped block sums tend to zero in operator norm. $\square$

6.2 Separated scales and representative norms

Let $\mathcal{C}$ be the span of strict diagonal words applied to Ω . For two positive scales r_1, r_2 , represent an elementary tensor $u_1 \otimes u_2$ of such words in $\mathcal{D}$ by replacing every argument in word i by $r_i w$, multiplying the two resulting words in that order, and applying the product to Ω . Extend this prescription to finite linear combinations and denote a chosen representative of u by u_{r_1, r_2} . Different word expressions for the same tensor have difference tending to zero in norm as $|\log(r_1/r_2)| \rightarrow \infty$. Indeed, Proposition 3.7, applied after adjunction, gives

$$\langle u_{r_1, r_2}, v_{r_1, r_2} \rangle \longrightarrow \langle u, v \rangle \quad (u, v \in \mathcal{C} \otimes \mathcal{C}). \quad (6.8)$$

The asymptotic diagonal RTT exchange permits the two separated word groups to be interchanged with an error tending to zero in norm. Consequently a fixed diagonal marked row at $r_i w$ acts in the limit on factor i alone. More precisely, write $(A^D(w))_i$ for the action of $A^D(w)$ on tensor factor i , where D is diagonal and w is strict. Commuting the row through the other word group when necessary gives

$$\|A^D(r_i w)u_{r_1, r_2} - ((A^D(w))_i u)_{r_1, r_2}\| \longrightarrow 0.$$

The error is a finite sum of bounded marked-row products with auxiliary coefficients tending to zero by RTT. Thus this is a norm limit, not only a statement about vacuum expectations.

We use convergence of representatives, not a bounded map from the entire tensor completion. Precisely, a family $v_\delta \in \mathcal{D}$ converges strongly to $v \in \mathcal{D} \otimes \mathcal{D}$ if, for every $\epsilon > 0$, there is a finite word tensor u with

$$\|u - v\| < \epsilon, \quad \limsup_{\delta \downarrow 0} \|v_\delta - u_\delta\| < \epsilon.$$

The scales defining u_δ will always be specified. Equation (6.8) makes this definition independent of the chosen word expressions.

Lemma 6.2. *Suppose T_δ are uniformly bounded operators and, on every finite word tensor, $T_\delta u_\delta$ converges strongly to Tu . Then T extends to a bounded operator on the tensor completion, and $T_\delta v_\delta \rightarrow Tv$ for every strongly converging family $v_\delta \rightarrow v$. Such operator limits may be composed, without a commutation assumption.*

A family v_δ converges strongly to v if its pairings with all core representatives converge to the pairings with v and its norms converge to $\|v\|$.

Proof. If $\|T_\delta\| \leq M$, then (6.8) implies $\|Tu\| \leq M\|u\|$ on the core. Approximate v_δ by u_δ as in the definition. The norm of $T_\delta(v_\delta - u_\delta)$ is at most M times that approximation error, while $T_\delta u_\delta$ has the asserted limit. Taking the approximation error to zero proves the first assertion and then composition. For the last assertion, use

$$\|v_\delta - u_\delta\|^2 = \|v_\delta\|^2 + \|u_\delta\|^2 - 2\operatorname{Re}\langle v_\delta, u_\delta \rangle \longrightarrow \|v - u\|^2$$

and approximate v by a core vector u . □

Proposition 6.3 (Two-scale spectral limit). *Take $r_1 = \delta^{-1}$ and $r_2 = \delta$, with $\delta \downarrow 0$ through arbitrary positive real values. On core representatives, the joint spectral functions $(f(\delta^{-1} \cdot), f(\delta \cdot))$ converge in law in the compact-open topology of the right half-plane to the two tensor-factor functions (F_1, F_2) . Jointly with these functions,*

$$\begin{aligned} \delta^{-1} H_0 &\longrightarrow A := a_+ \otimes I + I \otimes a_-, \\ \delta^{-1} \arg S &\longrightarrow -a_+ \otimes I + I \otimes a_-, \quad \operatorname{sgn} P(1) \longrightarrow I \end{aligned} \quad (6.9)$$

in spectral law. The argument is the principal argument near zero. There is an unbounded set of $R > 0$ for which

$$Q_\delta(R) := \mathbf{1}_{H_0/\delta \leq R} \longrightarrow Q(R) := \mathbf{1}_{A \leq R} \quad (6.10)$$

strongly on representatives. Bounded spectral multipliers obtained jointly from these limits have the same strong interpretation whenever their discontinuity sets have zero limiting state mass.

If $m_\delta \geq 0$ and l_δ are integers with $\delta m_\delta \rightarrow x \geq 0$ and $\delta l_\delta \rightarrow y$, then

$$P(1)^{m_\delta} S^{l_\delta} \longrightarrow e^{-(x+iy)a_+} \otimes e^{-(x-iy)a_-} \quad (6.11)$$

strongly on representatives.

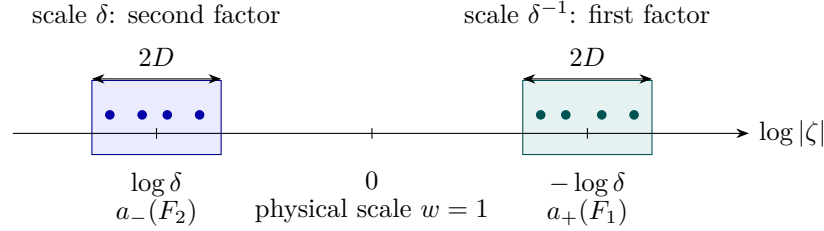


Figure 1: Schematic zero windows for the two factors: the window at $\log \delta$ belongs to factor 2, and the window at $-\log \delta$ belongs to factor 1. The proof fixes their half-width D and a bound on the number of zeros before sending δ to zero. It controls the state mass of configurations with any zero outside these windows before expanding at the intervening physical scale $w = 1$.

Proof. Spectral functions and exclusion of exterior zeros. The strong row actions preceding Lemma 6.2 give convergence of every polynomial in finitely many samples of the two spectral functions and their complex conjugates. The functions are bounded by one on the right half-plane; normal-family compactness therefore gives the joint compact-open convergence. The limiting functions are the original finite Blaschke spectral functions on the two factors, and are not identically zero. For a vector v , its state spectral measure is $\sigma_v(B) = \|\mathbf{1}_B v\|^2$ on measurable sets of spectral functions. These measures have total masses $\|u_\delta\|^2 \rightarrow \|u\|^2$; normalization to probability is harmless when $u \neq 0$.

The rescaled quantities in (6.9) require control of *every* zero, including zeros between the two scales. For a fixed core representative, the exterior RTT estimate from Lemma 4.4 gives, outside fixed word offsets,

$$\|(I - P(e^s))u_{r_1, r_2}\|^2 \leq C_u \sum_{i=1}^2 e^{-2\eta|s - \log r_i|}. \quad (6.12)$$

Every zero ζ forces a fixed-width interval about $\log |\zeta|$ on which $|1 - f(e^s)|$ is bounded below by a positive constant. This follows from the cone condition and the modulus loss of its single Blaschke factor; all other factors have modulus at most one. Integrating (6.12) outside slightly smaller windows thus yields

$$\begin{aligned} \sigma_{u_{r_1, r_2}} \left\{ \text{some zero lies outside } \bigcup_{i=1}^2 \{\zeta : |\log(|\zeta|/r_i)| \leq D\} \right\} \\ \leq C_u e^{-c_* D}. \end{aligned} \quad (6.13)$$

An additive fixed change in D absorbs the dip width and the word offsets. To justify the event bound explicitly, if any zero lies outside the enlarged windows, one entire dip interval lies in the integration region. The integral of $|1 - f(e^s)|^2$ there is then at least a fixed positive number. Markov's inequality after integration over the state spectral measure gives (6.13).

Complete zero clusters and the physical-scale expansion. Here is the order of truncations. Given a desired exceptional mass, first choose D large enough in (6.13) and so that the finite zero sets of F_1, F_2 lie in their annuli with correspondingly high probability. Choose their boundary circles to carry no limiting zero almost surely. Only countably many radii are excluded: first restrict to a bounded zero count, then apply this assertion to its finite intensity measure, and finally take the union over count bounds. On these fixed annuli, the cone keeps the zeros in a compact subset of the open right half-plane. The argument principle and compact-open convergence give joint convergence of the complete zero multisets, including multiplicities. Next choose a finite degree bound M so that larger limiting counts have arbitrarily small mass. Take the scale limit only after D and M have been fixed. Zero counts and locations now lie in a compact, bounded-count set, so every product expansion below is uniform. Finally let the allowed exceptional mass tend to zero, increasing D and M if necessary.

Each limiting product has even degree: conjugate zeros pair, real zeros contribute a factor -1 between the two ends, and both end values are one. On the preceding high-mass event the complete limiting zero counts occur in the chosen annuli. Multiplicity convergence therefore makes both finite-scale clusters even with probability tending to one. This proves the needed parity; it is not an assumption on a truncated cluster.

For clarity, write the large zeros as $\delta^{-1}\alpha_j$ and the small zeros as $\delta\beta_j$. On the retained event each normalized zero has modulus between e^{-D} and e^D , and there are at most M zeros. Normalize each factor to one at zero. The two even clusters then give, for $w = 1, i$,

$$f(w) = 1 - \delta \left(w \sum_j 2 \operatorname{Re} \alpha_j^{-1} + w^{-1} \sum_j 2 \operatorname{Re} \beta_j \right) + O_{D,M}(\delta^2). \quad (6.14)$$

The two sums converge jointly to $a_+(F_1)$ and $a_-(F_2)$. Taking logarithmic modulus at 1 and phase at i gives (6.9). The middle sign is positive on this event. Values of H_0/δ can be arbitrarily large on the discarded event; convergence in law and bounded multiplier convergence require no first-moment estimate for that quantity.

Strong multiplier convergence and propagators. We next strengthen bounded spectral convergence to convergence of vectors. Choose a countable total word core, including its rational linear combinations, and exclude the union of the atoms of A in all their state measures. That union is countable; any other R is admissible in (6.10). At such R , continuous cutoffs approximate the indicator in each state L^2 norm. More generally, let b_δ be a uniformly bounded spectral multiplier with the stated joint limit b . The algebra of polynomials in a countable dense set of interior samples and their conjugates generates the spectral functions and is dense in the state L^2 space. Given a state u , choose such a polynomial p with $\|(b - p)u\|$ small. Joint convergence gives

$$\|(b_\delta - p_\delta)u_\delta\|^2 \longrightarrow \|(b - p)u\|^2. \quad (6.15)$$

For indicators, take the continuous approximations first. Polynomial actions already converge strongly. Equation (6.15) proves strong convergence of the multiplier actions; polarization gives the corresponding mixed pairings. Lemma 6.2 extends these actions to all strongly converging families and permits their compositions. No assertion that a current preserves the exceptional-zero event is needed.

Finally, $|f(1)| \leq 1$ and $|f(i)| = 1$. The bounded multipliers $f(1)^{m_\delta} f(i)^{l_\delta}$ converge jointly by (6.9) to $e^{-xA+iy(-a_+ \otimes I + I \otimes a_-)}$, which is (6.11). The same vector argument applies. $\square$

The exceptional-event estimate is the reason that local spectral convergence can be used at the intervening physical scale $w = 1$. It also permits the energy cutoffs needed for current insertions. We next prove their convergence in norm, before removing those cutoffs by time damping.

6.3 Current insertions and damping

Lemma 6.4. *For every admissible $R > 0$ in Proposition 6.3 and every $|w| = 1$ with $\operatorname{Re} w \geq 0$,*

$$\delta^{-1} Q_\delta(R) N(w) Q_\delta(R) \longrightarrow Q(R) (w(j_+ \otimes I) + w^{-1}(I \otimes j_-)) Q(R) \quad (6.16)$$

strongly on representatives. All displayed operators are bounded at fixed R . Finite products with propagators from (6.11) converge when successive positive time gaps stay bounded below. In these vacuum products, the cutoffs on both sides of each current may be removed after the scale limit, with error tending to zero uniformly on compact sets of separated time coordinates.

Proof. Write $Q_\delta = Q_\delta(R)$ and $Q = Q(R)$. Glue $J_\delta(w) = Q_\delta J(w) Q_\delta$ to the full annulus

$$CR\delta < |w| < c_*/(R\delta)$$

using Lemma 4.5. Let $C_{k,\delta}$ be its Laurent coefficients. Fix $\rho > 0$ sufficiently small compared with R^{-1} . At the outer circle, the substitution $v = \delta w$ gives

$$\delta^{-1} C_{1,\delta} = \frac{1}{2\pi i} \int_{|v|=\rho} J_\delta(v/\delta) \frac{dv}{v^2}. \quad (6.17)$$

At the inner circle, put $w = \delta v$ to obtain

$$\delta^{-1} C_{-1,\delta} = \frac{1}{2\pi i} \int_{|v|=\rho^{-1}} J_\delta(\delta v) dv. \quad (6.18)$$

Both contours are counterclockwise. The rescaled contours are fixed, and both integrands are uniformly bounded in operator norm at fixed R, ρ .

On the strict right semicircle the marked rows converge on their respective tensor factors. The cutoffs converge strongly by (6.10). On the reflected semicircle the integrand is the product specified by reversal, for example

$$-(Q_\delta P(v/\delta)^{-1} Q_\delta) J_\delta(v/\delta) (Q_\delta P(v/\delta)^{-1} Q_\delta).$$

The inverse factors are uniformly bounded spectral multipliers on $Q_\delta \mathcal{D}$ only. Their joint limits follow from Proposition 6.3, since on $Q\mathcal{D}^{\otimes 2}$ the first energy is at most R and (6.6) applies. The second contour uses the analogous estimate at infinity. Lemma 6.2 therefore proves convergence of these noncommuting products at every strict contour point.

Here is the norm justification for passing through the contour integrals. For a representative u_δ , let $v_\delta(t)$ be the integrand applied to it, including the scalar contour weight, and let $v(t)$ be its limit. The vectors are bounded uniformly in t, δ . Core test pairings of $\int v_\delta(t) dt$ converge by a single dominated integral. Their squared norms converge because

$$\left\| \int v_\delta(t) dt \right\|^2 = \iint \langle v_\delta(t), v_\delta(s) \rangle dt ds \longrightarrow \iint \langle v(t), v(s) \rangle dt ds. \quad (6.19)$$

Pairings of strongly convergent representatives converge, so dominated convergence applies to this double integral. The two imaginary-axis points have contour measure zero. The norm criterion in Lemma 6.2 now gives strong convergence of each coefficient action. By the definitions of the chiral forms, the limits of (6.17) and (6.18) are respectively

$$Q(j_+ \otimes I)Q, \quad Q(I \otimes j_-)Q.$$

It remains to pass from these coefficients to N . Put $\Sigma = \text{sgn } P(1)$ and

$$T_\delta(w) = w\delta^{-1}C_{1,\delta} + w^{-1}\delta^{-1}C_{-1,\delta}.$$

The same-sign and opposite-sign estimates of Lemma 4.5 give, for some $\theta > 0$,

$$\delta^{-1}Q_\delta N(w)Q_\delta = \frac{1}{2}(\Sigma T_\delta(w) + T_\delta(w)\Sigma) + O_R(\delta^\theta) \quad (6.20)$$

in operator norm, uniformly on the unit semicircle. Indeed the anticommutator is $+T_\delta$ on the positive-sign block, $-T_\delta$ on the negative-sign block, and zero between opposite signs. On same-sign blocks the even Laurent part and all odd modes of order larger than one vanish after rescaling, with the stated remainder. The opposite-sign block of the rescaled N has the same vanishing bound. Proposition 6.3 gives $\Sigma \rightarrow I$ strongly on representatives. The coefficient operators are uniformly bounded and have the strong limits just proved. Composition in (6.20) proves (6.16). The low-band bounds include the endpoints $w = \pm i$.

We finish with a summable bound for removing cutoffs. For all $U > 0$ and all sufficiently small δ ,

$$\|\mathbf{1}_{H_0/\delta \leq U} \delta^{-1}N(w)\mathbf{1}_{H_0/\delta \leq U}\| \leq CU. \quad (6.21)$$

If $U\delta$ is small this is Lemma 4.5; otherwise the global bound on N gives it by $\delta^{-1} \leq CU$. Decompose H_0/δ into bands $[0, 1]$ and $(2^{q-1}, 2^q]$, $q \geq 1$. A current block between levels q, r costs at most $C \max(2^q, 2^r)$. Each internal propagator with $\delta m \geq b > 0$ supplies at level $q \geq 1$ the factor $e^{-b2^{q-1}}$, since $\|P(1)^m \mathbf{1}_{H_0/\delta > 2^{q-1}}\| \leq e^{-b2^{q-1}}$; unit shifts do not affect this norm. The outer states are the vacuum, of energy zero. For a product of k currents, a bound for the sum over its $k-1$ internal indices is a constant times

$$\sum_{q_1, \dots, q_{k-1} \geq 0} \prod_{j=1}^{k-1} (1 + 2^{q_j})^2 e^{-b'2^{q_j}}, \quad b' > 0. \quad (6.22)$$

This converges, and its tails with any $2^{q_j} > R$ tend to zero uniformly in the mesh. The limiting forms have the identical bound using A and (6.3). Fixed-cutoff convergence, followed by this tail estimate, proves the product assertion. The constants are uniform when the finitely many limiting time gaps have a fixed positive lower bound. $\square$

6.4 The limiting derivative correlations

For the centered differences introduced at the start of the section, Lemma 6.4 and (6.11) show that the insertions of $\delta^{-1}D_\delta^+$ and $\delta^{-1}D_\delta^-$ converge to $-j_+ \otimes I$ and $-I \otimes j_-$, respectively. For example, the two combinations follow from

$$N(1) \rightsquigarrow j_+ \otimes I + I \otimes j_-, \quad N(i) \rightsquigarrow ij_+ \otimes I - iI \otimes j_-.$$

Currents in different factors commute. Their propagators and the tensor vacuum factor as well. The limiting mixed correlations therefore factor into a pure $+$ correlation and a pure $-$ correlation.

Write $a = a_+$ and $j = j_+$ for the moment. For increasing real parts, $\operatorname{Re} z_{i_1} < \dots < \operatorname{Re} z_{i_k}$, define

$$F_k(z_1, \dots, z_k) = \left\langle \Omega, (-j) e^{-(z_{i_k} - z_{i_{k-1}})a} (-j) \dots e^{-(z_{i_2} - z_{i_1})a} (-j) \Omega \right\rangle. \quad (6.23)$$

Set $F_0 = 1$ and $F_1 = 0$. The products in (6.23) are defined by inserting positive real damping on both sides of each form; arbitrary extra end damping may be inserted because $a\Omega = 0$. Lemma 6.1 makes the resulting factors bounded, and the dyadic tail bound shows that the value is independent of this allocation. By reality of the height field, the pure $-$ answer is $\overline{F_k(z_1, \dots, z_k)}$. All the preceding convergence is uniform on compact subsets of the region of distinct real coordinates. Indeed, it holds along every sequence of moving mesh faces converging to such a configuration, and the cutoff and damping bounds are uniform on its compact subsets.

The two-point function is already explicit. If $\operatorname{Re} z_1 < \operatorname{Re} z_2$, skew-adjointness, the vacuum identity and (6.5) give

$$F_2(z_1, z_2) = - \int a(f)^2 e^{-(z_2 - z_1)a(f)} d\mu(f) = -\kappa \int_0^\infty t e^{-(z_2 - z_1)t} dt = -\frac{\kappa}{(z_2 - z_1)^2}.$$

The integrals converge because $\operatorname{Re}(z_2 - z_1) > 0$. For higher moments, we need this coefficient at a collision even between nonvacuum exterior states.

We now determine these holomorphic correlations from their singularities. The two axial orderings give analytic formulas away from a finite grid of possible punctures. We will bound the growth at every puncture, compute the coefficient when two physical points actually collide, and remove the remaining artificial punctures. These facts and decay at infinity will force Wick's recursion.

Lemma 6.5. *Let*

$$U_x = \{(z_1, \dots, z_k) : \operatorname{Re} z_i \neq \operatorname{Re} z_j \text{ for } i \neq j\}, \quad U_y = \{(z_1, \dots, z_k) : \operatorname{Im} z_i \neq \operatorname{Im} z_j \text{ for } i \neq j\}.$$

The function F_k has a symmetric, jointly holomorphic extension to $U_x \cup U_y$, and is the limit of the centered + derivative correlations on both sets.

Fix $z_2, \dots, z_k$ with distinct real coordinates and distinct imaginary coordinates. As a function of z_1 , the extension is holomorphic away from the grid points

$$p_{jh} = \operatorname{Re} z_j + i \operatorname{Im} z_h, \quad 2 \leq j, h \leq k. \quad (6.24)$$

Near any such point it is $O(|z_1 - p_{jh}|^{-2})$, locally uniformly as the fixed points vary while retaining these coordinate separations. For $k \geq 2$ it tends to zero as $|z_1| \rightarrow \infty$ with the other points fixed.

Proof. On a compact subset of an eastward ordering chamber, give each current a fixed positive amount of damping on both sides. The remaining semigroups have parameters with real parts bounded below. These semigroups are holomorphic in operator norm there: for instance $\|a^m e^{-sa}\| \leq m! s^{-m}$ after reducing s by a fixed factor. The resulting bounded product is jointly holomorphic in all its parameters. This proves joint holomorphy on U_x , rather than merely separate holomorphy.

By Proposition 4.3, northward quantization describes the same physical law. In its coordinates $z' = -iz$, the centered differences transform as $D_{x',\delta} = D_{y,\delta}$ and $D_{y',\delta} = -D_{x,\delta}$, hence $D_\delta^+ = -iD_\delta'^+$. The possible face-origin shift is one lattice spacing and is covered by moving-face convergence. The northward formula is thus holomorphic on U_y . On $U_x \cap U_y$ both formulas are limits of the same physical moments, so they agree and glue. Permuting the insertions does not change a physical product, which proves symmetry.

With the other points as in the statement, failure of both axial orderings occurs only at the grid points (6.24). Choose a small neighborhood of the fixed configuration in which all its real gaps and imaginary gaps remain at least $b > 0$. Near p_{jh} , the eastward representation has only one possibly small gap, $g_x = |\operatorname{Re}(z_1 - p_{jh})|$, between z_1 and z_j . The northward representation has only one possibly small gap, $g_y = |\operatorname{Im}(z_1 - p_{jh})|$, between z_1 and z_h . Use the ordering with $g = \max(g_x, g_y) \geq |z_1 - p_{jh}|/\sqrt{2}$. Only the two currents bordering that gap need shrinking damping. Give each damping $g/4$ on both sides: the inner dampings consume $g/2$, and the outer dampings come from neighboring fixed gaps or from a vacuum endpoint. All other currents receive fixed damping. Equation (6.4) gives a cost C/g for each of the two currents and a bounded cost for the rest. This proves the uniform quadratic bound. A point sharing the other coordinate is separated in the chosen ordering.

Finally, when $|z_1|$ is large, one coordinate makes z_1 extreme in the corresponding ordering, with a gap g of order $|z_1|$ to all other points. Give the isolated current damping of order g on both sides, using free damping on its vacuum side. Keep fixed damping around all remaining currents. The isolated factor costs C/g ; imaginary semigroup factors have norm one. Thus $F_k \rightarrow 0$ at infinity. $\square$

6.5 The collision operator

The remaining input for Gaussianity is the coefficient at a genuine collision. A second separation of scales shows that a short pair of chiral currents becomes a scalar, even between nonvacuum exterior states.

Lemma 6.6. *For $a = a_+$ and $j = j_+$, the bounded operators*

$$K_\epsilon = e^{-\epsilon a}(\epsilon j)e^{-\epsilon a}(\epsilon j)e^{-\epsilon a} \quad (6.25)$$

are uniformly bounded for $\epsilon > 0$ and satisfy

$$K_\epsilon \rightarrow -\kappa I \quad (\epsilon \downarrow 0). \quad (6.26)$$

Consequently, fix $z_2, \dots, z_k$ with pairwise distinct real coordinates and pairwise distinct imaginary coordinates, and let $2 \leq j \leq k$. Then

$$\lim_{\epsilon \downarrow 0} \epsilon^2 F_k(z_j + \epsilon, z_2, \dots, z_k) = -\kappa F_{k-2}(z_2, \dots, \widehat{z}_j, \dots, z_k). \quad (6.27)$$

The coefficient on the right is independent of both colliding positions.

Proof. Split the middle damping of (6.25) into two halves. Each of the resulting damped current factors has norm bounded by a constant, by (6.4) at scale $\epsilon/2$. This gives uniform boundedness.

Apply the representative construction with scales $r_1 = \epsilon$ and $r_2 = 1$. Its Gram and row limits use only separation of the two scales, so they are unchanged. The all-zero event estimate (6.13), with the same order of fixed windows, bounded degree, and then scale limit, gives

$$\epsilon a(f) \rightarrow a(F_1) \quad (6.28)$$

jointly with the two spectral functions. To see the scaling directly, a zero $\epsilon\alpha$ contributes $2 \operatorname{Re} \alpha^{-1}$ to ϵa , whereas a zero β at the unshifted scale contributes $2\epsilon \operatorname{Re} \beta^{-1}$. On the bounded-count annuli the latter sum tends uniformly to zero. The exceptional-zero event has arbitrarily small mass and is discarded before making this expansion.

Choose a non-atomic band endpoint and put $B_\epsilon = \mathbf{1}_{\epsilon a \leq R}$ and $B = \mathbf{1}_{a \otimes I \leq R}$. The spectral multiplier argument proves $B_\epsilon \rightarrow B$ strongly. The definition of j by its Taylor coefficient gives, for a sufficiently small fixed $\rho < c_*/R$,

$$\epsilon B_\epsilon j B_\epsilon = \frac{1}{2\pi i} \int_{|v|=\rho} B_\epsilon J(\epsilon v) B_\epsilon \frac{dv}{v^2} \longrightarrow B(j \otimes I)B. \quad (6.29)$$

Indeed, the unscaled band is $a \leq R/\epsilon$, whose Taylor disc has radius proportional to ϵ/R . On the right semicircle the row acts on the first factor; on the left semicircle the bounded compressed inverse and reversal apply. The single- and double-contour argument (6.19) proves convergence in representative norm. Thus (6.28) and (6.29) give the cutoff limit of K_ϵ .

These cutoffs can be removed in operator norm uniformly in ϵ . Decompose ϵa into $[0, 1]$ and the dyadic bands above one. A block of ϵj between levels q, r costs at most $C \max(2^q, 2^r)$. The three damping factors of K_ϵ give an absolutely summable majorant

$$C \sum_{q_0, q_1, q_2 \geq 0} \max(2^{q_0}, 2^{q_1}) \max(2^{q_1}, 2^{q_2}) e^{-b(2^{q_0} + 2^{q_1} + 2^{q_2})}.$$

Absorbing the lowest-band constants into C gives some fixed $b > 0$. The part with any $2^{q_i} > R$ tends to zero as $R \rightarrow \infty$. The same estimate holds for the limiting tensor product. Therefore, on separated-scale representatives,

$$K_\epsilon \longrightarrow (e^{-a} j e^{-a} j e^{-a}) \otimes I.$$

A fixed unboosted word vector u is exactly the representative of $\Omega \otimes u$: its small-scale word is the identity. Hence for fixed word vectors u, v the last limit contracts the first factor to

$$\langle \Omega, j e^{-a} j \Omega \rangle = - \int a(f)^2 e^{-a(f)} d\mu(f) = -\kappa \int_0^\infty t e^{-t} dt = -\kappa.$$

The sign uses the skew-adjointness of j ; the integrals are justified by (6.5). This proves (6.26) on a dense set of test vectors. Uniform boundedness extends it to every pair of vectors in $\mathcal{D}$.

For (6.27), take $z_1 = z_j + \epsilon$ in the eastward ordering. The two colliding currents are consecutive. After multiplication by ϵ^2 , their part of the product becomes K_ϵ , provided its two outer dampings are subtracted from the adjacent exterior gaps. Those gaps remain positive for small ϵ ; at an endpoint, insert damping freely on the vacuum. The two exterior vectors converge strongly, because all their damping gaps remain positive. They may therefore be replaced by their limits in the weak convergence of the uniformly bounded K_ϵ . The two insertion minus signs cancel. Omitting the colliding currents recombines the adjoining semigroups, giving exactly the right side of (6.27). No dependence on the collision position remains. $\square$

6.6 Removal of artificial poles and Wick recursion

Proposition 6.7. *Let $k \geq 1$, and let $z_1, \dots, z_k$ have pairwise distinct real coordinates or pairwise distinct imaginary coordinates. For $\varepsilon_i \in \{+, -\}$ and moving mesh faces $z_{i,\delta} \rightarrow z_i$,*

$$\lim_{\delta \downarrow 0} \mathbb{E} \prod_{i=1}^k \delta^{-1} D_\delta^{\varepsilon_i} h_\delta(z_{i,\delta}) = \sum_{\pi} \prod_{\{i,j\} \in \pi} C_{\varepsilon_i, \varepsilon_j}(z_i, z_j), \quad (6.30)$$

where the sum is over pairings, is zero for odd k , and

$$C_{++}(z, \xi) = -\frac{\kappa}{(z - \xi)^2}, \quad C_{--}(z, \xi) = -\frac{\kappa}{(\bar{z} - \bar{\xi})^2}, \quad C_{+-}(z, \xi) = C_{-+}(z, \xi) = 0. \quad (6.31)$$

Convergence is uniform on compact subsets of either separated-time region. By linearity the corresponding assertion holds for all products of the real centered derivatives $\delta^{-1}D_{x,\delta}$ and $\delta^{-1}D_{y,\delta}$. These are the off-diagonal derivatives of the Gaussian covariance

$$K_\kappa(z, \xi) = -2\kappa \log |z - \xi|. \quad (6.32)$$

Proof. The preceding lemmas already give the limits as products of pure chiral functions and their conjugates. It remains to determine F_k . Fix generic spectators $z_2, \dots, z_k$ as in Lemma 6.5. Its quadratic bound makes every grid puncture of F_k a pole of order at most two: the Cauchy bound for a Laurent coefficient of order $-m$, $m > 2$, is Cr^{m-2} on a circle of radius r , and tends to zero.

First consider an artificial puncture $p = p_{jh} = \operatorname{Re} z_j + i \operatorname{Im} z_h$ with $j \neq h$. Take a fixed small circle Γ about its position for the chosen spectators. In a sufficiently small complex neighborhood of the spectators, the circle encloses just this moving grid point and remains in $U_x \cup U_y$. The contour moments

$$m_\ell = \frac{1}{2\pi i} \int_\Gamma \zeta^\ell F_k(\zeta, z_2, \dots, z_k) d\zeta$$

are jointly holomorphic in the spectator variables, by the uniform bounds and joint holomorphy on the circle. If the principal part is $A/(\zeta - p)^2 + B/(\zeta - p)$, residue calculation gives

$$m_\ell = Bp^\ell + A\ell p^{\ell-1}, \quad m_0 m_2 - m_1^2 = -A^2, \quad m_0 m_3 - m_1 m_2 = -2pA^2. \quad (6.33)$$

If A is nonzero at any parameter value, the nonzero denominator persists locally and

$$p = \frac{m_0 m_3 - m_1 m_2}{2(m_0 m_2 - m_1^2)}$$

is holomorphic there. This is impossible, since

$$p = \frac{z_j + \bar{z}_j}{2} + \frac{z_h - \bar{z}_h}{2}, \quad \partial_{\bar{z}_j} p = \frac{1}{2}.$$

Thus A vanishes. If B is nonzero, $p = m_1/m_0$ gives the same contradiction. Every artificial puncture is removable. This removal is joint in the parameters: inside Γ , the Cauchy integral of $F_k(\eta, \mathbf{z})/(\eta - \zeta)$ supplies a jointly holomorphic extension and agrees with each one-variable removable extension.

At a true collision, put $\xi = (z_1 + z_j)/2$ and $d = z_1 - z_j$, retaining generic exterior points. Locally the union $U_x \cup U_y$ contains every $d \neq 0$. The function $H(\xi, d, \mathbf{z}) = d^2 F_k$ is uniformly bounded there and therefore extends jointly holomorphically to $d = 0$. One can see this directly by its Cauchy coefficients in d : boundedness removes the negative coefficients, uniformly on smaller spectator neighborhoods. Symmetry under exchanging the two colliding insertions makes H even in d . Lemma 6.6 identifies its constant term as

$$H(\xi, 0, \mathbf{z}) = -\kappa F_{k-2}(\mathbf{z}),$$

independent of ξ . Subtracting this term leaves an even holomorphic function with zero constant term, hence a function divisible by d^2 . It follows that

$$F_k + \frac{\kappa F_{k-2}(\mathbf{z})}{(z_1 - z_j)^2}$$

is holomorphic at this diagonal. In particular there is no simple residue. Independence of the collision center is needed here: a varying coefficient of d^{-2} could otherwise create a simple term when z_j is held fixed.

Subtract all these true double-pole terms as a function of z_1 . The artificial points are removable, the true points have no remaining principal parts, and the resulting function is entire. Lemma 6.5 gives decay at infinity, and the subtracted rational terms also decay there. Liouville's theorem yields

$$F_k(z_1, \dots, z_k) = -\kappa \sum_{j=2}^k \frac{F_{k-2}(z_2, \dots, \widehat{z}_j, \dots, z_k)}{(z_1 - z_j)^2}. \quad (6.34)$$

Together with $F_0 = 1$ and $F_1 = 0$, this is Wick's recursion with contraction $-\kappa/(z_i - z_j)^2$. Continuity extends it from the generic spectators to every strict axial chamber. Factorization of the two chiral sectors and reality then give (6.30).

Finally, $\partial_z \partial_\xi K_\kappa(z, \xi) = -\kappa/(z - \xi)^2$ and $\partial_z \partial_{\bar{\xi}} K_\kappa(z, \xi) = 0$ away from collisions. The conjugate identity gives the other pure contraction. This proves the covariance assertion and, by linearity, all the real derivative formulas. The uniform moving-face convergence established above gives the asserted local uniformity. Estimates across a time-ordering wall and for undifferentiated fields will be supplied in Section 7; none is needed for this proposition. $\square$

7 Moment bounds and convergence of the height field

The current limits of Section 6 concern insertions at separated times. We now pass to the height field itself. We first identify its law on smooth neutral tests, then remove endpoint smoothings to obtain the separated increment moments. The logarithmic variogram bound extends this identification to finite-energy signed measures. After these three steps, we prove tightness in negative regularity, treating ordinary restriction spaces, the outward test convention, and intrinsic dual spaces separately.

Two estimates drive these passages. An energy-commutator argument gives uniform ordinary moments for neutral spatial averages. A small neutral factor separated from all other factors has a correlation that tends to zero as a positive power of its diameter. The latter estimate makes the successive removal of endpoint smoothings summable.

Throughout the angular argument, fix $0 < c < 2$, and retain the plane state, κ , the vacuum Ω , the shift $S = P(i)$, and the nonnegative endpoint Hamiltonian E from Section 4. The Gaussian field appearing before its normalization is identified has covariance

$$K_\kappa(x, y) = -2\kappa \log |x - y|. \quad (7.1)$$

We denote this field modulo constants by $\mathcal{H}_\kappa$. All constants below may depend on the fixed parameter and the stated moment order. None depends on an integer approximation to δ^{-1} : $\delta > 0$ is always an arbitrary real mesh size.

7.1 Ordinary moments from the endpoint Hamiltonian

On an equal-time column, Z_l is the diagonal spin at site l , with values ± 1 . These are self-adjoint site operators; they should be distinguished from marked time rows. A path height increment is the negative of its step spin. Vacuum moments of the commuting site spins are their ordinary moments under $\mathbb{P}_c$.

Lemma 7.1 (Ordinary spatial moments). *Fix $0 < A < \infty$. Let $R \geq 1$, and let $g = (g_l)_{l \in \mathbb{Z}}$ be a finitely supported real sequence such that*

$$\sum_l |g_l| \leq AR, \quad \sum_l |g_{l+1} - g_l| \leq A, \quad \sum_l |g_{l+1} - g_l|^2 \leq A/R. \quad (7.2)$$

Then, for every $1 \leq b < \infty$,

$$\left\| \sum_l g_l Z_l \right\|_{L^b(\mathbb{P}_c)} \leq C_{A,b}. \quad (7.3)$$

In particular the conclusion holds if g has support of length at most AR , $|g_l| \leq A$, and $|g_{l+1} - g_l| \leq A/R$. The same assertions hold with the two axes interchanged.

Proof. Work throughout this proof in $\mathcal{H}_{\text{spin}}$, the closed span of $O\Omega$ with O a finite local spin matrix. By Lemma 4.6, this subspace contains Ω and reduces E . Every finite local spin matrix and its adjoint preserve the local generating vectors and hence their closed span. In particular $D = D_g = \sum_l g_l Z_l$ is bounded and self-adjoint on $\mathcal{H}_{\text{spin}}$. The shift preserves this subspace in both directions, since it translates local matrices and fixes Ω . In this proof E denotes its self-adjoint restriction to $\mathcal{H}_{\text{spin}}$, $B_M = \mathbf{1}_{\{E \leq M/R\}}$ is its spectral projection there, and all Hilbert-space norms are taken on this subspace.

First we prove a two-sided compression bound. For fixed M , Lemma 4.6 places the range of B_M inside the ambient spectral band $H_0 \leq CM/R$, when R is sufficiently large. The full-space estimates of Lemma 4.5 therefore apply to these compressions without increasing their norms. The identity

$$N(i) = \frac{1}{2} S^2(Z_0 + Z_1)$$

and translation by the unitary shift therefore give $\|B_M(Z_l + Z_{l+1})B_M\| \leq C_M/R$. Pair the consecutive sites in the sum for D . The coefficients of the paired sums have total absolute value at most AR ; the remaining single-site terms have total absolute coefficient at most $\sum_l |g_{l+1} - g_l|$. Consequently

$$\|B_M D B_M\| \leq C_{A,M}. \quad (7.4)$$

For bounded R , this follows instead from $\|D\| \leq \sum_l |g_l|$.

The local-link description of E in Lemma 4.6 gives

$$\|[D, [E, D]]\| \leq C \sum_l |g_{l+1} - g_l|^2 \leq C_A/R. \quad (7.5)$$

Indeed the diagonal part of a link commutes with D , and its spin exchange preserves $Z_l + Z_{l+1}$; only the coefficient difference $g_l - g_{l+1}$ enters each commutator. These identities initially hold on the local analytic core in $\mathcal{H}_{\text{spin}}$, which is a core for the restricted E . On that core, $[E, D]$ is a finite sum of bounded local operators. To extend it, approximate a vector in $\text{Dom}(E) \subset \mathcal{H}_{\text{spin}}$ in the graph norm by local core vectors and use $EDu = DEu + [E, D]u$ there. Boundedness of D and of this commutator, followed by closedness of E , shows that D preserves the restricted domain and that the identity extends to it. The same holds for every fixed local power of D , justifying the following energy-projection and vacuum calculations on this subspace.

Set $m_* = \|DB_1\|$, which is finite since D is bounded. For a unit vector $u \in B_1\mathcal{H}_{\text{spin}}$, expanding the double commutator yields

$$\langle Du, EDu \rangle = \frac{1}{2} \langle u, [D, [E, D]]u \rangle + \text{Re} \langle Du, DEu \rangle \leq \frac{C_A + m_*^2}{R}.$$

Here $Eu \in B_1\mathcal{H}_{\text{spin}}$ and $\|Eu\| \leq R^{-1}$. The component of Du above energy $2/R$ thus has squared norm at most $(C_A + m_*^2)/2$. Its component below $2/R$ is bounded by (7.4), with $M = 2$. Taking a supremum gives

$$m_*^2 \leq C_A + \frac{1}{2} m_*^2, \quad \text{and therefore} \quad \|DB_1\| = \|B_1 D\| \leq C_A.$$

We next bound powers of D on the vacuum. Since $E\Omega = 0$,

$$2\langle D^k\Omega, ED^k\Omega\rangle = \langle\Omega, [D^k, [E, D^k]]\Omega\rangle.$$

In the site-diagonal representation, let $D^{\text{sw},l}$ be the value of D after exchanging the spins at $l, l+1$. An exchange term in this double commutator has, up to a bounded link coefficient and sign, the multiplier $(D^k - (D^{\text{sw},l})^k)^2$. This nonnegative multiplier commutes with the exchange. Furthermore,

$$|D - D^{\text{sw},l}| \leq 2|g_l - g_{l+1}|, \quad |D^k - (D^{\text{sw},l})^k|^2 \leq C_{A,k}|g_l - g_{l+1}|^2(1 + |D|^{2k-2}).$$

For the second inequality use the factorization of $a^k - b^k$ and $|a - b| \leq 2A$. Taking the absolute value of an exchange expectation is bounded by the expectation of its nonnegative multiplier, since the exchange is a norm-one involution commuting with that multiplier. Summing the links proves

$$\langle D^k\Omega, ED^k\Omega\rangle \leq \frac{C_{A,k}}{R}(1 + \|D^{k-1}\Omega\|^2). \quad (7.6)$$

The low-energy component satisfies $\|B_1 D^k\Omega\| \leq C_A \|D^{k-1}\Omega\|$; on its complement, positivity of $E - R^{-1}$ bounds the squared norm by $R\langle D^k\Omega, ED^k\Omega\rangle$. Induction in k now bounds every $\|D^k\Omega\|$. These are the even ordinary moments of D ; interpolation gives all finite orders. Axial symmetry gives the other orientation. $\square$

Corollary 7.2 (Neutral blocks). *Let $b \in \mathbb{R}^2$, $r > 0$, and let f be a measurable real density supported in a square of side Ar centered within distance Ar of b , with*

$$\int f = 0, \quad |f| \leq Ar^{-2}.$$

For every finite q ,

$$\|\langle h_\delta, f \rangle\|_{L^q} \leq C_{A,q} \quad (7.7)$$

uniformly in b, r, δ . Products of finitely many such blocks have uniform absolute moment bounds, even when their supports overlap.

Proof. Suppose first $r/\delta \geq 1$, and put $R = r/\delta$. Integrate f on mesh cells, giving masses a_{jl} of size $O(R^{-2})$, with $O(R)$ possible indices in each direction. Let $m_j = \sum_l a_{jl}$, and choose a mass-one discrete profile (p_l) , of width $O(R)$, bounded by C/R . The masses $a_{jl} - m_j p_l$ are neutral in each column. Their discrete primitives are bounded by C/R , with consecutive differences $O(R^{-2})$. Summation by parts and Lemma 7.1, applied after multiplication by R , bound the L^q norm of each column contribution by C_q/R . Their sum is bounded by Minkowski's inequality. For the remaining masses $m_j p_l$, note that $\sum_j m_j = 0$, $|m_j| \leq C/R$; sum by parts in the other axis, then average against p_l . This gives the same bound. If $r < \delta$, the support meets only a bounded number of neighboring faces. Neutrality, bounded total variation, and unit height jumps give a deterministic bound. Finally apply Hölder's inequality to products, using the required higher moments. No disjointness is used. $\square$

7.2 Smooth test functions

Proposition 7.3. *For every finite family of smooth compactly supported densities $\phi_1, \dots, \phi_m$ of integral zero, the vector $(\langle h_\delta, \phi_i \rangle)_i$ converges jointly in law and in every joint moment to the corresponding pairings of $\mathcal{H}_\kappa$. The moment convergence is uniform when the tests vary in a compact smooth family with a common compact support.*

Proof. First assume that $\int \phi(x, y) dy = 0$ for every x . For fixed continuum time x , put $F(x, y) = \int_{-\infty}^y \phi(x, t) dt$. Discrete summation by parts writes the line pairing as

$$\int h_\delta(x, y) \phi(x, y) dy = - \sum_{l \in \mathbb{Z}} F(x, (l+1)\delta) (h(\lfloor x/\delta \rfloor, l+1) - h(\lfloor x/\delta \rfloor, l)).$$

The primitive is compactly supported, bounded, and changes by $O(\delta)$ at consecutive sites. Lemma 7.1 therefore bounds this pairing in every L^q , uniformly in x and δ . The same remains true for any finite product of line pairings.

Replace the forward height differences in this expression by the centered differences of Proposition 6.7, shifting the coefficients accordingly. The coefficient error has size $O(\delta)$ and consecutive differences $O(\delta^2)$, on $O(\delta^{-1})$ sites. Dividing this error by δ puts it under Lemma 7.1; the resulting L^q error is $O(\delta)$, uniformly in the line coordinate.

Expand a joint moment as an integral over its time coordinates. For distinct fixed times, Proposition 6.7 applies to the transverse Riemann sums, locally uniformly on their compact supports and including the rounding of faces. Their limit is Wick's formula for transverse derivatives of K_κ . Integration by parts against the compact primitives gives the undifferentiated test densities and the kernel (7.1). The line moment bounds dominate the integrand uniformly. Equal-time walls have Lebesgue measure zero, so dominated convergence completes the time integrations. The resulting products of logarithmic kernels are integrable: in each Wick pairing the two-point variables are disjoint from those of every other pair, and the logarithm is locally integrable in two dimensions. Gaussian moments determine the law. Axial symmetry gives the same result for horizontally line-neutral tests.

For an arbitrary smooth neutral ϕ , let

$$q(x) = \int \phi(x, y) dy, \quad p_R(y) = R^{-1} p(y/R), \quad \phi_R(x, y) = \phi(x, y) - q(x)p_R(y),$$

where p is a fixed smooth compact probability density. Then ϕ_R is vertically line-neutral, while qp_R is horizontally line-neutral. For fixed R , the already proved result gives

$$\lim_{\delta \downarrow 0} \mathbb{E} |\langle h_\delta, qp_R \rangle|^2 = \iint K_\kappa(z, z') q(x) p_R(y) q(x') p_R(y') dz dz'.$$

This tends to zero as $R \rightarrow \infty$. To see this, set $y = Rt$, $y' = Rt'$. The additive $-2\kappa \log R$ contributes zero because $\int q = 0$. The remaining logarithm is $\log \sqrt{((x-x')/R)^2 + (t-t')^2}$, which converges almost everywhere to $\log |t-t'|$, independently of x, x' . On the fixed supports it is bounded in absolute value by $C + \log_+(1/|t-t'|)$. Dominated convergence and neutrality of q prove the assertion. The identical approximation holds for $\mathcal{H}_\kappa$. Thus first taking the mesh limit at fixed R , and only then taking $R \rightarrow \infty$, proves joint convergence in law.

For any fixed compact support, Corollary 7.2 bounds all higher moments of each fixed smooth pairing. Uniform integrability therefore upgrades convergence in law to all joint moments. Finally, small changes in a smooth test on a fixed support give small L^q changes by the same corollary, after dividing by the sup norm of the change. A finite covering of a compact family proves the asserted uniformity. $\square$

Smooth neutral tests now have the claimed Gaussian limit and all its joint moments. To recover point increments, we compare each endpoint with a smooth average at successively smaller scales. The next estimate controls the correlation contributed by the smallest scale in such a product.

7.3 Removing smoothings at separated endpoints

Let ϱ be a smooth probability density supported in a small unit square, and define

$$H_{\delta,r}(b) = \int h_{\delta}(x) r^{-2} \varrho((x-b)/r) dx.$$

Differences $H_{\delta,r}(b) - H_{\delta,2r}(b)$ obey Corollary 7.2. At $r \asymp \delta$, the residual $h_{\delta}(b) - H_{\delta,r}(b)$ is deterministically bounded by unit height jumps. We will use both kinds of differences as individual endpoint factors.

Lemma 7.4 (A separated endpoint factor). *Fix an integer $k \geq 2$, constants $A, c_0, c_1 > 0$, a separation $d_0 > 0$, and a bounded region containing all factor centers. Choose $r_0 = a_k d_0$, with $a_k > 0$ sufficiently small in terms of these fixed data. Let $n \leq k$ and let*

$$A_i = \langle h_{\delta}, \mu_i \rangle, \quad 1 \leq i \leq n,$$

where each μ_i is a finite signed measure of mass zero, total variation at most A , and support in a square of side $A r_i$ about b_i . Assume $\|A_i\|_{L^{2k}} \leq A$. These hypotheses include the neutral density blocks of Corollary 7.2 and the terminal residuals just defined.

Suppose one selected factor has scale $c_0 \delta \leq r \leq r_0$, all other scales lie in $[r, r_0]$, and all other centers are at distance at least $c_1 d_0$ from its center. Then, for some $\tau > 0$,

$$\left| \mathbb{E} \prod_{i=1}^n A_i \right| \leq C(r/d_0)^{\tau}. \quad (7.8)$$

The constants are uniform in the centers, the measures, the mesh and the scales subject to these bounds. The other centers need not be mutually separated.

Proof. If empty time slabs separate the selected factor from all the others, the intervening rows damp its short height paths. General endpoint geometry need not have such a slab in either axial ordering. We will create one by translating the groups lying to the left and right of the selected factor's horizontal cluster. Those groups remain separated from the cluster by horizontal gaps; bounded analytic continuation across these gaps then returns them, one at a time, to their original positions.

The estimate with empty time slabs. First suppose that the selected support has an empty axial slab on each side, of width at least $c_2 d_0$, separating it from all other factors. Insert time cuts in these two slabs. The vectors defined by the exterior factors have bounded Hilbert norms. Here is the connection with ordinary moments. Read all heights on one side of a cut relative to a common face, along a common seam and spatial paths contained on that side. Spatial steps use equal-time diagonal spins; each time-seam step uses a time spin, with repeated occurrences allowed when several paths share the step. In a finite homogeneous system, reflection across the cut transposes the entire product and complex-conjugates its scalar coefficients. The crossing identity reverses the time-step spins and preserves the spatial-step spins, exactly as for the reflected height differences. The squared norm is therefore the ordinary expectation of the exterior product times its conjugate reflected product. This finite identity passes to the plane by Proposition 4.3. Hölder's inequality and the ordinary moments of the individual factors bound that expectation. The factors on a side may overlap; Hölder's inequality requires no separation among them.

Inside the slab, use neutrality to express the selected factor as an average of differences from a fixed reference face in its support. Each difference has a path of $O(r/\delta)$ steps, with a bounded number of straight segments. Choose the direction of the difference so that time runs forward;

spatial steps may have either sign. There remain $m \geq c_3 d_0 / \delta$ unmarked time rows on both sides. Pair consecutive steps along each straight segment. The paired current identity and Lemma 4.5 give the sandwiched norm bounds

$$\begin{aligned} \|P(1)^m(\text{two consecutive step sum})P(1)^m\| &\leq C m^{-1}, \\ \|P(1)^m(\text{one step})P(1)^m\| &\leq C m^{-\theta_0} \end{aligned} \quad (7.9)$$

for some $0 < \theta_0 \leq 1$. To verify the passage from band bounds to these inequalities, split both sides into the bands $H_0 \leq m^{-1}$ and $2^{j-1}/m < H_0 \leq 2^j/m$. A block between two such bands has current bound a power of the larger cutoff, while the time rows contribute $\exp(-c2^j - c2^{j'})$. The resulting double geometric-exponential sum is $O(m^{-1})$ for a paired current. For a single current use the equal-sign $O(u)$ and opposite-sign $O(u^\theta)$ estimates, with $\theta_0 = \min(1, \theta)$. Bands above a fixed cutoff are exponentially damped and the uncompressed row norms are bounded. Unit shifts commute with the damping and do not enlarge these norms.

There are $O(r/\delta)$ pairs and only a bounded number of unpaired steps. The bounded total variation of the selected factor and the exterior norm bounds give

$$\left| \mathbb{E} \prod_i A_i \right| \leq C \left(\frac{r}{d_0} + \left(\frac{\delta}{d_0} \right)^{\theta_0} \right) \leq C(r/d_0)^{\tau_0}, \quad \tau_0 = \min(1, \theta_0).$$

This includes a terminal residual: its comparison paths have bounded length and the same damping margins. Rounding the cuts changes their widths by only a bounded number of sites.

Creating a slab by two translations. Cluster the centers by their first-coordinate projections: consecutive projections belong to the same cluster if their gap is at most ad_0/k , for a sufficiently small fixed a . Every cluster has width at most ad_0 . Choose a_k much smaller than a/k , so that the supports preserve a gap of size $c_k d_0$ at each cluster boundary. In the cluster containing the selected center, every other center has second-coordinate distance at least $c_k d_0$ from it. Indeed its first-coordinate distance is at most ad_0 , whereas its Euclidean distance has the lower bound in the statement.

Translate the clusters strictly to the left as one group vertically, and do the same independently for the clusters strictly to the right. For each group there is an interval of translation amounts, of length at least $c_k d_0$ and at distance at most $C_k d_0$ from zero, such that every translation in the interval leaves a vertical gap of size $c_k d_0$ around the selected support. To construct one, exclude the intervals of translations putting any of its at most k centers within a fixed small multiple of d_0 of the selected height. Their total length is bounded by $C_k d_0$; choose a larger bounded translation range and take a complementary interval of the asserted size. The choices for the two groups are independent. Restrict to translations by even numbers of lattice sites. Lattice stationarity preserves each individual L^{2k} bound under these translations, so Hölder's inequality still controls the exterior products. Throughout the resulting rectangle of choices the selected factor has a free slab in the northward time ordering, so the preceding estimate applies.

Returning the two groups. The first-coordinate gaps have remained open throughout these vertical translations. They supply the damping needed to transfer the estimate back to zero translation. Write

$$K = \text{Arg}(S^2), \quad -\pi \leq K \leq \pi.$$

This is bounded spectral functional calculus. The low-band estimate $S = \pm 1 + O(H_0)$ gives $|K| \leq CH_0$ for small H_0 ; on the complement the same inequality follows by increasing C , since $|K| \leq \pi$. Thus

$$|K| \leq CH_0, \quad \|P(1)^N e^{izK}\| \leq 1 \quad \text{if} \quad |\text{Im } z| \leq N/C. \quad (7.10)$$

For integer m , $e^{imK} = S^{2m}$. Across a first-coordinate gap containing $N \geq c_k d_0 / \delta$ unmarked rows, translating an entire exterior group by $2m$ sites therefore inserts $P(1)^N S^{\pm 2m}$ between fixed exterior vectors. This identity follows by conjugating every site and seam insertion on that side by the site shift, which fixes the vacuum. Reflection and the ordinary moment estimates bound both exterior vectors uniformly. Equation (7.10) extends the pairing to a bounded holomorphic function of $z/(d_0/\delta)$ on a strip of fixed positive width. If one of the exterior groups is absent there is no restoration step for that group.

For completeness, the elementary analytic estimate used here is the following. A holomorphic function bounded by one on a fixed-width strip, whose absolute value is at most $a \leq 1$ on a fixed-length real interval within a fixed bounded distance of zero, has absolute value at zero at most Ca^ϑ , where $\vartheta > 0$ depends only on those geometric constants. Take a half-disc based on a shorter part of the interval. The two-constants theorem bounds the function by a^ω on an interior disc, since the harmonic measure of the base there is bounded below by $\omega > 0$. Cover the remaining path to zero by finitely many overlapping discs inside the strip. The three-circles inequality transfers a positive power bound at each step. The number and radii of these discs depend only on the fixed strip, interval, and distance bounds.

Our free-slab estimate initially holds only at the allowed integer values of m . Cauchy's estimate on a slightly smaller strip bounds the derivative in the normalized real coordinate. The gap between successive allowed values is $O(\delta/d_0)$, so throughout a shorter real interval the bound becomes

$$C(r/d_0)^{\tau_0} + C\delta/d_0 \leq C'(r/d_0)^{\tau_0},$$

using $r \geq c_0\delta$ and $\tau_0 \leq 1$. First restore the left group while keeping the right translation at any one of its allowed values. The analytic estimate gives the same bound with a possibly smaller positive exponent, uniformly over those right translations. Then restore the right group. The two reductions of the exponent still leave a fixed $\tau > 0$. This proves (7.8) in the original geometry. $\square$

Proposition 7.5. *For every $k \geq 1$, all k -point height-increment moments converge to the Wick increment moments of $\mathcal{H}_\kappa$, uniformly on compact sets of tuples whose endpoint sets belonging to different increments are disjoint.*

Proof. The first moment is zero by arrow-reversal symmetry. For $k \geq 2$, fix such a compact set of tuples and let $d_0 > 0$ be a lower bound on all cross-pair endpoint distances. Endpoints within a single pair may coincide. Choose $r_0 = a_k d_0$ as in Lemma 7.4, and put $r_j = 2^{-j} r_0$. For each endpoint, telescope its bump averages down to the last scale $r_j \asymp \delta$, and include the residual from that average to the specified face value. This is a finite exact identity, including for points on grid lines under the fixed floor convention.

Expand the product of increments. A factor is either a coarse increment at scale r_0 , or one of the two endpoint blocks at a dyadic scale, including the terminal residual. All individual blocks have uniformly bounded moments. The coarse increments also do: join their endpoints by a bounded-length chain of overlapping scale- r_0 bumps and telescope, applying Corollary 7.2 to each difference. The number of links is bounded by the fixed compact size and separation data.

In a summand containing an endpoint block, select one of minimum scale r . Decompose every other coarse increment by a bump chain avoiding a disc of radius $d_0/3$ about the selected endpoint. Such a chain exists with bounded length because both endpoints of that other pair are at least d_0 away; a polygonal detour around the disc has length bounded in terms of the same fixed data. After subdividing into steps of size $O(r_0)$, every factor other than the selected one has center at distance at least cd_0 from it. There are still at most k factors in each resulting product; the number

of resulting products is bounded independently of the mesh and dyadic levels. Apply Lemma 7.4 to each of them.

At fixed deepest dyadic index j , there are at most $C_k(1+j)^{k-1}$ choices of the other levels and endpoint labels. Consequently the error from keeping only scales coarser than r_{J_0} is bounded by

$$C \sum_{j \geq J_0} (1+j)^{k-1} 2^{-\tau j}, \quad (7.11)$$

uniformly once the mesh is fine enough to reach that level. This tends to zero as $J_0 \rightarrow \infty$. Notice that the terminal residual obeys the same bound; dropping it before applying the estimate would not justify point evaluation.

For fixed J_0 , Proposition 7.3 gives the joint moments of the smoothed increments, uniformly in their centers. The corresponding Gaussian Wick expressions converge uniformly as the smoothing radius tends to zero: each paired kernel involves endpoints from two different increments, so its arguments stay a positive distance apart. Combine this with (7.11). If the two endpoints within a pair coincide, that increment and its smoothed versions are identically zero; no additional exclusion is needed. $\square$

7.4 Ordinary finite-energy signed measures

Lemma 7.6. *Let μ be a compactly supported finite signed measure for which $\iint G(x, y) \mu(dx) \mu(dy)$ is an ordinary finite signed-measure integral. Then μ has no atoms, and*

$$\iint \log_+ \frac{1}{|x-y|} |\mu|(dx) |\mu|(dy) < \infty. \quad (7.12)$$

If μ, ν both satisfy this assumption, the corresponding mixed absolute logarithmic integral is finite as well.

Proof. The total variation of $\mu \otimes \mu$ is $|\mu| \otimes |\mu|$. Ordinary integrability is integrability with respect to this variation. On a fixed compact support, the negative part of $-\log|x-y|$ is bounded, which proves (7.12). An atom would give positive product mass to the logarithmic diagonal singularity, so there are no atoms.

For the mixed assertion take dyadic squares of side 2^{-j} in a box containing both supports. Up to an additive bounded kernel, $\log_+(1/|x-y|)$ is bounded above by a constant times $\sum_{j \geq 0} \mathbf{1}_{\{|x-y| \leq C2^{-j}\}}$. Write $a = |\mu|$, $b = |\nu|$. At a fixed level the mixed mass of these nearby pairs is at most $\sum_Q \sum_{Q' \sim Q} a(Q)b(Q')$, where each square has a bounded number of neighbors. Cauchy–Schwarz bounds this by

$$C \left(\sum_Q a(Q)^2 \right)^{1/2} \left(\sum_Q b(Q)^2 \right)^{1/2}.$$

The sums of $\sum_Q a(Q)^2$ and $\sum_Q b(Q)^2$ over all levels are finite by their self-integrability: pairs in one square are within $\sqrt{2}2^{-j}$, and summing their indicators is bounded by $C(1 + \log_+(1/|x-y|))$. A second Cauchy–Schwarz inequality over j proves the claim. $\square$

Proposition 7.7. *For every finite family of compactly supported, neutral, finite signed measures of the ordinary finite-energy class in Theorem 1.1, the associated height pairings converge jointly in law to the centered Gaussian vector with covariance obtained by integrating K_κ against the pairs of measures.*

Proof. Let $V(a) = \mathbb{E}(h(a) - h(0))^2$ for lattice-face displacement $a \in \mathbb{Z}^2$, and put $V(0) = 0$. Proposition 4.8 and axial symmetry imply

$$V(a) = 4\kappa \log(1 \vee |a|) + O(1),$$

with a uniform bounded error over all lattice displacements. For distinct continuum points x, y in a fixed compact set, write

$$V_\delta(x, y) = V(\lfloor x/\delta \rfloor - \lfloor y/\delta \rfloor).$$

For $0 < \delta \leq 1$ this gives

$$\left| V_\delta(x, y) - 4\kappa \log \delta^{-1} \right| \leq C(1 + |\log |x - y||). \quad (7.13)$$

Indeed when $|x - y|$ exceeds a fixed multiple of δ , the lattice separation is comparable to $|x - y|/\delta$. In the other case its size is bounded, and $\log \delta^{-1} \leq C + |\log |x - y||$. This latter case includes distinct points rounded to the same face.

For any four points, stationarity of increments and zero mean give

$$\begin{aligned} & \mathbb{E}[(h_\delta(x') - h_\delta(x))(h_\delta(y') - h_\delta(y))] \\ &= \tfrac{1}{2}(V_\delta(x', y) + V_\delta(x, y') - V_\delta(x', y') - V_\delta(x, y)). \end{aligned} \quad (7.14)$$

The four copies of $4\kappa \log \delta^{-1}$ cancel exactly. Equation (7.13) therefore supplies a majorant by a constant and four continuum logarithmic singularities.

For a neutral $\mu \neq 0$, its Jordan parts have the same mass $M > 0$. The measure $M^{-1}\mu^+(dx')\mu^-(dx)$ represents its pairing as an integral of $h_\delta(x') - h_\delta(x)$. Use this representation for two pairings. By Lemma 7.6, the four-logarithm majorant is integrable against the resulting product measure. Geometric coincidences of endpoints belonging to different pairings have measure zero because the Jordan parts are nonatomic. Distinct endpoints rounded to a common lattice face remain covered by the same majorant. Proposition 7.5 for two increments and dominated convergence thus give, for every such μ, ν ,

$$\lim_{\delta \downarrow 0} \mathbb{E}[\langle h_\delta, \mu \rangle \langle h_\delta, \nu \rangle] = \iint K_\kappa(x, y) \mu(dx) \nu(dy). \quad (7.15)$$

The zero measure causes no exception.

It remains to identify laws beyond second moments. Let $\mu_\epsilon = \mu * \varrho_\epsilon$, where the mollifier is smooth, nonnegative and compactly supported. These measures are smooth and neutral. Mollification converges in logarithmic energy:

$$\iint K_\kappa(x, y) (\mu - \mu_\epsilon)(dx) (\mu - \mu_\epsilon)(dy) \longrightarrow 0. \quad (7.16)$$

To check this for the present ordinary integral, the logarithm convolved in one or both variables converges pointwise off the diagonal. On the fixed enlarged compact support its absolute value is bounded by $C(1 + |\log |x - y||)$, uniformly in small ϵ . For $|x - y| \geq C\epsilon$ this follows by comparison of distances; for $|x - y| < C\epsilon$, scaling the bounded compact mollifier gives $C + |\log \epsilon| \leq C'(1 + |\log |x - y||)$. Lemma 7.6 permits dominated convergence in each of the four energy terms, proving (7.16).

Apply (7.15) to $\mu - \mu_\epsilon$. It gives

$$\lim_{\epsilon \downarrow 0} \limsup_{\delta \downarrow 0} \|\langle h_\delta, \mu - \mu_\epsilon \rangle\|_{L^2} = 0.$$

At fixed ϵ , Proposition 7.3 identifies the Gaussian limit. The same energy estimate approximates the Gaussian pairing. Applying this argument to a finite family, or to each of its linear combinations, proves the asserted joint law and cross covariance. $\square$

The scalar pairings and separated increments are now identified. We next strengthen convergence of individual pairings to convergence in norms. The ordinary restriction norms use fields localized on a fixed box. The source's outward seminorm instead tests the global field over an increasing sequence of boxes; it requires a separate tail estimate.

7.5 Negative regularity and the outward test convention

Fix a smooth compactly supported probability density ρ , and write $X_\delta = h_\delta - \langle h_\delta, \rho \rangle$. For a compactly supported smooth cutoff χ , put $X_\delta^\chi = \chi X_\delta$.

Lemma 7.8 (Pinned averages). *Let $0 < \ell \leq 1$, and let $\psi_{b,\ell}$ be supported in a square of side $A\ell$ about b , with $|\psi_{b,\ell}| \leq A\ell^{-2}$. For centers b in a fixed compact set and every finite q ,*

$$\|\langle X_\delta^\chi, \psi_{b,\ell} \rangle\|_{L^q} \leq C_{A,q,\chi,\rho}(1 + |\log \ell|). \quad (7.17)$$

The estimate is uniform in $\delta > 0$.

Proof. Let $m = \int \chi \psi_{b,\ell}$, so $|m| \leq C$. Subtract m times a smooth mass-one bump of scale ℓ at b . The difference obeys Corollary 7.2. Next telescope that bump through the scales $\ell, 2\ell, \dots, 1$, and finally compare the scale-one bump with ρ . Each difference is neutral, has bounded total variation, and is supported and bounded at its stated scale. There are $O(1 + |\log \ell|)$ differences. The same argument works below the mesh scale because the final clause of Corollary 7.2 has no lower bound on ℓ/δ . $\square$

Proposition 7.9. *For every bounded open $U \subset \mathbb{R}^2$ and every $-1 < \alpha < 0$, the pinned fields converge in law, with tightness, in the ordinary restriction spaces*

$$C_{\text{res}}^\alpha(U) := B_{\infty,\infty}^\alpha(U), \quad B_{p,q}^\alpha(U), \quad W^{\alpha,p}(U), \quad 1 \leq p < \infty, \quad 1 \leq q \leq \infty,$$

to the corresponding pinning of $\mathcal{H}_\kappa$. Here the Hölder laws are supported on the separable closure of smooth functions in the displayed norm. Convergence also holds in the literal outward-scaling completion defined by (16) of [7, Definition 2.6(iii)]. No boundary regularity of U is assumed. An intrinsic dual interpretation of the negative Sobolev norms is treated at the end of this section.

Proof. Choose a compactly supported χ equal to one on a box containing $\bar{U}$ and $\text{supp } \rho$. Use compactly supported wavelets of sufficient finite regularity, including their coarse-scale scaling functions [2, Section 4.C]. We use the full-space Besov coefficient characterization in [14, Proposition 2.3]. Normalize the scale- 2^{-j} test functions as densities of size $O(2^{2j})$, and denote their averages of X_δ^χ by $a_{j,k}$. There are $O(2^{2j})$ relevant indices k . For $\alpha < \beta < 0$, choose a moment order b with $2 + b\beta < 0$. Lemma 7.8 gives

$$\mathbb{E} \sum_{j \geq 0} \sum_k |2^{j\beta} a_{j,k}|^b \leq C_b \sum_{j \geq 0} 2^{(2+b\beta)j} (1+j)^b < \infty. \quad (7.18)$$

The coarse block has the same bound. This bounds the stronger $B_{\infty,\infty}^\beta$ norm in probability. On a fixed compact support its bounded sets have uniformly vanishing tails in $B_{\infty,\infty}^\alpha$, since

$$\sup_{j > J, k} 2^{j\alpha} |a_{j,k}| \leq 2^{(\alpha-\beta)J} \sup_{j,k} 2^{j\beta} |a_{j,k}|.$$

For finitely many levels only finitely many coefficients remain. This proves compactness by finite-dimensional truncation, and also puts the fields and limits in the separable smooth closure.

The same gain proves the other asserted norm compactness. Indeed a compactly supported distribution bounded in $B_{\infty,\infty}^\beta$ has scale blocks satisfying $\|\Delta_j X\|_{L^p} \leq C 2^{-j\beta}$ for every finite p . Thus

$$\sum_{j>J} 2^{j\alpha} \|\Delta_j X\|_{L^p} \leq C \sum_{j>J} 2^{j(\alpha-\beta)} \longrightarrow 0.$$

This controls every ℓ^q Besov block norm, including $q = \infty$. It also controls the negative Bessel-potential norm: on a scale block, $(1 - \Delta)^{\alpha/2}$ has an L^1 -bounded convolution kernel after extracting $2^{j\alpha}$, so summing these bounds works for all $p \geq 1$. The same summable block bound controls the usual full-space negative Slobodeckij realization. Passing to its restriction uses the defining quotient norm. Restricting a global extension to U is continuous by definition; no extension operator from U is being invoked. Proposition 7.3 identifies every subsequential limit, and hence proves the full mesh convergence.

For clarity, retain also the source's printed outward convention. Let $Q_0 = (-1, 1)^2$, and let $\mathcal{T}_1(Q_0)$ consist of compactly supported neutral densities whose zero extension is one-Lipschitz. For smooth functions its seminorm is

$$\|f\|_{\text{out},\alpha,U} = \sup_{\substack{0 < \varepsilon \leq 1, \\ \varphi \in \mathcal{T}_1(Q_0) \\ \text{supp } \varphi \subset U/\varepsilon}} \varepsilon^{-\alpha} \left| \int f(x/\varepsilon) \varphi(x) dx \right|. \quad (7.19)$$

The absolute value is immaterial for the symmetric test family. Constants vanish in this seminorm. Its completion is understood modulo its kernel, as in the source.

Write $\varepsilon = 2^{-j}a$, $1/2 \leq a \leq 1$. The exact mesh identity, including the floor convention, is

$$h_\delta(x/\varepsilon) = h_{2^{-j}\delta}(x/a).$$

After the change of variables $x = ay$, the test becomes $a^2\varphi(ay)$, supported in $2Q_0$, neutral, with a uniformly bounded Lipschitz norm. The condition involving U only reduces this family. For any fixed $-1 < \gamma < 0$, local $\mathcal{C}^\gamma$ control bounds all these pairings. One can see this directly from a wavelet expansion: cancellation of fine wavelets against a Lipschitz test gains 2^{-j} , and the resulting series converges because $1 + \gamma > 0$. The preceding moment argument, with a still stronger negative exponent and arbitrarily high moment order, therefore gives random variables $Z_{j,\delta}$ such that

$$\sup_{\delta} \|Z_{j,\delta}\|_{L^b} \leq C_b, \quad \sup_{2^{-j-1} \leq \varepsilon \leq 2^{-j}} (\text{quantity in (7.19)}) \leq C 2^{j\alpha} Z_{j,\delta}.$$

A local pin may be inserted in the rescaled field since every test is neutral. In particular the supremum over $j \geq J$ tends to zero in L^b , uniformly in δ , by the summability of $\sum_{j \geq J} 2^{j\alpha}$. On finitely many bands, the tested original coordinates lie in one bounded box. Their test norms are bounded by constants depending on the number of bands, so the stronger local compactness already proved applies.

These estimates also give the required smooth completion, rather than only a bound on a larger space of functionals. Take a smooth compact cutoff χ_0 equal to one on a sufficiently large fixed box and a compact smooth mollifier θ_η , and use

$$f_{\delta,J,\eta}(x) = \chi_0(2^{-J}x)(X_\delta * \theta_\eta)(x).$$

Convolution preserves the local negative Hölder bound, uniformly for $0 < \eta \leq 1$, since its rescaled radius on band j is at most 2^{-j} . For the outward bands $j \geq J$, the cutoff becomes a bump of scale at most 2^{J-j} in the rescaled coordinates. The local $\mathcal{C}^\gamma$ test bound gives its contribution the

additional factor $2^{(J-j)(2+\gamma)} \leq 1$; comparison of the fixed pin with the rescaled pin costs at most $C(1+j)$ by the same dyadic telescoping as in Lemma 7.8. Thus the cutoff approximations have tails bounded by a constant times $\sum_{j \geq J} (1+j)2^{j\alpha}$, uniformly in probability and in the mollification parameter. On finitely many bands send $\eta \downarrow 0$; then send $J \rightarrow \infty$. These smooth approximations and the vanishing tail prove membership and tightness in the completion of (7.19). Its limits agree with the smooth-test limit on each finite band. This proves the literal outward assertion too. $\square$

7.6 Intrinsic negative Sobolev norms on an arbitrary open set

The restriction-space proof above already requires no regularity of ∂U . We give a further argument to cover the convention in which a negative Sobolev norm is the norm dual to an intrinsic positive Slobodeckij norm on U . The distinction matters for irregular open sets, where an extension theorem is unavailable.

For $0 < s < 1$, $1 < q < \infty$, write

$$\|\phi\|_{W^{s,q}(U)}^q = \|\phi\|_{L^q(U)}^q + \iint_{U \times U} \frac{|\phi(x) - \phi(y)|^q}{|x - y|^{2+sq}} dx dy.$$

For $q = \infty$, use the sup norm and the essential supremum of $|\phi(x) - \phi(y)|/|x - y|^s$. The cells in the following argument are intersections of ordinary dyadic squares with U ; they may have arbitrarily small relative area and arbitrarily irregular shapes.

Lemma 7.10 (Averages on small-area cells). *Let Q be a square of side $0 < \ell \leq 1$ in a fixed bounded box, and let $A \subset Q$ be measurable with $|A| > 0$. Put $t = |A|/\ell^2$. Then for every finite b ,*

$$\left\| \frac{1}{|A|} \int_A X_\delta(x) dx \right\|_{L^b} \leq C_b(1 + |\log \ell| + |\log t|) \quad (7.20)$$

uniformly in δ and in the shape of A .

Proof. We first refine the comparison of line weights. Suppose nonnegative weights a_l , supported on $O(R)$ consecutive sites, satisfy

$$\sum_l a_l = m > 0, \quad a_l \leq M/R, \quad R \geq 1.$$

Let b_l be a mass-one profile bounded by C/R , and assume that the union of the supports of a and b lies in one interval of length CR . We claim

$$\left\| \sum_l (a_l - mb_l) h_l \right\|_{L^b} \leq C_b m (1 + \log_+(M/m)). \quad (7.21)$$

Increasing a fixed constant in M if needed, assume $M \geq m$. For dyadic integer $v \geq 1$, let $A_v a$ replace the weights in each consecutive block of v sites by their block average. The difference $d = A_v a - A_{2v} a$ is neutral in every $2v$ -block. For its normalized primitive $g_l = m^{-1} \sum_{i \leq l} d_i$, this gives

$$\sum_l |g_l| \leq Cv, \quad \sum_l |g_{l+1} - g_l| \leq C, \quad \sum_l |g_{l+1} - g_l|^2 \leq C/v.$$

The first bound sums a block length times its variation mass. For the last, use $\|d\|_\infty \leq Cm/v$ and $\|d\|_1 \leq 2m$. Lemma 7.1 and summation by parts give cost at most $C_b m$ for each flattening step. This use of the lemma explains why its statement retained the three summability conditions, without a support-diameter hypothesis.

If $\min(R, Rm/M) > 1$, choose a dyadic v_0 comparable from below to that number; otherwise put $v_0 = 1$. When $v_0 > 1$, one can start directly with $a - A_{v_0}a$. Its primitive has the same first two bounds, and

$$m^{-2}\|a - A_{v_0}a\|_2^2 \leq Cm^{-2}\|a\|_2^2 \leq CM/(Rm) \leq C/v_0.$$

When $v_0 = 1$, this initial difference vanishes. Continue flattening until $v \asymp R$, where comparison with mb has bounded cost by the original width- R case of the lemma. The number of steps is $O(1 + \log_+(M/m))$, proving (7.21).

Now rescale the square Q to a unit square. For any probability density $f \leq M$ on it, set $m(x) = \int f(x, y) dy$. Apply (7.21) to its mesh masses on each vertical line, comparing them with $m(x)$ times one bounded profile. Minkowski's inequality bounds the total cost by

$$C_b \int m(x)(1 + \log(M/m(x))) dx \leq C_b(1 + \log M).$$

The last inequality is Jensen's inequality for the concave function $u \mapsto u \log(M/u)$, with $\int m = 1$ on an interval of length one. Compare the remaining horizontal marginal with a bounded profile by the same line estimate. We have therefore compared the original density with a fixed scale- ℓ product smoothing, at cost $C_b(1 + \log M)$. Telescoping that smoothing to scale one and then to the pin costs $C_b(1 + |\log \ell|)$, as in Lemma 7.8. If $\ell < \delta$, all comparisons inside a comparable square use only a bounded number of faces and have a deterministic bound; the remaining pin comparison is unchanged. Taking $f = \ell^2|A|^{-1}\mathbf{1}_A$ in rescaled coordinates gives $M = t^{-1}$ and proves (7.20). $\square$

Proposition 7.11 (Intrinsic dual convergence). *Let $U \subset \mathbb{R}^2$ be bounded and open, let $0 < s < 1$, and let $1 \leq p < \infty$, with conjugate exponent $q = p'$. The pinned fields converge in law, with norm tightness, in $(W^{s,q}(U))^*$ to the canonically pinned Gaussian field with kernel K_κ . The laws are supported on a separable closed subspace of this dual. No extension property or regularity of ∂U is required.*

Proof. Take a fixed square containing U , subdivided dyadically at side lengths $\ell_j \asymp 2^{-j}$. For a square Q at level j , put

$$U_Q = Q \cap U, \quad t_Q = |U_Q|/|Q|, \quad X_{\delta,Q} = |U_Q|^{-1} \int_{U_Q} X_\delta.$$

Ignore cells of zero area. Let

$$Z_{\delta,j} = \sum_{Q \text{ at level } j} X_{\delta,Q} \mathbf{1}_{U_Q}, \quad F_{\delta,j}(\phi) = \int_U Z_{\delta,j} \phi,$$

and let $P_j \phi$ denote cell averaging of ϕ on the same partition. Consistency of averaging implies

$$(F_{\delta,j+1} - F_{\delta,j})(\phi) = \int_U Z_{\delta,j+1}(\phi - P_j \phi). \quad (7.22)$$

Suppose first $p > 1$, so $q < \infty$, and choose $0 < \zeta < sq$. Call a parent P of side $\ell = \ell_j$ thick when $t_P \geq \ell^\zeta$. Jensen's inequality gives

$$\begin{aligned} \int_{U_P} |\phi - \phi_P|^q &\leq |U_P|^{-1} \iint_{U_P \times U_P} |\phi(x) - \phi(y)|^q dx dy \\ &\leq C \ell^{sq} t_P^{-1} \iint_{U_P \times U_P} \frac{|\phi(x) - \phi(y)|^q}{|x - y|^{2+sq}} dx dy. \end{aligned}$$

Summing over thick parents yields

$$\|\phi - P_j \phi\|_{L^q(\text{thick})} \leq C \ell^{s-\zeta/q} [\phi]_{W^{s,q}(U)}. \quad (7.23)$$

On all cells, averaging is an L^q contraction, so the same difference has L^q norm at most $2\|\phi\|_{L^q}$.

Lemma 7.10, with $b = p$, implies

$$\mathbb{E} \|Z_{\delta,j+1}\|_{L^p(U)}^p \leq C_p (1+j)^p.$$

For this inequality sum $|Q|t_Q(1+j+\log(1/t_Q))^p$ over children, and use $t(1+\log(1/t))^p \leq C_p$ for $0 < t \leq 1$. A child of a thin parent has $t_Q \leq 4\ell^\zeta$. The sharper elementary inequality

$$t(A + \log(1/t))^p \leq C_p A^p t^{1/2}, \quad A \geq 1,$$

therefore gives

$$\mathbb{E} \|Z_{\delta,j+1} \mathbf{1}_{\text{thin}}\|_{L^p(U)}^p \leq C_p (1+j)^p \ell^{\zeta/2}. \quad (7.24)$$

The sum of the areas $|Q|$ is bounded by the area of the surrounding square, independently of the geometry of U . Equations (7.22)–(7.24) and Hölder's inequality now give

$$\mathbb{E} \|F_{\delta,j+1} - F_{\delta,j}\|_{(W^{s,q}(U))^*} \leq C(1+j) \left(\ell_j^{s-\zeta/q} + \ell_j^{\zeta/(2p)} \right). \quad (7.25)$$

Both exponents are positive.

At $p = 1$, $q = \infty$, the intrinsic Hölder seminorm gives directly

$$\|\phi - P_j \phi\|_\infty \leq C \ell_j^s [\phi]_{C^s(U)}.$$

There is no need to distinguish thin parents in this case. The first moment version of the cell estimate gives

$$\mathbb{E} \|F_{\delta,j+1} - F_{\delta,j}\|_{(W^{s,\infty}(U))^*} \leq C(1+j) \ell_j^s. \quad (7.26)$$

Thus at every finite p the right-hand side, denoted by a_j , is summable over j , uniformly in δ .

Let $\mathcal{E}_0$ be the closed linear span in $\mathcal{E} = (W^{s,q}(U))^*$ of the countable collection of cell indicator pairing functionals. It is a separable Banach space. For each fixed mesh, X_δ is a bounded function on the bounded set U , and its conditional cell averages converge to it in $L^p(U)$. Since $L^p(U)$ maps continuously into $\mathcal{E}$, they also converge in $\mathcal{E}_0$. Summing the preceding estimates proves

$$\sup_\delta \mathbb{E} \|X_\delta - F_{\delta,J}\|_{\mathcal{E}} \leq \sum_{j \geq J} a_j \longrightarrow 0. \quad (7.27)$$

Every coefficient $X_{\delta,Q}$ is the pairing with the neutral measure

$$\mu_Q = |U_Q|^{-1} \mathbf{1}_{U_Q} dx - \rho dx.$$

This bounded compact density has ordinary finite logarithmic energy, regardless of the shape or area of U_Q . Proposition 7.7 therefore gives joint convergence of the coefficients at each finite level. Couple their Gaussian limits using the countable family of Gaussian pairings with μ_Q , and denote the corresponding finite-level elements of $\mathcal{E}_0$ by F_J . Finite-dimensional convergence and the lower semicontinuity of the norm transfer the bound $\mathbb{E} \|F_{j+1} - F_j\|_{\mathcal{E}} \leq a_j$. The sum of these norms is almost surely finite and integrable, so F_J converges almost surely and in $L^1(\mathcal{E}_0)$ to an $\mathcal{E}_0$ -valued random variable F . Finite-level convergence and (7.27) then prove $X_\delta \Rightarrow F$ in $\mathcal{E}_0$, including norm tightness.

We finally identify the limit on the whole intrinsic test space, not only on interior smooth tests. For $\phi \in W^{s,q}(U)$, the measure

$$\mu_\phi = \phi \mathbf{1}_U dx - \left(\int_U \phi \right) \rho dx$$

has ordinary finite logarithmic energy. Indeed $q > 1$ or $q = \infty$, and on a fixed compact set Hölder's inequality for the logarithmic kernel gives

$$\iint |\log |x - y|| |f(x)f(y)| dx dy \leq C_q \|f\|_{L^1} \|f\|_{L^q}.$$

For finite q , $P_J \phi \rightarrow \phi$ in $L^q(U)$; for $q = \infty$ the intrinsic Hölder bound gives uniform essential convergence. The associated neutral measures consequently converge in logarithmic energy. Hence $F_J(\phi)$ converges to the canonical pinned Gaussian pairing with μ_ϕ . This identifies F on every intrinsic test and rules out an unidentified boundary functional in the larger dual. The proof is complete. $\square$

Remark 7.12 (Other difference-norm conventions). If a negative Besov convention is dual to a positive intrinsic Besov difference norm, choose $0 < t < s < 1$. The positive test norm of order s controls the intrinsic $W^{t,q}$ norm: in a dyadic difference expansion, the gain $2^{-j(s-t)}$ is summable, for every sequence exponent including its endpoints. The convergence just proved for $(W^{t,q}(U))^*$ therefore gives the corresponding negative Besov dual convergence of order $-s$. This is in addition to the ordinary restriction Besov conclusion of Proposition 7.9.

For $0 < c < 2$, the section has established all three convergence mechanisms with covariance K_κ . Since $K_\kappa = 4\pi\kappa G$, the squared multiplier is $4\pi\kappa$. Proposition 5.1 computes it for $c < 1$; the conditional normalization theorem quoted in the introduction applies after these convergence statements have been proved for $1 \leq c < 2$.

7.7 The endpoint topology implication

The endpoint argument uses a different route to Gaussian convergence. We isolate the additional implication needed for the ordinary restriction spaces and arbitrary-domain intrinsic duals, rather than identify these spaces with the outward seminorm in [7].

Proposition 7.13 (Endpoint topology implication). *At $c = 2$, suppose that the height pairings with every finite family of ordinary finite-energy neutral measures converge jointly to those of $\sigma\Gamma$. Then the commonly pinned fields also converge, with tightness, in all the negative-regularity spaces stated in Propositions 7.9 and 7.11. This includes the ordinary restriction Hölder, Besov and Sobolev spaces on every bounded open set, the outward-scaling completion, and the explicitly defined intrinsic Sobolev duals.*

Proof. We obtain the required moment bounds directly from [7, Theorem 4.5 and equations (28)–(30)], which applies at $c = 2$. For $2n$ increments whose continuum endpoints lie in a fixed bounded box, that theorem implies, for $0 < \delta \leq 1$,

$$\left| \mathbb{E}_2 \prod_{i=1}^{2n} (h_\delta(x_i) - h_\delta(y_i)) \right| \leq C_n \sum_\pi \prod_{\{i,j\} \in \pi} \left(1 + \sum_{u \in \{x_i, y_i\}} \sum_{v \in \{x_j, y_j\}} \log_+ \frac{1}{|u - v|} \right). \quad (7.28)$$

Here and below inequalities with a logarithmic singularity are used off the corresponding coincidence diagonals. To check the mesh cutoff, let d_{ij} be the minimum cross-endpoint distance after rounding

to lattice faces, and D_{ij} the continuum distance. Rounding gives $D_{ij} \leq \delta(d_{ij} + C)$, while each lattice increment length is at most C/δ . The logarithmic branch of the cited bound therefore satisfies

$$1 + \log_+ \frac{C/\delta}{1 \vee d_{ij}} \leq C' \left(1 + \log_+ \frac{1}{D_{ij}} \right).$$

Its other branch is at most one. The logarithm of the minimum of four distances is bounded by the sum of their positive logarithms, giving (7.28). If a rounded increment is zero, the entire moment vanishes; otherwise every scale ratio is well-defined. Thus the estimate also handles distinct continuum endpoints that round to the same face.

Let f be a probability density supported in a square of side $0 < \ell \leq 1$ and bounded by $C/(\ell^2 t)$, where $0 < t \leq 1$. For every point v , the disc-mass bound gives

$$\begin{aligned} \int \log_+ \frac{1}{|u - v|} f(u) du &= \int_0^\infty \left(\int_{|u-v| < e^{-s}} f(u) du \right) ds \\ &\leq \int_0^\infty \min \left(1, \frac{C e^{-2s}}{\ell^2 t} \right) ds \leq C' (1 + |\log \ell| + |\log t|). \end{aligned}$$

Write $\langle X_\delta, f \rangle$ as the integral of $h_\delta(x) - h_\delta(y)$ against $f(x)\rho(y) dx dy$. After expanding its $2n$ -th moment, each pair in (7.28) involves endpoint variables disjoint from those in every other pair. The pair integrals therefore multiply. The fixed smooth pin has a uniform logarithmic integral, and the last display bounds the other ones. Consequently

$$\mathbb{E}_2 |\langle X_\delta, f \rangle|^{2n} \leq C_n (1 + |\log \ell| + |\log t|)^n.$$

This is stronger than the bound needed in (7.20). Taking $f = |A|^{-1} \mathbf{1}_A$, for an arbitrary positive-area subset A of the square, proves that estimate at the endpoint. Signed densities bounded by $C\ell^{-2}$ are dominated by a constant times the uniform probability density on their supporting square. The same moment expansion therefore proves (7.17), including its smooth cutoff. Interpolation gives all finite moment orders. Coarser meshes have uniform deterministic bounds on the fixed box.

We also need scale-uniform neutral blocks for the outward completion. If both endpoints of every increment lie in a square of side r , the preceding rounding argument replaces $1 + \log_+(1/D_{ij})$ by $1 + \log_+(r/D_{ij})$. Integration against densities bounded by Cr^{-2} gives a constant independent of r and δ when $\delta \leq r$. When $\delta > r$, unit jumps give the same bound deterministically. For a neutral signed density g on that square, choose a probability density $p \leq Cr^{-2}$ there and use the exact representation

$$\langle h_\delta, g \rangle = \iint (h_\delta(x) - h_\delta(y)) g(x) p(y) dx dy.$$

Domination by $|g(x)|p(y) \leq Cr^{-4}$ gives the required moment bound without dividing by a possibly small Jordan mass. This proves the neutral-block estimate (7.7) at $c = 2$.

We may now repeat the wavelet proof of Proposition 7.9: for $\alpha < \beta < 0$ choose b with $2 + b\beta < 0$. The endpoint pinned-average bound gives (7.18); finite-level truncations and the gain $2^{j(\alpha-\beta)}$ yield compactness in every asserted restriction norm. Smooth neutral pairings identify the limits by the hypothesis. The global localization and continuous restriction to U require no boundary extension theorem. The neutral-block bound and the exact mesh-rescaling identity give the outward tails and smooth approximations by the same proof.

Finally the endpoint cell-average estimate gives every thick- and thin-cell bound in the proof of Proposition 7.11, including the case $p = 1$, $q = \infty$. Its finite-level coefficients are pairings with ordinary finite-energy measures, so their joint convergence is part of the hypothesis. The summable dual norm tail proves convergence in the same separable closed subspace; energy approximation identifies the limit on the entire intrinsic test space. This proves the proposition. $\square$

8 Completion of the proof

Proof of Theorem 1.1. At $c = 2$, Proposition A.5 proves the result. Its independent loop argument supplies the rotational-invariance premise of the public endpoint proof, and its final step retains the ordinary restriction and intrinsic topologies of the present theorem.

Fix $0 < c < 2$. Sections 2 and 3 construct the limits of every finite word of rows and marks. Proposition 4.3 identifies the physical words with local arrow probabilities of the prescribed plane law. For $c < 1$ the construction uses the exactly balanced sector throughout and proves existence in the required iterated order. Where an auxiliary near-balanced limit is used for $c \geq 1$, the equilibrium identification gives the already existing balanced law. The same proposition identifies the two axial descriptions of its local observables.

The angular representation of Section 4 has orbit mass $\kappa < \infty$ and height variogram $4\kappa \log |x| + O(1)$. Proposition 6.7 establishes the holomorphic and antiholomorphic Wick rules, with logarithmic covariance $-2\kappa \log |x - y|$. Propositions 7.3, 7.5 and 7.7 identify the smooth tests, separated increments and finite-energy measures. Propositions 7.9 and 7.11 give the restriction, outward and intrinsic norm convergences. In each case the squared GFF multiplier is $4\pi\kappa$. Their estimates apply to arbitrary real mesh sizes and identify every subsequential limit; hence the limits hold along the full $\delta \downarrow 0$.

If $0 < c < 1$, Proposition 5.1 gives

$$4\pi\kappa = \frac{2}{\pi - \lambda} = \frac{2}{\arccos \Delta}.$$

For $1 \leq c < 2$, we have now proved a subsequential GFF limit in the required senses for the balanced law, so the hypothesis of [7, Theorem 4.3] is satisfied. That theorem identifies its squared multiplier as $-1/f''(0)$, where f is the source's free-energy function. By [7, Theorem 4.4], $f''(0) = -\frac{1}{2} \arccos \Delta$, giving the same multiplier. This use of the conditional theorem comes after, and does not supply, the convergence argument.

Finally, ordinary finite signed-energy tests have the joint covariance in the statement by Proposition 7.7. All pinning choices differ by an additive constant, the negative-regularity assertions use the common chosen pin, and increments cancel it. The uniformity in (1.3) and the arbitrary-open-set conclusions have already been proved in their respective propositions. This completes the theorem. $\square$

A Rotation of the fair-signed height at $q = 4$

At $c = 2$, the height can be represented by critical $q = 4$ FK loops with independent fair orientations. We use this representation to prove isometry invariance of its separated increment correlations, the symmetry input in the endpoint GFF argument of [7].

Averaging over orientations leaves only loops that separate at least two of the prescribed endpoint pairs. Such loops have a positive macroscopic diameter. The homotopy coupling of [5, Sections 3.5–3.6 and 4.2] matches these loops with their multiplicities and preserves the topological information from which we recover their endpoint separations. Qualitative bulk RSW estimates control the unmatched exceptional events. We use the coupling itself because the loop-set distance in the rotation theorem does not retain multiplicities.

For $k \geq 2$, let

$$\mathcal{D}_k = \left\{ \mathbf{u} = (u_1, u'_1, \dots, u_k, u'_k) : \{u_i, u'_i\} \cap \{u_j, u'_j\} = \emptyset \quad (i \neq j) \right\}.$$

Coincidence within one pair is allowed. Give the critical $q = 4$ FK loops independent fair signs and let h_δ be the resulting height, with jumps of size one and an immaterial additive constant. At a continuum point use any fixed face representative within $O(\delta)$. Put

$$\Phi_{k,\delta}(\mathbf{u}) = \mathbb{E} \prod_{i=1}^k (h_\delta(u'_i) - h_\delta(u_i)). \quad (\text{A.1})$$

Theorem A.1 (Isometry invariance of the endpoint correlations). *For every even $k \geq 2$ and every compact $K \subset \mathcal{D}_k$,*

$$\lim_{\delta \downarrow 0} \sup_{\mathbf{u} \in K} \sup_{I \in \text{Isom}(\mathbb{R}^2)} |\Phi_{k,\delta}(\mathbf{u}) - \Phi_{k,\delta}(I\mathbf{u})| = 0. \quad (\text{A.2})$$

We first identify this fair-signed field with the balanced plane law of Theorem 1.1. The torus loop weight and the torus FK weight differ, so the passage to the plane is part of the argument.

Lemma A.2 (The endpoint plane correspondence). *After the fixed similarity identifying the medial square grid with $\mathbb{Z}^2$, the fair unit-jump height of critical plane FK₄ has the same gradient law as the zero-slope six-vertex height with $\mathbf{a} = \mathbf{b} = 1$ and $\mathbf{c} = 2$.*

Proof. On an even square torus, give equal weight to each pair consisting of noncrossing local switch pairings and an orientation of each resulting closed loop. Erasing orientations gives weight 2^ℓ to a configuration with ℓ loops. Erasing pairings gives an ice-rule arrow configuration: each of the four a - and b -type vertices admits one compatible pairing, and each of the two c -type vertices admits two. Thus the arrow marginal has weights $(1, 1, 2)$, and the loop orientations, conditionally on the unoriented loops, are independent and fair. This is the endpoint specialization of the local calculation in [24, Section 2, Equations (2.1)–(2.3)].

The torus weight 2^ℓ need not equal the torus FK weight. Precisely the needed plane passage is supplied by [24, Lemma 3.1]: its unoriented loop law converges locally to critical plane FK₄. Fix a finite set of observed medial edges. Bulk annular RSW implies that every plane FK loop meeting this set is finite almost surely. For any $\varepsilon > 0$, choose a box containing all these loops with probability at least $1 - \varepsilon$. The event that the observed loops close inside this box is determined by its finitely many pairings. On that event, local convergence transfers their full loop identities, and hence their independent fair orientations. Letting the box grow proves convergence of the local arrow law to independently oriented plane FK loops.

On the other hand, the unrestricted six-vertex torus laws have the zero-slope plane limit used in [7, Theorem 2.2 and the paragraph following it]. That paragraph identifies the unrestricted torus limit with the slope-zero law defined through balanced cylinders. This is a lattice Gibbs-law fact preceding the GFF argument. The two descriptions of the arrow limit agree, and arrows determine unit height differences. Additive constants do not affect the gradient law. This proves the lemma. $\square$

Consequently Theorem A.1 supplies exactly the $\mathbf{c} = 2$ case of the symmetry input [7, Theorem 4.1]. The proof uses the lattice correspondence and qualitative FK estimates, before identifying any Gaussian limit.

A.1 Even moments retain only macroscopic loops

Smooth the discrete contours locally into disjoint Jordan loops. This changes their location by $O(\delta)$. For such a loop L set

$$D_i(L) = \mathbf{1}_{\text{int } L}(u'_i) - \mathbf{1}_{\text{int } L}(u_i).$$

A reference orientation may multiply all the $D_i(L)$ by one sign depending on L ; the fair sign absorbs that choice. Let $\Pi_{\text{even}}(k)$ denote the partitions of $\{1, \dots, k\}$ whose blocks all have even size. Conditional expectation over the signs gives the exact identity

$$P_k(\mathcal{L}; \mathbf{u}) := \mathbb{E} \left[\prod_i (h_\delta(u'_i) - h_\delta(u_i)) \middle| \mathcal{L} \right] = \sum_{\pi \in \Pi_{\text{even}}(k)} \sum_{\substack{f: \pi \rightarrow \mathcal{L} \\ f \text{ injective}}} \prod_{B \in \pi} \prod_{i \in B} D_i(f(B)). \quad (\text{A.3})$$

At positive mesh only finitely many loops separate any prescribed pair: each such loop crosses a fixed finite lattice path between its endpoints. Hence the expansion is a finite identity. In particular, there is no limiting interchange in its derivation.

Call a loop relevant if at least two coordinates of $(D_1(L), \dots, D_k(L))$ are nonzero. Every loop in a nonzero summand of (A.3) is relevant. If endpoints belonging to distinct slots are separated by $d > 0$, a relevant loop surrounds one endpoint from each of two distinct slots. Therefore

$$\text{diam } L \geq d. \quad (\text{A.4})$$

Here we used that the closure of a Jordan domain is contained in the convex hull of its boundary. Also, a separating loop intersects the segment joining the separated endpoints. Thus, if the endpoints lie in $B(0, M)$, every relevant loop meets that disk. For rounded continuum endpoints, replace d by $d - O(\delta)$.

We record the quantitative estimates needed below. They hold also for the two-angle mixed isoradial lattices in the rotation construction. For any fixed choice of the two angles, their mesh geometry has bounded distortion and their bulk conditional RSW bounds are available from [5, Theorem 2.2, Remark 2.3 and Proposition 2.12].

Lemma A.3 (Counts, escape, and deterministic-point avoidance). *Let $N(a, M)$ be the number of loops of diameter at least a meeting $B(0, M)$. For δ sufficiently smaller than a , and every fixed $p < \infty$,*

$$\|N(a, M)\|_{L^p} \leq C_p(1 + M/a)^2. \quad (\text{A.5})$$

There are $C, \gamma > 0$ such that, with harmless fixed changes of the radii,

$$\mathbb{P}\{\exists L : L \cap B(0, M) \neq \emptyset, L \not\subset B(0, R)\} \leq C(M/R)^\gamma, \quad R \geq 10M, \quad (\text{A.6})$$

$$\mathbb{P}\{\exists L : \text{diam } L \geq a, \text{dist}(z, L) \leq r\} \leq C((r + C\delta)/a)^\gamma, \quad 0 < r < a/100. \quad (\text{A.7})$$

The constants are independent of the deterministic center z .

Proof. Bulk RSW in a sequence of buffered annuli gives a polynomial bound for an actual primal or dual arm between their extreme rims: each annulus has a uniformly positive conditional probability of a blocking circuit of the opposite color.

We also need a count consequence of RSW. In an annulus, an *actual component* consists of vertices connected by open paths inside the annulus; boundary wiring alone does not identify its vertices. The number of such components crossing a fixed-aspect annulus has a geometric tail. To see this, explore complete actual components from the inner rim and reveal all their closed exiting edges. Remove the explored components, including noncrossing ones, until a crossing component has been found. The domain Markov property leaves on the surviving graph only the boundary identifications inherited from the original rims: the closed exiting edges prevent an explored component from wiring surviving bulk vertices. Extend the surviving configuration by closed edges on the deleted graph. Its conditional law is stochastically dominated by FK on the original annulus with every

vertex on both original rims wired together. Even under this maximal boundary condition, bulk RSW gives a dual circuit in a buffered middle annulus with probability bounded below. Such a circuit blocks a further actual primal crossing, irrespective of the identifications at the original rims. Repeating this exploration bounds the conditional chance of each additional crossing by a fixed number below one. Apply the same argument with the colors exchanged.

A loop has a primal cluster on one side and a dual cluster on the other. The cluster on its bounded side is its *interior-side cluster*; an open path in this cluster running alongside an arc of the contour is an *interior bank*. Each loop is the unique exterior contour of its interior-side cluster, so distinct loops give distinct colored clusters. Cover $B(0, M)$ by $O((1 + M/a)^2)$ balls of radius $a/100$. A loop of diameter at least a meeting one of these balls has an interior bank crossing an annulus about the ball's center whose two radii are fixed multiples of a . Choosing one such annular component for each loop gives distinct actual components for distinct global clusters. The preceding geometric tail and Minkowski's inequality prove (A.5).

For (A.6), a bank of an escaping loop gives an actual arm from scale M to scale R . For (A.7), a bank of the indicated loop gives an arm from scale $r + C\delta$ about z to a fixed fraction of a . The polynomial arm bound proves both assertions. $\square$

Fix a compact $K \subset \mathcal{D}_k$, write $d_K > 0$ for its minimum cross-slot endpoint distance, and choose M containing every endpoint in K . If $k = 2m$, an even partition has at most m blocks, so (A.3) implies

$$|P_{2m}(\mathcal{L}; \mathbf{u})| \leq C_m N(d_K/2, M+1)^m, \quad \mathbf{u} \in K, \quad (\text{A.8})$$

for sufficiently small mesh. All fixed moments of the right-hand side are uniformly bounded by Lemma A.3. These are bounds for the *sign-averaged polynomial*; no uniform absolute-moment assertion about a product of pointwise height differences is being made.

Two consequences will be useful. First, removing loops not contained in $B(0, R)$ changes the expectation of the polynomial by at most $C_K(M/R)^{\gamma/2}$, by (A.6), (A.8), and Cauchy–Schwarz. Second, if corresponding endpoints of two tuples in a compact subset of $\mathcal{D}_k$ move by at most r , evaluate both polynomials in the same loop sample. Off the events in (A.7) at those endpoints, every relevant loop has unchanged endpoint indicators. Hence

$$|\Phi_{k,\delta}(\mathbf{u}) - \Phi_{k,\delta}(\mathbf{v})| \leq C_K((r + C\delta)/d_K)^{\gamma/2}. \quad (\text{A.9})$$

This estimate is uniform in the absolute location of the tuple. It covers within-slot coincidences and changes of the rounding convention. The supremum over K is a supremum of deterministic tuple estimates; we do not require one avoidance event to hold simultaneously at all points occurring in K .

A.2 The FK coupling and its marked clusters

Let $\mathbb{L}(\alpha)$ be the rectangular isoradial lattice in [5, Section 1.4]; $\mathbb{L}(\pi/2)$ is a fixed rotation and dilation of the square lattice. The next lemma extracts the precise part of its coupling that we use. Its assertion is about loop multiplicities, not merely about proximity of loop sets.

Lemma A.4 (Homotopy coupling with dense marked clusters). *Fix $\alpha \in (0, \pi)$. As the mesh tends to zero, the construction of [5, Section 3.6] gives a coupling of two auxiliary loop samples. Their restrictions to a deterministic window exhausting the plane have, unconditionally, the respective homogeneous laws on $\delta\mathbb{L}(\pi/2)$ and $\delta\mathbb{L}(\alpha)$.*

Outside an event of probability tending to zero, there are correspondingly indexed, pairwise disjoint marked clusters, called nails, whose mesh tends to zero and whose window exhausts the

plane. Corresponding nails have diameter tending to zero and lie at distance tending to zero from the same grid point. There is a bijection ψ between all loops surrounding at least two but not all nails in the two samples. It preserves their oriented reduced crossing words relative to the nail grid.

Proof. This is the construction preceding the metric conclusion, with its parameters retained. In the notation of [5, Definition 3.8 and Section 3.6, pp. 49–50], choose $\eta = N^{-b}$ and $t = N^{-b^2}$, for $b > 0$ sufficiently small. On an event of probability $1 - O(N^{-c})$, the selected nail clusters survive the whole extrema-resampling and track-exchange chain. The proof in the cited section retains a bijection of the non-trivial loops throughout this chain. During an extrema-resampling step, each such loop either avoids the resampled boxes or the configuration remains unchanged. During a track exchange, the local changes preserve the loops without breaking them and preserve their homotopy classes relative to the surviving nails. Thus the oriented reduced words are unchanged at every step, for both colors of contour.

Initially, the nail grid spacing is $t^{1/2}N$, nail diameters are $O(tN)$, and the nail window has size comparable to $t^{1/4}N$. The terminal nail positions differ by these errors from their images under $M_{\pi/2, \alpha}$. The independent FK argument in [5, Section 4.2, proof of Theorem 1.7] proves $M_{\pi/2, \alpha} = \text{id}$. We may therefore use the already constructed coupling with corresponding nails at the same limiting positions.

For clarity, these scales can be arranged for every small physical mesh δ . Set $a = b^2$ and choose $N = 2\lfloor \frac{1}{2}\delta^{-1/(1-3a/8)} \rfloor$, so N is even as required by the track construction. Then the physical grid spacing, nail diameter, and nail-window size are respectively of orders

$$N^{-a/8}, \quad N^{-5a/8}, \quad N^{a/8}.$$

The homogeneous observation window has scale δN , which also diverges. Center the nail window by translating each endpoint sample by a period vector of its own homogeneous lattice nearest the same centering vector. The two chosen translations differ by $O(\delta)$; this only adds a vanishing error to corresponding nail positions. The endpoint restrictions have exactly the homogeneous laws, by [5, Proposition 2.11]. After centering, both the nail window and the homogeneous observation windows exhaust the plane. The probability $O(N^{-c})$ tends to zero. This proves the stated extraction; no conclusion about curve-set convergence is used to recover the bijection. $\square$

Here is the topological information in a crossing word that will matter. Orient each loop by the reference convention and let $s_L \in \{-1, 1\}$ be its winding number at an interior point. For any two nails, join chosen points in them by a path of nail-grid edges, with transverse perturbations if necessary. The algebraic intersection number of L with this path equals the difference of its winding numbers at the two nail points. It is read from the oriented crossing word. Cyclic permutation and deletion of an adjacent inverse pair do not change it. Consequently, if n, n' and $\tilde{n}, \tilde{n}'$ are corresponding nail points,

$$s_L(\mathbf{1}_{\text{int } L}(n') - \mathbf{1}_{\text{int } L}(n)) = s_{\psi(L)}(\mathbf{1}_{\text{int } \psi(L)}(\tilde{n}') - \mathbf{1}_{\text{int } \psi(L)}(\tilde{n})). \quad (\text{A.10})$$

We need only these differences. In particular, no inference of an absolute inside set from a finite grid word is required.

A.3 Transfer of the correlation polynomial

Fix $\alpha \in (0, \pi)$ and compact $K \subset \mathcal{D}_k$. Choose $0 < r < d_K/1000$. Apply Lemma A.4 and, for each endpoint, select an indexed nail close to it. For all small enough meshes both corresponding nails lie within $r/2$ of that endpoint. Nails selected for endpoints in distinct slots are distinct. In both

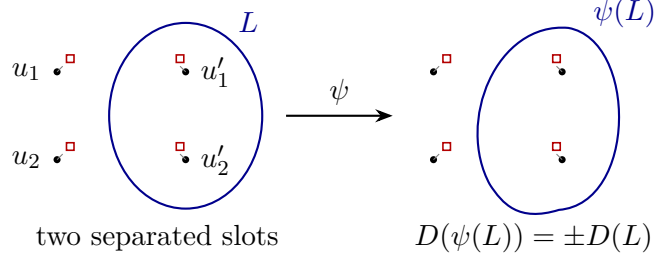


Figure 2: Nearby nails (squares) detect the separations of endpoint pairs (dots). The homotopy bijection preserves oriented winding differences. A single sign change of a loop's whole separation vector cancels when that loop occurs an even number of times.

samples discard the event that a loop of diameter at least $d_K/4$ comes within r of an endpoint. Lemma A.3 bounds the probability of these exceptions by $C_K(r/d_K)^\gamma + o(1)$, uniformly for the individual tuples in K .

On the remaining event, a relevant loop L surrounds the nails near one endpoint of each of two distinct slots, and excludes a nail near the other endpoint of a separated slot. It is therefore non-trivial in the exact sense of Lemma A.4, so $\psi(L)$ is defined. For those two slots the winding differences at the corresponding nails are nonzero by (A.10). Thus $\psi(L)$ separates each of the two nail pairs. In particular, it encloses one nail from each of two distinct slots, whose distance is at least $d_K - O(r + \delta)$. This proves that $\psi(L)$ itself has diameter at least $d_K/4$. Its endpoint-avoidance event is consequently applicable; smallness of the partner has not been assumed away.

Endpoint indicators and nearby-nail indicators agree for both loops, as illustrated in Figure 2. Equation (A.10) now gives, for *every* slot,

$$D_i(\psi(L)) = \frac{s_L}{s_{\psi(L)}} D_i(L). \quad (\text{A.11})$$

The inverse bijection gives the same statement starting with a relevant loop in the second sample. Hence it is a bijection between all relevant loops, preserving their entire separation vectors up to one sign per loop. For a block B in an even partition,

$$\prod_{i \in B} D_i(\psi(L)) = \left(\frac{s_L}{s_{\psi(L)}} \right)^{|B|} \prod_{i \in B} D_i(L) = \prod_{i \in B} D_i(L).$$

Transporting the injective labels in (A.3) therefore makes the two polynomials *exactly equal* on the good event. This uses the actual bijection, so distinct loops are never identified with one another. Cauchy–Schwarz and (A.8) bound the expectation error by $C_K(r/d_K)^{\gamma/2} + o(1)$.

It remains to pass from the auxiliary mixed samples to the intended homogeneous ones. Fix $R > 10M$ and couple their restrictions identically in $B(0, R)$ using the local-law assertion of Lemma A.4; for small enough mesh this disk lies in both homogeneous windows. Off the escaping-loop event in either sample, all relevant complete loops are identical. Equations (A.6) and (A.8) bound the expectation error by $C_K(M/R)^{\gamma/2}$. Let the mesh tend to zero, then $r \downarrow 0$ and $R \uparrow \infty$. Writing $\Phi_{k,\delta}^\alpha$ for the fair-signed correlation on $\delta\mathbb{L}(\alpha)$, we obtain

$$\lim_{\delta \downarrow 0} \sup_{\mathbf{u} \in K} \left| \Phi_{k,\delta}^{\pi/2}(\mathbf{u}) - \Phi_{k,\delta}^\alpha(\mathbf{u}) \right| = 0. \quad (\text{A.12})$$

No diameter threshold needs to be a continuity point: all small loops were removed by the even-moment identity itself.

Proof of Theorem A.1. The rectangular model on $\mathbb{L}(\alpha)$ is invariant under reflection $S_{\alpha/2}$ about $e^{i\alpha/2}\mathbb{R}$ [5, Remark 1.8]. Its fair-sign height correlations inherit that symmetry. If the chosen endpoint rounding is not reflection equivariant, the reflected representatives differ by $O(\delta)$, which is harmless by (A.9). Apply (A.12) at the original and reflected tuples. The square model is consequently asymptotically invariant under $S_{\alpha/2}$, uniformly on every compact subset of $\mathcal{D}_k$. Composing with its exact horizontal-reflection symmetry yields invariance under rotation through α . Square symmetries cover the endpoints and the other half of the circle. This proves the assertion for each fixed rotation.

The estimate (A.9) upgrades this to uniformity in the rotation angle. Indeed, the orthogonal images of K form a compact subset of $\mathcal{D}_k$, and a finite angular net approximates every rotation with endpoint displacement at most C_K times its angular mesh. First use convergence at the finitely many net angles, and then let the angular mesh decrease. Reflections follow by composition with an exact square symmetry. This argument requires no estimate uniform in a degenerating isoradial angle.

Finally, any translation differs from a period translation of the mesh- δ square lattice by a vector of length $O(\delta)$. Exact period invariance and the location-uniform estimate (A.9) handle this residual displacement, uniformly over *all* translation vectors. Every plane isometry is a translation followed by an orthogonal map. The fixed rotation and dilation identifying $\mathbb{L}(\pi/2)$ with $\mathbb{Z}^2$ merely change the mesh convention and compact set. These observations prove (A.2) for the original square lattice. $\square$

A.4 The endpoint field and all test classes

Proposition A.5 (Completion at the endpoint). *The conclusions of Theorem 1.1 hold at $c = 2$, with squared multiplier $2/\pi$.*

Proof. Lemma A.2 identifies the fair unit-jump height with the prescribed balanced plane law. Theorem A.1 supplies [7, Theorem 4.1] at $c = 2$; odd correlations vanish by arrow reversal. We can therefore follow the isotropic argument of that paper with its symmetry premise now proved here.

Specifically, [7, Theorems 6.1, 7.1 and 8.1] give this alternative: either the field converges to a fixed GFF multiple in all three senses of its Definitions 2.5–2.7, or two mesh sequences give GFF limits with distinct nonnegative multipliers. These senses include separated correlations, finite-energy tests and the outward completion. The free-energy identification in its Theorems 4.3–4.4 gives every possible squared multiplier as

$$\sigma^2 = -\frac{1}{f''(0)} = \frac{2}{\arccos(-1)} = \frac{2}{\pi}.$$

Thus the alternative with two distinct multipliers is excluded, and convergence holds along the full mesh limit.

Proposition 7.13 now applies to the finite-energy convergence. Using the lattice-cutoff estimate of [7, Theorem 4.5], it proves convergence and tightness in the ordinary restriction and intrinsic negative topologies on every bounded open set, with the common chosen pin. Its proof retains endpoints rounded to the same face and arbitrarily small positive-area cells at the boundary.

Finally, the fixed similarity identifying the FK medial lattice with $\mathbb{Z}^2$ changes the logarithmic Green kernel only by an additive constant. Neutrality cancels that constant, so the coefficient and test conclusions agree with Theorem 1.1. $\square$

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